MAGNETOM

Replacement of Parts System

RF System

Document Version

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This chapter describes replacement procedures for the FRUs of the **RF system**.

Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance <= 100 mOhm with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. RFPA) and the protective earth wire connected to the "main component" (e.g. ACC).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

Tab. 1 RFSU

RFSU					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tools**
Synthesizer D10 (73 88 734) D120 (7T/3T) (10018540) Avanto: >26124 Espree: > 30292	80 (110) min.	(→ Synthesizer, Modulator, Re-ceiv-er / Page 23)	■ TU/ Tuning Calibration ■ TU/ BC Tuning ■ TU/ RF Characteristic ■ TU/ RF Characteristic LC (if MNO option is installed) ■ TU/ Coil Power Losses or BC Power Losses ■ TU/ Abs.Rec.Path Calibration or BC Image Brightness	■ QA/ Abs. Rec. Path Calibration Check or BC Image Check ■ QA/ BC Tuning Check ■ QA/ RF Verify/ RF Verify LC (if MNO option is installed) ■ QA/ Coil Power Losses Check or BC Power Losses Check ■ QA/ Stability Check ■ QA/ Synthesizer Check ■ QA / Stability_Long Term Check (if available, add. 30 min.)	■ Normal tool kit ■ Screwdriver Torx TX8, TX10 ■ ESD equipment ■ BNC 2m Service cable for step TU/ Tuning Calibration (8733479) ■ only if MNO option is installed: Service Plug kit (07715720) ■ Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration

RFSU					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tools**
Transmitter D111 (100 92 367) (replaces D101 and D10/120)	80 (110) min.	(→ Synthesizer, Modulator, Receiv- er / Page 23)	▪ TU/Tuning Calibration ▪ TU/ BC Tuning ▪ TU/ RF Characteristic ▪ TU/ RF Characteristic LC (if MNO option is installed) ▪ TU/ Coil Power Losses or BC Power Losses Check ▪ TU/ Abs.Rec.Path Calibration or BC Image Brightness	▪ QA/ Abs. Rec. Path Calibration Check or BC Image Check ▪ QA/ BC Tuning Check ▪ QA/ RF Verify/ RF Verify LC (if MNO option is installed) ▪ QA/ Coil Power Losses Check or BC Power Losses Check ▪ QA/ Stability Check ▪ QA/ Synthesizer Check ▪ QA / Stability_Long Term Check (if available, add. 30 min.)	▪ Normal tool kit ▪ Screwdriver Torx TX8, TX10 ▪ ESD equipment ▪ BNC 2m Service cable for step TU/ Tuning Calibration (8733479) ▪ only if MNO option is installed: Service Plug kit (07715720) ▪ Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration

RFSU					
FRU	Time	Instructions	Adjustments	Final checks	Tools
Modulator D101 (73 89 062) with MKO: Modulator 7T (100 18 301) (Usable for 1.5 / 3 T or other systems depending on the DIP switch setting) Note label on the modulator.	80 (110) min.	(→ Synthesizer, Modulator, Receiver / Page 23) Check DIP switch position on modulator board: 1.5/3T: Off other/7T: On	■ TU/ Tuning Calibration ■ TU/ BC Tuning ■ TU/ RF Characteristic ■ TU/ RF Characteristic LC (if MNO option is installed) ■ TU/ Coil Power Losses or BC Power Losses ■ TU/ Abs.Rec.Path Calibration or BC Image Brightness	■ QA/ Abs. Rec. Path Calibration Check or BC Image Check ■ QA/ BC Tuning Check ■ QA/ RF Verify/ RF Verify LC (if MNO option is installed) ■ QA/ Coil Power Losses Check or BC Power Losses Check ■ QA/ Stability Check ■ QA / Stability_Long Term Check (if available, add. 30 min.)	■ Normal tool kit ■ Screwdriver Torx TX8, TX10 ■ ESD equipment ■ BNC 2m Service cable for step TU/ Tuning Calibration (8733479) ■ only if MNO option is installed: Service Plug kit (07715720) ■ Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration

RFSU					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tools**
Receiver D112 (73 89 047) Receiver D122 (10276813) Avanto: >26410 Espree: >30439	80 (110) min.	(➔ Synthesizer, Modulator, Receiv-er / Page 23)	▪ TU/ Tuning Calibration ▪ TU/ BC Tuning ▪ TU/ RF Characteristic ▪ TU/ RF Characteristic LC (if MNO option is installed) ▪ TU/ Coil Power Losses or BC Power Losses ▪ TU/ Rel. Rec.Path Calibration or RX Gain Calibration ▪ TU/ Abs.Rec.Path Calibration or BC Image Brightness	▪ QA/ Rel. Receive Path Calibration Check or RX Gain Calibration Check ▪ QA/ Abs. Rec. Path Calibration Check or BC Image Check ▪ QA/ BC Tuning Check ▪ QA/ RF Verify/ RF Verify LC (if MNO option is installed) ▪ QA/ Coil Power Losses Check or BC Power Losses Check ▪ QA/ Stability Check ▪ QA / Stability_Long Term Check (if available, add. 30 min.)	▪ Normal tool kit ▪ Screwdriver Torx TX8, TX10 ▪ ESD equipment ▪ BNC 2m Service cable for step TU/ Tuning Calibration (8733479) ▪ only if MNO option is installed: Service Plug kit (07715720) ▪ Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration
Power Supply (30 95 783)	40 (70) min.	(➔ Power supply (PSU) / Page 31)	n.a.	▪ QA/ Stability Check ▪ MR image ▪ QA/ Stability_Long Term Check (if available, add. 30 min.) ▪ In case of noise in patient images: Run QA/Exp/RFNoise Check	▪ Normal tool kit ▪ Screwdriver Torx TX8, TX10

RFSU

FRU	Time	Instructions	Adjustments	Final checks	Tools
Backplane D20 (73 89 138)	40 (70) min.	(→ RFSU Frame, Back-plane / Page 34)	n.a.	■ QA/ Stability Check ■ MR image ■ QA/ Stability_Long Term Check (if available, add. 30 min.) ■ In case of noise in patient images: Run QA/Exp/RFNoise Check	■ Normal tool kit ■ Screwdriver Torx TX8, TX10
RFSU frame (73 91 001)	40 min.	(→ RFSU Frame, Back-plane / Page 34)	If replacing electronic RF components: Refer to the specific RF components	■ QA/ Stability Check ■ MR image	■ Normal tool kit ■ Screwdriver Torx TX8, TX10
ACC internal cable	40 min.	n.a.		■ QA/ RF Verify	Torque wrench
RFSU internal cable (e.g., 73 91 456)	40 min.	n.a.		■ QA/ Stability Check	Torque wrench

Tab. 2 RFAS

RFAS					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tools**
TALES (73 91 886)	70 min.	(→ TALES, BCCS / Page 45)	▪ TU/ Tuning Calibration ▪ TU/ BC Tuning ▪ TU/ RF Characteristic ▪ TU/ RF Characteristic LC (if MNO option is installed) ▪ TU/ Coil Power Losses or BC Power Losses ▪ TU/ Abs.Rec.Path Calibration or BC Image Brightness	▪ QA/ Abs. Rec. Path Calibration Check or BC Image Check ▪ QA/ BC Tuning Check ▪ QA/ RF Verify/ RF Verify LC (if MNO option is installed) ▪ QA/ Coil Power Losses Check or BC Power Losses ▪ QA/ Expert/ RF Path Attenuation Check	▪ Non-magnetic tool kit ▪ Non-magnetic torque wrench ▪ ESD equipment ▪ BNC 2m Service cable for step TU/ Tuning Calibration (8733479) ▪ only if MNO option is installed: Service Plug kit (07715720) ▪ Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration
BCCS_63 (Avanto) (47 53 229) BCCS_63P (Espree) (100 17 953)	70 min.	(→ TALES, BCCS / Page 45)			
RCCS D11 (73 89 013) Avanto: <27081 Espree: <30785 RCCS_2 D11 (102 77 196) Avanto: >27080 Espree: >30784	70 min.	(→ RCCS / Page 46)	▪ TU/ Rel. Receive Path Calibration or RX Gain Calibration ▪ TU/ Abs.Rec.Path Calibration or BC Image Brightness	▪ QA/ Abs. Rec. Path Calibration Check or BC Image Check ▪ QA/ Rel. Receive Path Calibration Check or RX Gain Calibration Check ▪ QA/ Stability Check	▪ Non-magnetic tool kit ▪ ESD equipment

RFAS					
FRU	Time	Instructions	Adjustments	Final checks	Tools
TAS_3TM (73 92 058) Avanto: <27127 Espree: all TAS_CM (102 77 008) Avanto: >27126 Espree: not used	120 min.	(→ TAS_3TM - TAS_CM / Page 56)	▪ TU / RF Characteristics ▪ TU/ RF Characteristic LC (if MNO option is installed) ▪ TU/ Coil Power Losses or BC Power Losses ▪ TU/ Abs.Rec.Path Calibration or BC Image Brightness	▪ QA/ Abs. Receive Path Cal. Check or BC Image Check ▪ QA/ RF Verify/ RF Verify LC (if MNO option is installed) ▪ QA/ Coil Power Losses Check or BC Power Losses Check ▪ QA/ Exp/ RF Path Attenuation Check ▪ MR measurement with local TX/RX coil (e.g., Extremity coil)	▪ Normal tool kit ▪ Open-ended wrench SW 36, 42 ▪ Two pieces of water hose for depressurizing the secondary water cooling circuit ▪ Torque wrench ▪ only if MNO option is installed: Service Plug kit (07715720)
SAMI (73 83 123) (only for systems with MKO)	60 min.	Refer to Options/ Installation of MKO	n.a.	▪ MR measurement with Body coil and local TX/RX coil.	▪ Normal tool kit ▪ Open-ended wrench SW 36, 42 ▪ Torque wrench
RF cable between TALES and Body Coil	40 min.	n.a.	n.a.	n.a.	▪ Non-magnetic tool kit ▪ Non-magnetic torque wrench

Tab. 3 RFIS

RFIS					
FRU	Time	Instructions	Adjustments	Final checks	Tools
RFIS Mother-board (73 89 195)	90 min.	(→ RFIS / Page 66)	n.a.	n.a.	▪ Non-magnetic tool kit ▪ Non-magnetic torque wrench **Espree:** Special short antimagnetic screwdriver (part no. 5137083)
DYSCON_BC (73 89 377)	45 min.	(→ RFIS / Page 66)	n.a.	n.a.	Non-magnetic tool kit
DYSCON_16 (73 88 775)	45 min.	(→ RFIS / Page 66)	n.a.	n.a.	Non-magnetic tool kit
CAN_AND_CODE_CTRL (73 89 351)	45 min.	(→ RFIS / Page 66)	n.a.	n.a.	Non-magnetic tool kit
Power supply E4A (30 95 700)	45 min.	(→ RFIS / Page 66)	n.a.	▪ Check Error LEDs on RFIS ▪ Run QA/ Coil Check for Body Coil. ▪ Run Coil QA/ Coil Check with a local coil connected to the ap-propriate LC Plug(s).	Normal tool kit

Tab. 4 RFPA

RFPA					
FRU	Time	Instructions	Adjustments	Final checks	Tools
RFPA (Avanto/ DORA) (47 53 062) RFPA (Espree/ PORA) (73 92 470)	90 (110) min.	(→ RFPA / Page 77)	■ TU/ RF Characteristics ■ TU/ RF Characteristic LC (if MNO option is installed) ■ TU/ Coil Power Losses or BC Power Losses ■ TU/ Abs.Rec.Path Calibration or BC Image Brightness	■ QA/ Abs. Rec. Path Calibration Check or BC Image Check ■ QA/ RF Verify/ RF Verify LC (if MNO option is installed) ■ QA/ Coil Power Losses Check or BC Power Losses Check ■ QA/ Stability_Long Term Check (if available, add. 30 min.)	■ Normal tool kit ■ only if MNO option is installed: Service Plug kit (07715720)
Systems with MKO: (Avanto/CORA) (73 83 339)	60 min.	refer to Options/MKO/ Installation	■ TU/ RF Characteristic LC	■ QA/ RF Verify LC	■ Normal tool kit ■ Service Plug kit (07715720)

Tab. 5 Body Coil

Body Coil					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tools**
Body Coil Avanto (81 20 540)	5.5 hr.	(→ Body Coil (BC Avan-to) / Page 82)	■ TU/ Tuning Calibration ■ TU/ BC Tuning ■ TU/ RF Characteristic ■ TU/ Coil Power Losses or BC Power Losses ■ TU/ Abs.Rec.Path Cali-bration or BC Image Brightness	■ QA/ Abs. Rec. Path Cal-ibr. Check or BC Image Check ■ QA/ BC Tuning Check ■ QA/ RF Verify ■ QA/ Coil Power Losses Check or BC Coil Power Losses Check ■ QA/ Spike Check ■ SAR monitor test (VB17A and higher)	■ Non-magnetic tool kit ■ Non-magnetic measuring tape or gauge ■ Non-magnetic spirit level ■ Mechanical service support (part no. 8590841) ■ Open-end wrench SW 18 (part no. 8120904) - (delivered with new BC) ■ Screwdriver (73 66 771) for BC trimmer capacitor adjust-ment ■ BNC 2m Service cable for step TU/ Tuning Calibration (8733479) ■ Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration

Body Coil					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tools**
Upgrade Espree straight bore Gradient/Body coil to Slimline	5 days	(→ Upgrade Espree straight bore Gradient/ Body Coil to Slim-line / Page 106)	■ Complete reshim ■ Complete TuneUp/QA 1.5 days	■ General Quality Assurance ■ QA/ Exp/ BC Tuning ■ QA/ Exp/ Coil Check BC	■ Standard tool kit ■ Non-magnetic steel tool set ■ Torque wrench 20 - 50 Nm ■ Accessory kit torque wrench ■ Spirit level ■ Measuring tape or gauge ■ Ethylene glycol and antifreeze tester ■ Service aid back cover (if needed) ■ Tools for hammering in the wedges (→ Fig. 76 Page 121) ■ Lifting device for GC/BC with accessory parts, 1225 short beam, 4 m ■ Coil support (sliding carriages with 2 support beams) ■ Magnet Power Supply ■ Espree Array Shim Device ■ Shim plates, left over from first installation ■ Helium fill kit ■ Screwdriver (73 66 771) for BC trim. capacitor adjustment

Body Coil					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tools**
Upgrade Slimline continued					■ only if MNO option is installed: Service Plug kit (07715720) ■ BNC 2m Service cable for step TU/ Tuning Calibration (8733479) ■ Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration ■ Double hexagon 12-point socket ■ Plumber's tape (1 to 1.5" white sand cloth (local purchase)) ■ Scotch Brite pad (local purchase) ■ Heat gun

Body Coil					
FRU	Time	Instructions	Adjustments	Final checks	Tools
Body Coil Slim 70 Espree (10048071)	5.5 hr.	(→ Body Coil (Slim Line) Espree / Page 140)	■ TU/ Tuning Calibration ■ TU/ BC Tuning ■ TU/ RF Characteristic ■ TU/ Coil Power Losses ■ TU/ Abs.Rec.Path Calibration or BC Image Brightness	■ QA/ Abs. Rec. Path Calibr. Check or BC Image Check ■ QA/ BC Tuning Check ■ QA/ RF Verify ■ QA/ Coil Power Losses Check or BC Coil Power Losses Check ■ QA/ Spike Check ■ SAR monitor test (VB17A and higher)	■ Non-magnetic tool kit ■ Non-magnetic measuring tape or gauge ■ Non-magnetic spirit level ■ Screwdriver (73 66 771) for BC trimmer capacitor adjustment ■ BNC 2m Service cable for step TU/ Tuning Calibration (8733479) ■ Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration

Tab. 6 Local Coil

Local Coils					
FRU	Time	Instructions	Adjustments	Final checks	Tools
Local Coils	30 min.	(→ Local Coils / Page 158)	n.a.	■ Coil QA/ Coil Check for Local Coil	Phantoms and Phantom Holders

2.1 General

CAUTION

Strong magnetic field.

Magnetic parts may cause injury.

» Maintain a safe distance to the magnetic field.

CAUTION

Before returning a replaced RFPA, drain the water from the pipes of the device by opening both ends.

» There is the potential for irreversible damage to the device if the internal water pipes freeze during shipping.

CAUTION

Never unplug the quick connectors from the RFPA when the water pump in the ACS (SEP) or the cooling circuit (IFP) is in operation (because of high impact pressure).

» Switch off the water pump via cutter switch F3 in the ACS cabinet (SEP) or switch off the cooling circuit via F1 in ACC (IFP).

Please ensure that the system is powered on for at least 20 minutes before performing any (service) measurements. This is to stabilize the temperature in the control cabinet.

Adhere to special safety guidelines when working with heavy loads.

(Refer to the Installation Manual).

Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance <= 100 mOhm with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. GPA) and the protective earth wire connected to the "main component" (e.g. ACC).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

Fig. 1: RFSU Frame

(1) Position of frame mounting nuts, accessible from back
(2) Frame mounting screws

3.1 Synthesizer, Modulator, Receiver

(D10/120, D101, D112/122;

transmitter D111 = combination of modulator and synthesizer instead of D101 and D10/120)

The transmitter D111 is located in the slot of former modulator D101.

Please double check the RFSU configuration at your system before ordering the spare parts.

3.1.1 Tools and time required

3.1.1.1 Time

80 (110) min. for each RFSU component

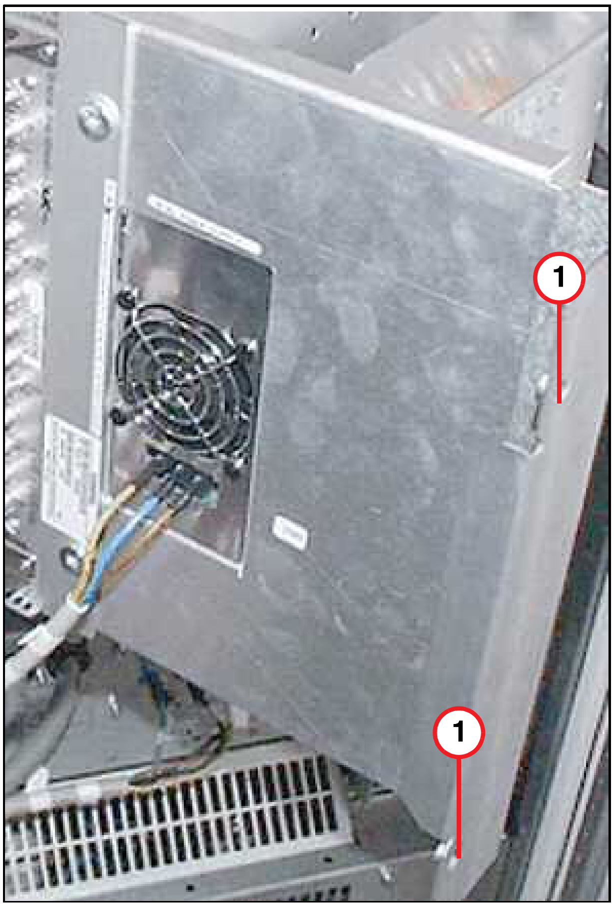
3.1.1.2 Tools

- Normal tool kit
- Screwdriver Torx TX8, TX10
- ESD equipment
- BNC 2m Service cable for step TU/ Tuning Calibration (8733479)
- Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration
- only if MNO option is installed: Service Plug kit (07715720)

3.1.2 Preliminary

Be careful not to crush or damage cables and/or FOCs when swinging in/out the RFSU swivel frame.

Fig. 2: RFSU swivel frame



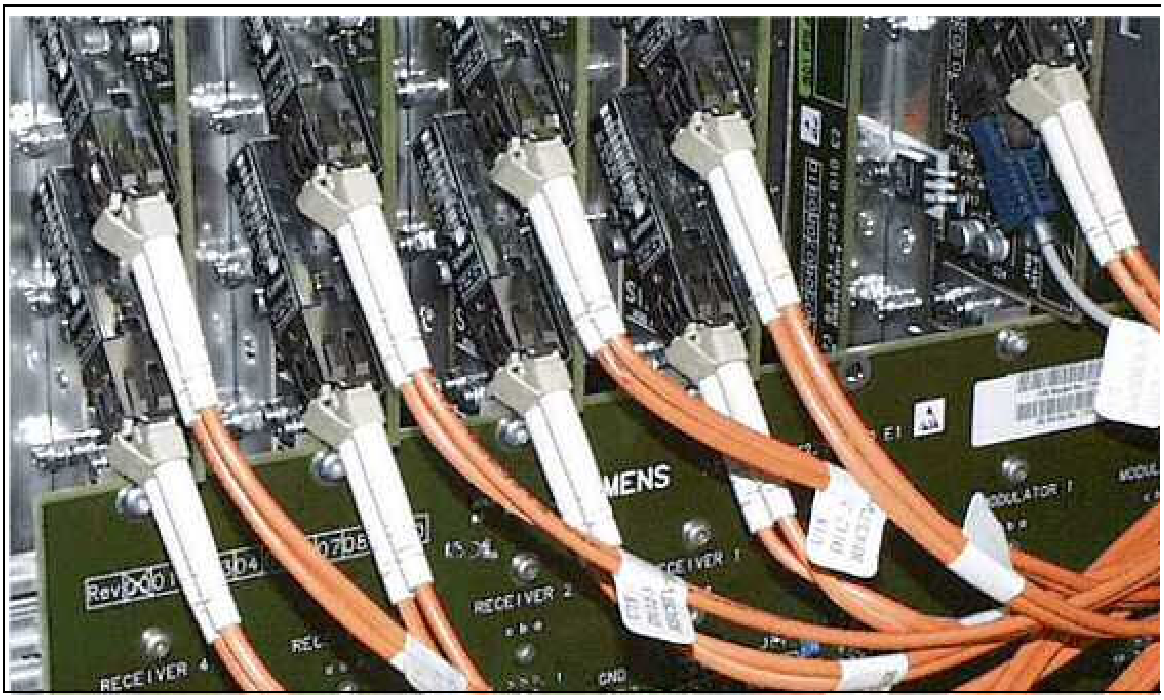
(1) Position of securing screws

1. Switch off ACC/F23.
2. Remove the two securing screws Pos. 1 at the right side of the RFSU swivel frame (right swivel frame side in the ACC).
3. Swing out the RFSU swivel frame, lifting the frame slightly.

3.1.3 Disassembly

Before dismounting the RFSU components, remove the FOCs at the back.

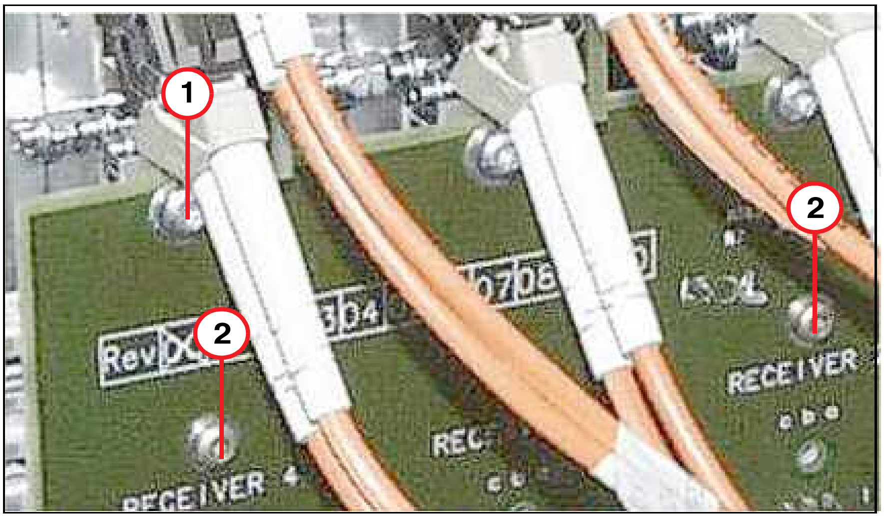
Fig. 3: FOC connections at back of RFSU components



- Disconnect FOC in back of the RFSU component.
- Remove the securing screw (TORX 10) just below the lower FOC connection (see (➔ 1/Fig. 4 Page 25)) at the back for the RFSU component/board that has to be replaced only.

  The two rows of 7 screws (see (➔ 2/Fig. 4 Page 25)) only have to be removed when replacing the backplane. These screws hold the backplane in the correct position.

Fig. 4: RFSU backplane



(1) RFSU board securing screws (TORX 10)

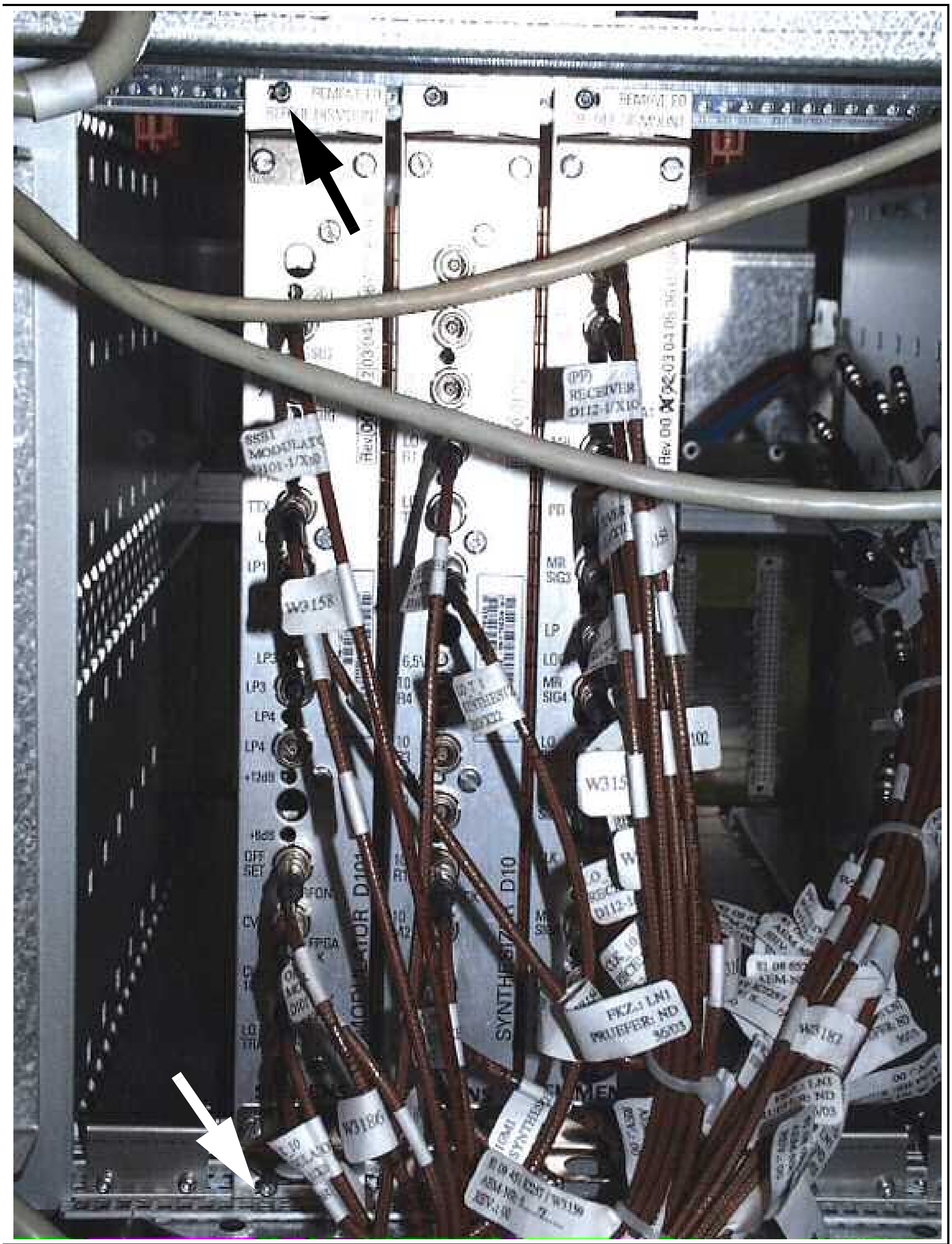
(2) Backplane securing screws (TORX 8, upper row of seven screws)

- Remove all cables from the front of the RFSU component.

  For synthesizers on 8 and 16-channel systems:

  Remove the 50 Ohm terminators from the unused outputs.
- Remove the two mounting screws (TORX 8) at front side of the RFSU component.

Fig. 5: Modulator - Synthesizer - Receiver (example, 8-channel system) Mounting screws of RFSU components



- Carefully pull the RFSU component out of the frame.

3.1.4 Assembly

CAUTION

When replacing the TX/RX modules, carefully as well as gently push in the new TX/RX modules.

» **Otherwise, you may damage the connectors of the modules.**

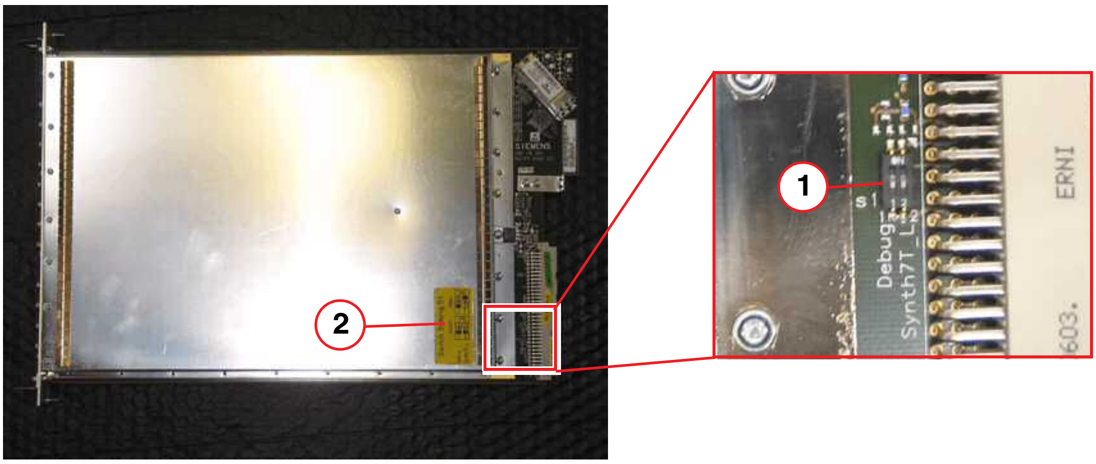
For systems with spectroscopy option before assembling the Modulator:

Check the DIP switch setting on modulator board on backside below:

Note the label attached on the side cover of the modulator near the tiny DIP switch S1:

For 1.5/3T operation: Switch off; for other/7T operation: switch on.

Fig. 6: Modulator 7 T



(1) Switch S1
(2) Label: Switch Setting S1 (for 1.5/3 T or Other)

Fig. 7: RF sealing on the lateral surfaces of the RFSU components

- Attach the new RF sealing (delivered with the spare part) on the sides of the new RFSU component according to the old component.

  After reinstalling the board, it must contact the neighboring board.

Fig. 8: RF sealings between RFSU components (rear side)

(→ Fig. 8 Page 28) shows an example of damaged RF sealing to the left and a well-mounted sealing to the right.

- Carefully and gently push in the new component.
- Mount the two screws at the front and the securing screw at the back.
- Reconnect all cables at the front and the FOCs at the back. **For synthesizers on 8 and 16-channel systems:**

  Attach the 50 Ohm terminators to the unused outputs.

3.1.5 Concluding steps

- Swing in the swivel frame and install the securing screws on the right side.
- Switch on ACC/F23.

3.1.6 Adjustments

3.1.6.1 Synthesizer, Modulator (Transmitter)

- Perform TU/ Tuning Calibration
- Perform TU/ BC Tuning
- Perform TU/ RF Characteristic
- Perform TU/ RF Characteristic LC (if MNO option is installed)
- Perform TU/ Coil Power Losses or BC Power Losses
- Perform TU/ Abs. Rec. Path Calibration or BC Image Brightness
- Perform TU/ Image Brightness or BC Image Brightness

3.1.6.2 Receiver

- Perform TU/ Tuning Calibration
- Perform TU/ BC Tuning
- Perform TU/ RF Characteristic
- Perform TU/ RF Characteristic LC (if MNO option is installed)
- Perform TU/ Coil Power Losses or BC Power Losses
- Perform TU/ Rel. Rec.Path Calibration or RX Gain Calibration
- Perform TU/ Abs. Rec.Path Calibration or BC Image Brightness
- Perform TU/ Image Brightness or BC Image Brightness

3.1.7 Final checks

3.1.7.1 Synthesizer

- Perform QA/ Abs. Rec.Path Calibration Check or BC Image Check
- Perform QA/ BC Tuning Check
- Perform QA/ RF Verify/ RF Verify LC (if MNO option is installed)
- Perform QA/ Coil Power Losses Check or BC Power Losses Check
- Perform QA/ Stability Check
- Perform QA/ Synthesizer Check
- Perform QA/ Stability_Long Term Check (if available)

3.1.7.2 Modulator (Transmitter)

- Perform QA/ Abs. Rec. Path Calibration Check or BC Image Check

- Perform QA/ BC Tuning Check
- Perform QA/ RF Verify/ RF Verify LC (if MNO option is installed)
- Perform QA/ Coil Power Losses Check or BC Power Losses Check
- Perform QA/ Stability Check
- Perform QA/ Synthesizer Check (only for Transmitter)
- Perform QA/ Stability_Long Term Check (if available)

3.1.7.3 Receiver

- Perform QA/ Rel. Receive Path Calibration Check.or RX Gain Calibration Check
- Perform QA/ Abs.Receive Path Calibration Check or BC Image Check
- Perform QA/ BC Tuning Check
- Perform QA/ RF Verify/ RF Verify LC (if MNO option is installed)
- Perform QA/ Coil Power Losses Check or BC Power Losses Check
- Perform QA/ Stability Check
- Perform QA/ Stability_Long Term Check (if available)

3.2 Power supply (PSU)

(Part no. 30 95 783)

3.2.1 Tools and time required

3.2.1.1 Time

40 (70) min.

3.2.1.2 Tools

Normal tool kit

Screwdriver TORX 10, shaft length 7.5 cm

3.2.2 Preliminary

1. Switch off ACC/F23.
2. Remove the securing screws at the right side of the RFSU swivel frame (right swivel frame side in the ACC).
3. Swivel out the RFSU frame, lifting the frame slightly.

3.2.3 Disassembly

- Remove the power lines at front side of the PSU.

- Remove the supply cables from back of the PSU.

Fig. 9: RFSU - PSU connections on back

- Remove the 4 mounting screws on the bottom and remove the old PSU (➔ Fig. 10 Page 33).

Fig. 10: RFSU power supply - mounting screws

3.2.4 Assembly

- Install the new PSU with the 4 screws in the swivel frame.
- Reconnect the supply and power lines.

3.2.5 Concluding steps

- Swing in the swivel frame and install the securing screws on the right side.
- Switch on ACC/F23.

3.2.6 Final checks

- Perform QA/ Stability Check.
- Perform an MR measurement.
- Perform QA/ Stability_Long Term Check (if available).
- In case of noise in patient images: Run QA/Exp/RFNoise Check

3.3 RFSU Frame, Backplane

(Part no. 73 91 001, 73 89 138)

3.3.1 Tools and time required

3.3.1.1 Time

40 (70) min.

3.3.1.2 Tools

Normal tool kit

Screwdriver TORX 8, 10

3.3.2 Preliminary

1. Switch off ACC/F23.
2. Remove the securing screws at the right side of the RFSU swivel frame (right swivel frame side in the ACC).
3. Swivel out the RFSU frame, lifting the frame slightly.

3.3.3 Disassembly

- Remove all cables and FOCs from the components in the RFSU frame and the back-plane.
- Remove all RFSU components from the RFSU frame.

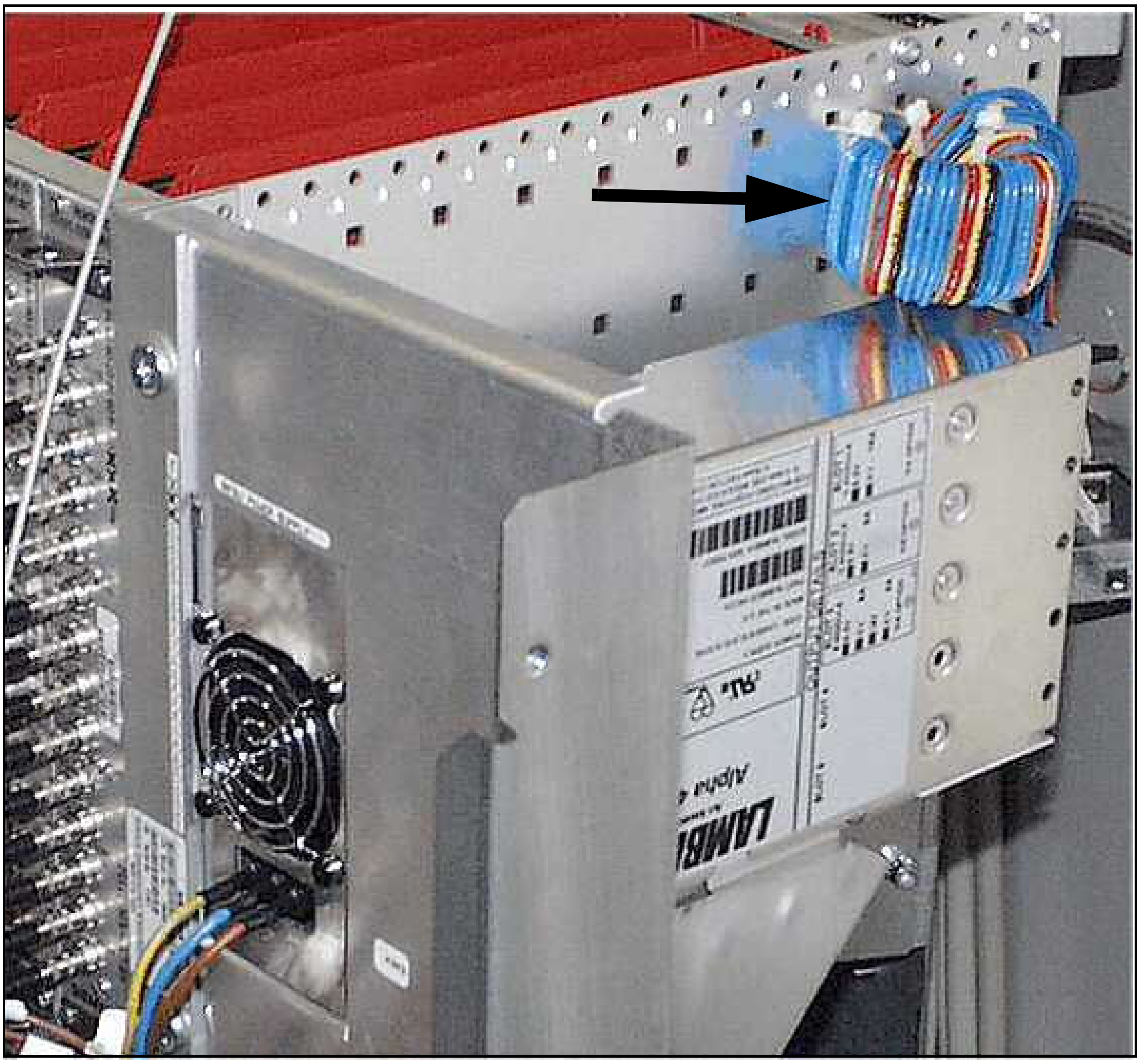
3.3.3.1 Disassembling RFSU Frame

- Remove the two mounting screws on the right side of the RFSU frame (front) Pos. 2 and the two nuts on the left side of the RFSU frame Pos. 1 (accessible from the back when the frame is swiveled out).

Fig. 11: RFSU Frame

(1) Position of frame mounting nuts, accessible from back
(2) Frame mounting screws

Fig. 12: Choke coil with power cables

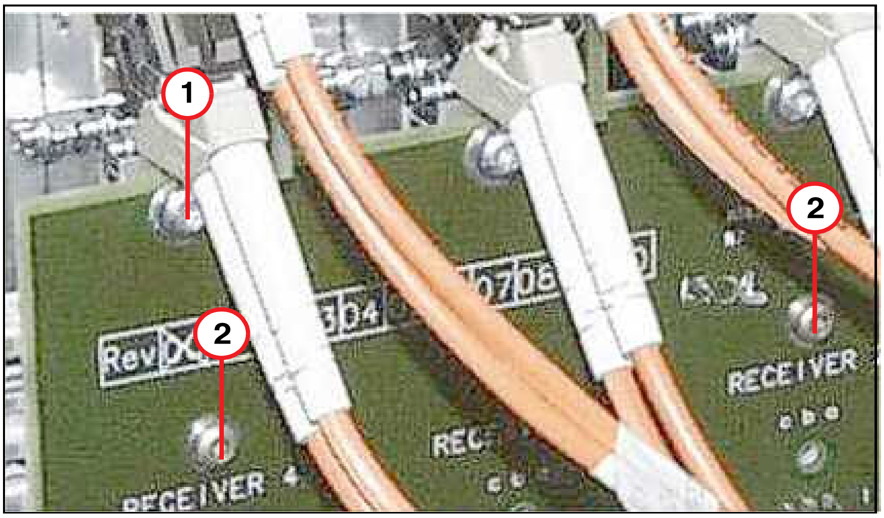


3.3.3.2 Backplane

- Remove the 2x7 mounting screws (TORX 8) on the backplane only if you are replacing the backplane.

  The next figure shows the board securing screws Pos.1 (TORX 10) above (next to the lower FOC-connections) and the upper row of backplane mounting screws Pos. 2 (TORX 8).

Fig. 13: RFSU backplane



(1) RFSU board securing screws (TORX 10)
(2) Backplane securing screws (TORX 8, upper row of seven screws)

3.3.4 Assembly

- Mount the new backplane or the new RFSU frame.
- Push in the RFSU components.
- Connect all cables and FOCs.

3.3.5 Concluding steps

- Secure cables with cable ties, as necessary.
- Swing in the swivel frame and install the securing screws on the right side.
- Switch on ACC/F23.

3.3.6 Final checks

- Perform QA/ Stability Check.
- Perform a MR measurement.
- Perform QA/ Stability_Long Term Check (if available).
- In case of noise in patient images: Run QA/ Exp/ RFNoise Check

- **In case of problems with lines and excessive noise in patient images:**

  Lines or excessive noise in patient images may be caused by interference from an incorrectly mounted backplane (RF sealing) or the power supply.

  Running the service sequence rf_noise:

  Connect spine matrix coil and body matrix coil to the table plugs. Put a small bottle phantom on spine array element one. Center the light localizer on element one and send to the center. This is to match the adjustments.

  Leave the body matrix outside the bore, no phantom is needed for the body array. Use the exam explorer and navigate to

  SIEMENS/Sequence Region/Service Sequences/Default protocols. Scroll down to the rf_noise sequence and drag it down into the lower left portion of the screen.

  Open the sequence, turn on spine matrix element 1 and all of the body matrix elements.

  Start the scan and review the images. Noise interference will now be visible.

CAUTION

Strong magnetic field.

Magnetic parts may cause injury.

» **Maintain a safe distance to the magnetic field.**

4.1 TALES, BCCS (Avanto), BCCS_63P (Espree)

TALES D6 (material number 73 91 886)

BCCS Avanto (material number 47 53 229)

BCCS_63P Espree (material number 100 17 953)

Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad (TDK equipment Type 8005 from 3M Deutschland GmbH).

4.1.1 Tools and time required

4.1.1.1 Time

70 min. for each component

4.1.1.2 Tools

- Non-magnetic tool kit
- Non-magnetic torque wrench
- ESD equipment
- BNC 2m Service cable for step TU/ Tuning Calibration (8733479)
- only if MNO option is installed: Service Plug kit (07715720)
- Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration

4.1.2 Preliminary

1. Switch off ACC/F24.
2. Remove the left side of the magnet cover (seen from patient end).

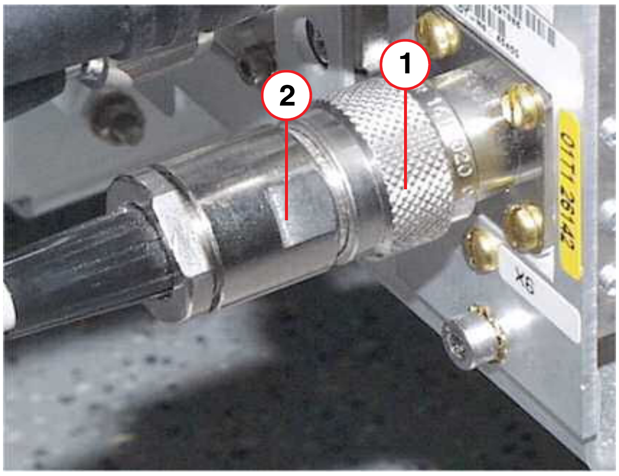
4.1.3 Disassembly

- Remove the RF cables from the front of the old component.

When removing or connecting the connector X6, hold the back part of the connector(2) in a fixed position while unscrewing the joint (1). (→ Fig. 14 Page 41)

Otherwise the RF cable will be twisted and the cable shield can break off. High rf-reflection will be the consequence!

Fig. 14: TALES connector X6 (TX->LC)



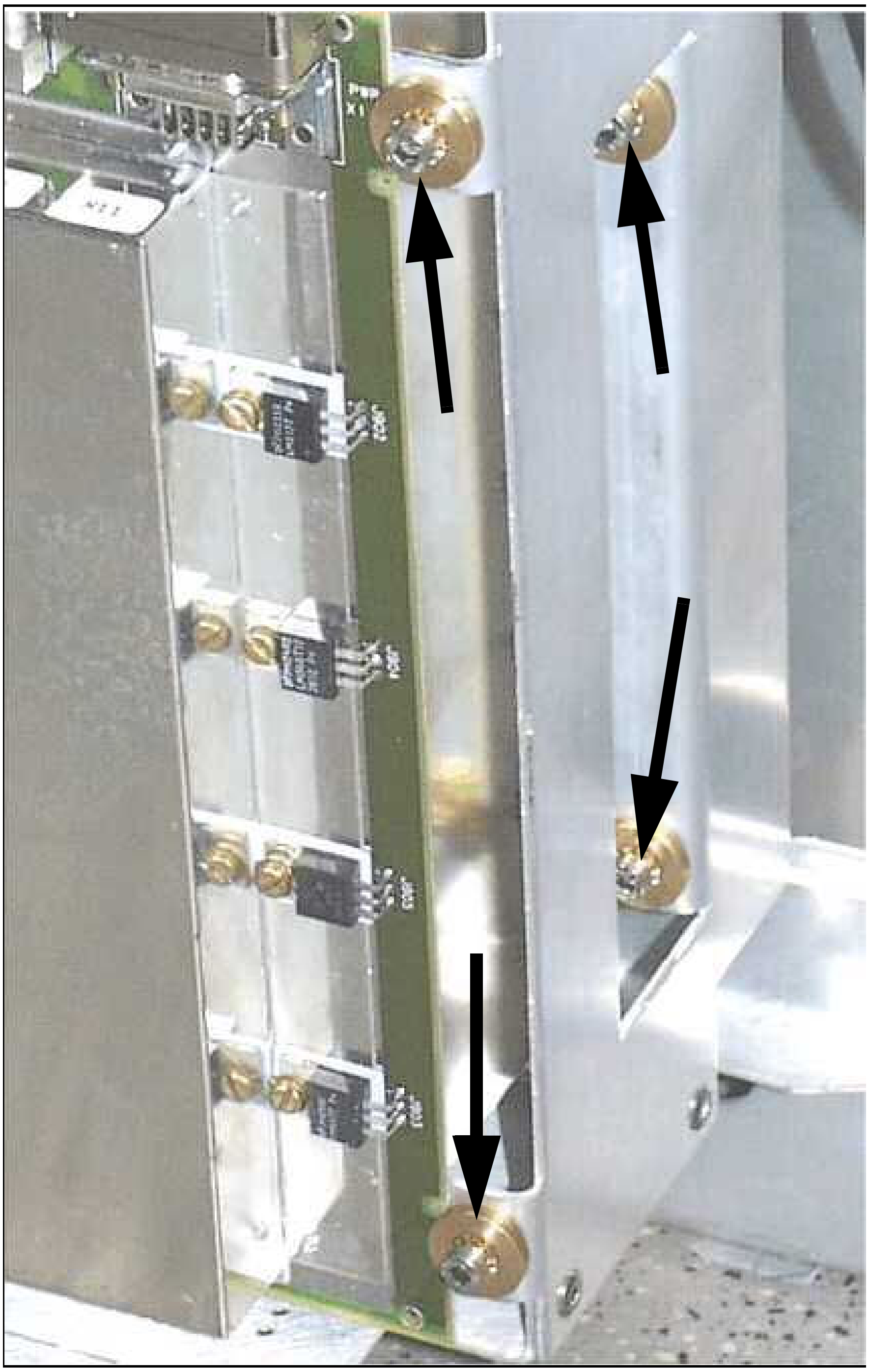
(1) Screwed joint of N connector
(2) Back part of N connector

Fig. 15: BCCS, TALES front side, mounting screws

- Remove the FOCs and supply lines from the old component.

- Remove the screws at the front of the old component (➔ Fig. 15 Page 42).

Fig. 16: TALES, BCCS back



- Remove the screws at the back of the old component (➔ Fig. 16 Page 43).
- Remove QLA connectors (if present) from the top of the old component.

- **For BCCS:** Disconnect the RF cable (7/16) at connector X1 and dismount the connector at the BCCS cable from the mounting plate (➔ Fig. 17 Page 44).

Fig. 17: BCCS: RF input connector X1

- **For BCCS:** Disconnect the QLA connectors (cables to RCCS) on top of the BCCS.
- Remove the old component.

4.1.4 Assembly

- Position the new component in the frame
- **For BCCS:** Connect the QLA connectors (cables to RCCS) on top of the BCCS.
- **For BCCS:** Install the connector (X1) on the mounting plate.
- Attach the new component with the screws.
- Reconnect all QLA/supply cables and FOCs to the new component.
- Connect RF cables and tighten the RF connectors with the following torque:

  7/16: 25 Nm (SW 27 or 32)

  HN: 6 Nm (4 -8 Nm, SW 22)

  N: 1.7 Nm (1-2 Nm, (strongly) tightened by hand)

When removing or connecting the connector X6, hold the back part of the connector(2) in a fixed position while unscrewing the joint (1). (→ Fig. 14 Page 41)

Otherwise the RF cable will be twisted and the cable shield can break off. High rf-reflection will be the consequence!

4.1.5 Concluding Steps

- Secure cables with cable ties, as necessary.
- Reassemble the magnet cover.
- Switch on ACC/F24.

4.1.6 Adjustments

4.1.6.1 TALES, BCCS

- Perform TU/ Tuning Calibration.
- Perform TU/ BC Tuning
- Perform TU/ RF Characteristic
- Perform TU/ RF Characteristic LC (if MNO option is installed)
- Perform TU/ Coil Power Losses or BC Power Losses.
- Perform TU/ Abs.Rec.Path Calibration or BC Image Brightness

4.1.7 Final Checks

4.1.7.1 TALES, BCCS

- Perform QA/ Abs. Rec.Path Calibration Check or BC Image Check
- Perform QA/ BC Tuning Check
- Perform QA/ RF Verify
- Perform QA/ RF Verify LC (if MNO option is installed)
- Perform QA/ Coil Power Losses Check or BC Power Losses Check
- Perform QA/ Expert/ RF Path Attenuation Check

4.2 RCCS

RCCS D11 (material number 73 89 013)

RCCS_2 D11 (material number 102 77 196)

Adhere to ESD regulations and use the appropriate tool set: anti-static wristband and grounded pad (TDK equipment Type 8005 from 3M Deutschland GmbH).

(→ Fig. 18 Page 46) shows the component RCCS with the six mounting holes (arrows) on lower side; there are six corresponding mounting holes at the top side.

The RCCS is mounted at the magnet side on a mounting plate on twelve thread studs with Allen screws (size 3).

Fig. 18: RCCS

4.2.1 Tools and time required

4.2.1.1 Time

70 min.

4.2.1.2 Tools

- Non-magnetic tool kit
- Non-magnetic torque wrench

4.2.2 Preliminary

1. Switch off ACC/F24 (RFIS PSU).
2. Remove the left side of the magnet cover (seen from patient end).

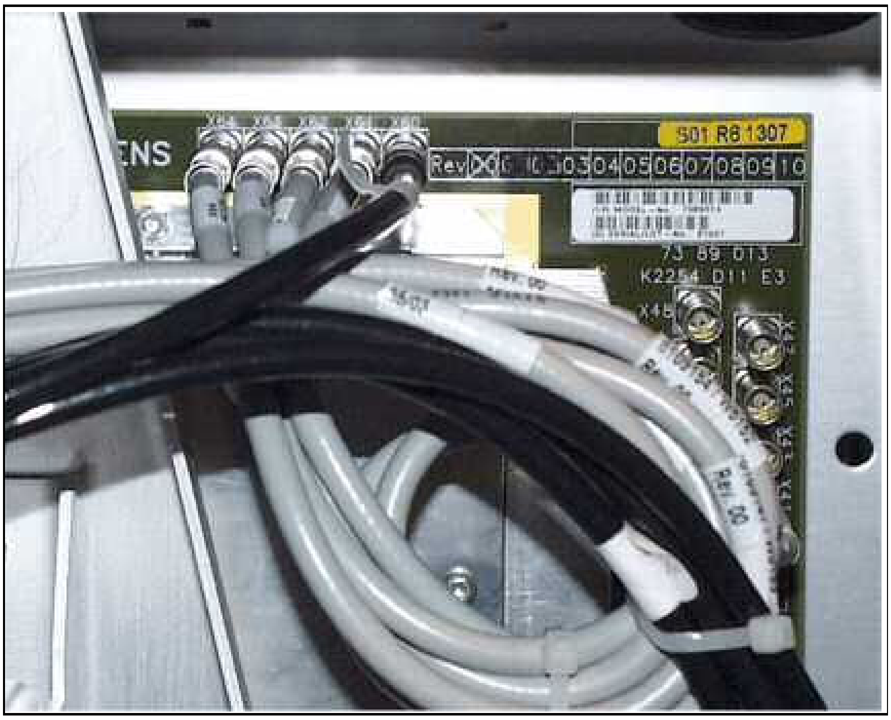
4.2.3 Disassembly

- Dismount the clamp rail for the RX cables from the RCCS to the RFSU: First remove the upper two nuts (SW10) on both sides, then the lower two nuts (SW7).

Fig. 19: RCCS: Clamp rail for RX cables

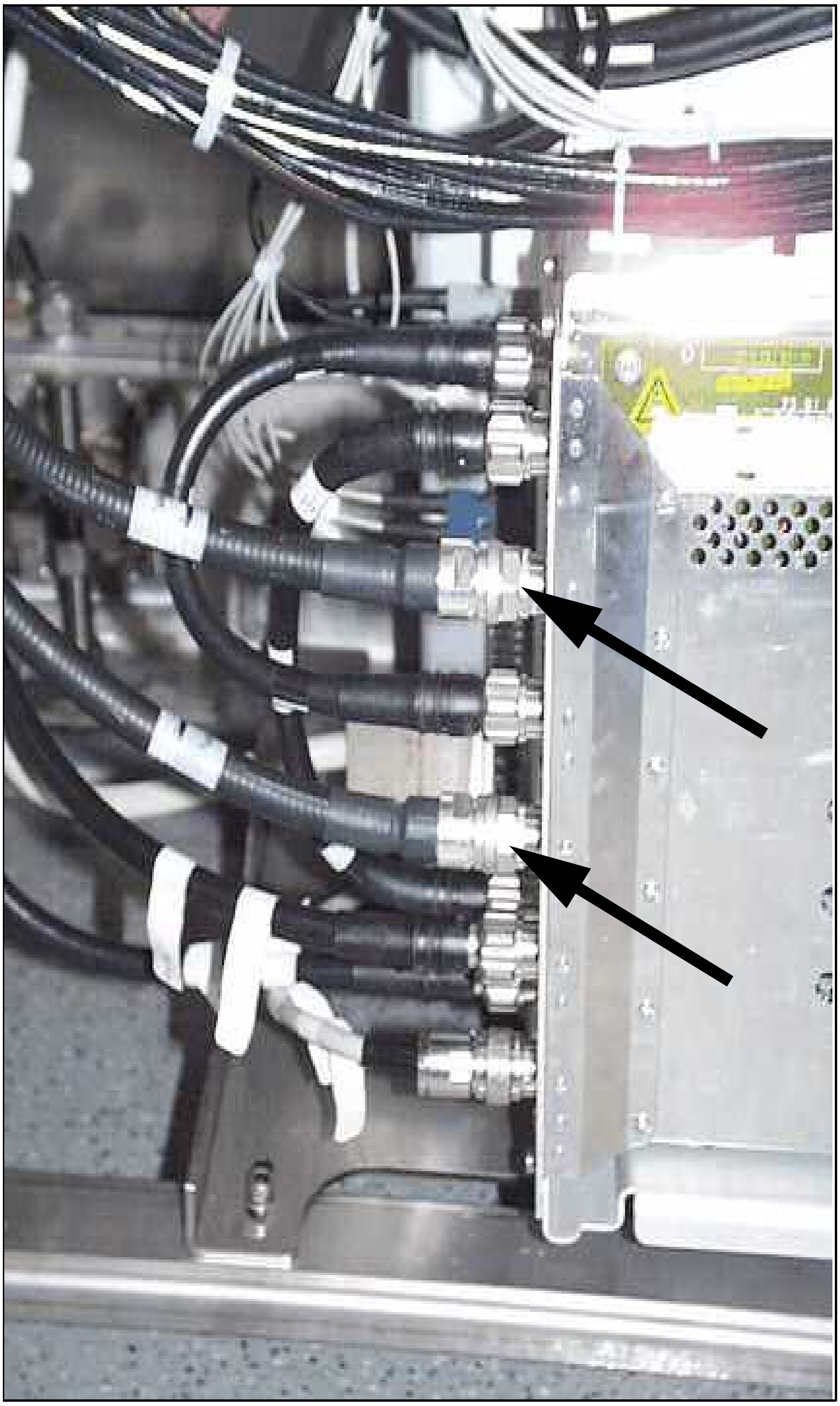
- Disconnect the RX-cables/QLA connectors (to RFSU and BCCS) at top side of the RCCS (→ Fig. 20 Page 48).

Fig. 20: RCCS - BCCS cable connections



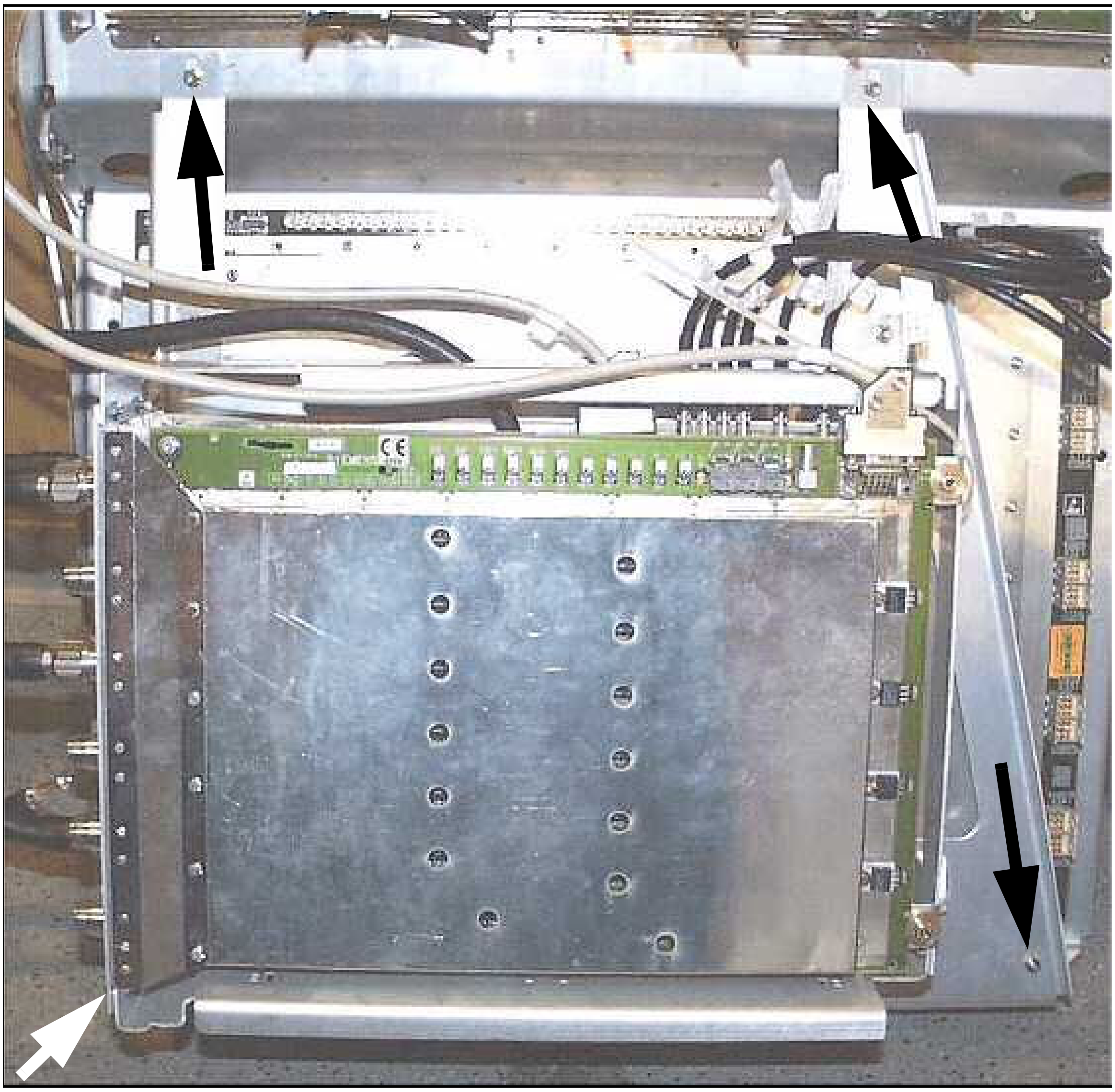
- From TALES remove the Body Coil TX cables X3 and X4 (→ Fig. 21 Page 49).

Fig. 21: TALES RF connectors - BC connections



- Dismount the mounting frame for TALES and BCCS, removing four nuts at the top and bottom of the frame (➔ Fig. 22 Page 50).

Fig. 22: Mounting frame TALES, BCCS



- **Carefully** disconnect the local coil receive cable connectors on the right side of RCCS (➔ Fig. 23 Page 51).

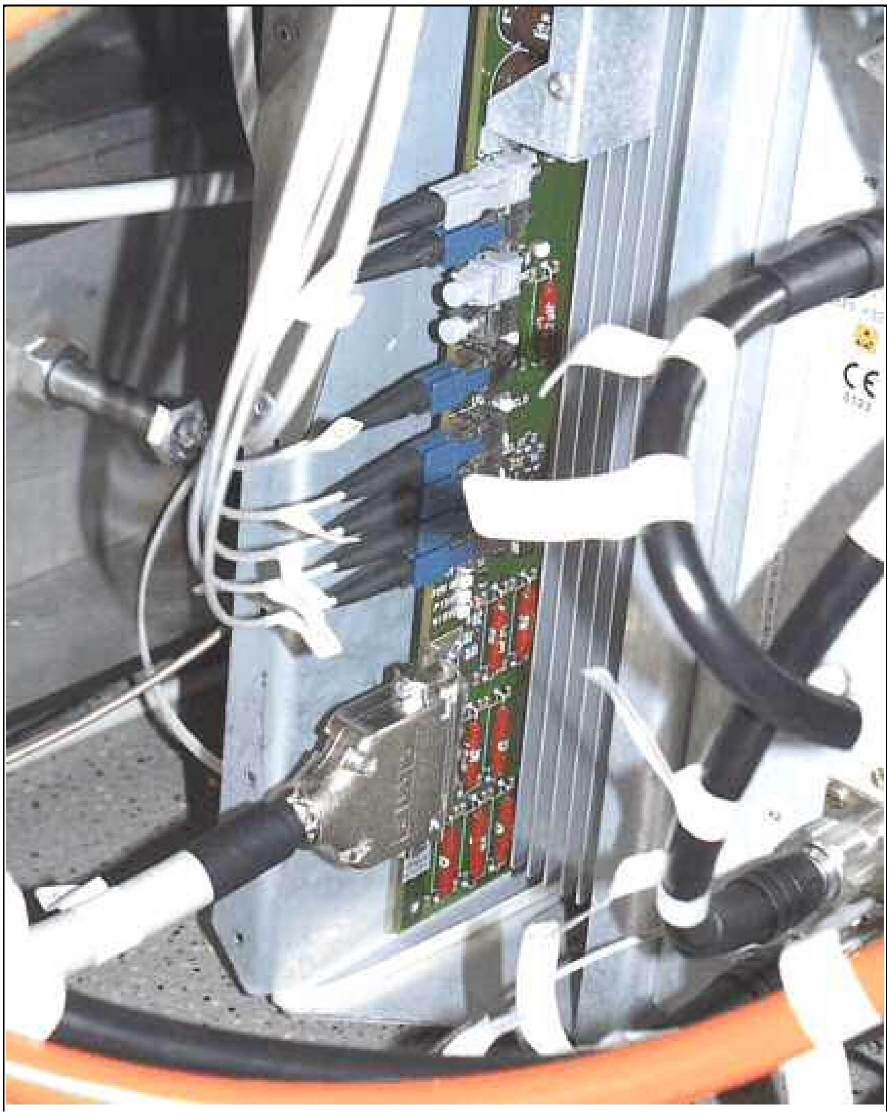
> Proceed with care when disconnecting these plugs. The locking assembly may break off, if handled improperly.

Fig. 23: RCCS: Local coil RX cable connections

- To access the RCCS screws, push aside the mounting frame of the TALES and BCCS.

Fig. 24: Frame of TALES/BCCS, partially removed

Fig. 25: RCCS: FOC and power supply connections



- Remove the FOCs and power cable from the left side of the RCCS.

- Dismount the RCCS from the mounting plate, loosening the six Allen screws (size 3) at the top and bottom of the RCCS (➔ Fig. 26 Page 54).

Fig. 26: Mounting plate - RCCS removed

4.2.4 Assembly

- Install the new RCCS on the mounting plate.
- Connect the FOCs and the power cable at the left side of the RCCS.
- Mount the frame of the TALES and BCCS.
- Connect the local coil plugs at the right side of the RCCS.
- Connect the RX cables/QLA connectors (to RFSU and BCCS) at top of the RCCS.
- Mount the clamp rail for the RX cables from the RCCS to the RFSU: First install the lower two nuts (SW7) on both sides with the RX cables in place, then the upper two nuts on the frame (SW10).
- Reconnect the TX cables to BC at TALES X3 and X4 (with torque 1.7 Nm).

4.2.5 Concluding Steps

- Reassemble the magnet cover.
- Switch off ACC/F24 (RFIS PSU).

4.2.6 Adjustments

- Perform TU/ Rel.Receive Path Calibration.or RX Gain Calibration
- Perform TU/ Abs.Receive Path Calibration or BC Image Brightness

4.2.7 Final Checks

- Perform QA/ Abs. Rec. Path Calibration Check or BC Image Check
- Perform QA/ Rel.Receive Path Calibration Check. or RX Gain Calibration Check
- Perform QA/ Stability Check.

4.3 TAS_3TM - TAS_CM

TAS_3TM (material number 73 92 058)

TAS_CM (material number 102 77 008)

4.3.1 Tools and time required

4.3.1.1 Time

120 min.

4.3.1.2 Tools

- Normal tool kit
- Open-ended wrench SW 36, 42
- Two pieces of water hose for depressurization of the secondary water cooling circuit
- Torque wrench
- only if MNO option is installed: Service Plug kit (07715720)

4.3.2 Preliminary

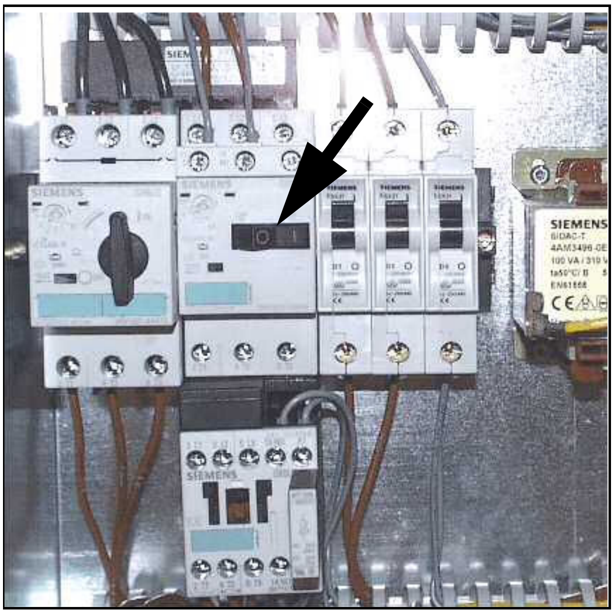
1. Switch off ACC/F24 (RFIS PSU).
2. Switch off and drain the cooling water circuit (see below).

4.3.2.1 Switch off Cooling Water Circuit

(For additional information, refer to the section **Cooling**)

SEP

Fig. 27: SEP: Power switch water pump



- Switch off ACC/F102 (line connection box)
- Switch off the cutter switch in the SEP (first remove the cover plate of the control cabinet).

Fig. 28: SEP: Control cabinet above, compensation vessel with valve below

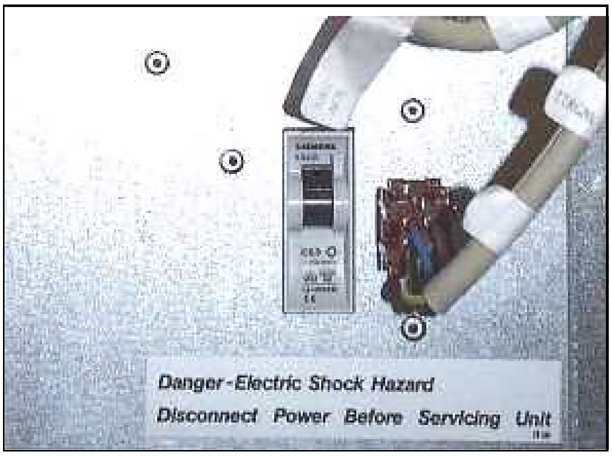
- Close the valve of the compensation vessel in the SEP (➔ Fig. 28 Page 57).
- Close the inflow and return valves on the roof of the ACC (using an open-ended wrench).

Fig. 29: SEP: Drain/Fill connections

- Depressurize the secondary water cooling circuit via the valves in the ACC below: Connect pieces of water hoses attached to the adapters to the valves and drain the water in a bucket (➔ Fig. 29 Page 58).

IFP

Fig. 30: IFP: Switch F1 in the ACC



- Switch off F1 in the ACC (➔ Fig. 30 Page 58).
- Close the inflow and return valves on the roof of the ACC (using an open-ended wrench).
- Depressurize the secondary water cooling circuit via the valves in the ACC below: Connect pieces of water hoses attached to the adapters to the valves and drain the water in a bucket.
- After depressurization, close the valves again.

4.3.3 Disassembly

Be careful when disconnecting the chilled water connection at the TAS. Use plastic foil to cover the electronics cabinets below, and have a bucket and paper towels ready to catch the water from the hoses.

Chilled water containing ethylene glycol should be collected while draining. This water should be refilled into the cooling circuit.

If drained cooling fluid (38% ethylene glycol water mixture) requires refilling, use a hand pump (service tool) or water pump with drill.

CAUTION

The water circuit from the chiller to the MR system (ACS) via the IFP is filled with 38% ethylene glycol. Therefore the cooling fluid is conductive.

If the cooling fluid splashed over connectors at the top of the ACC (especially RF connectors), this may result in severe problems such as sporadic spikes. It may be necessary to exchange the connectors.

» **Take extreme care not to spill cooling fluid on the electronics assemblies when disconnecting the water hoses from the TAS.**

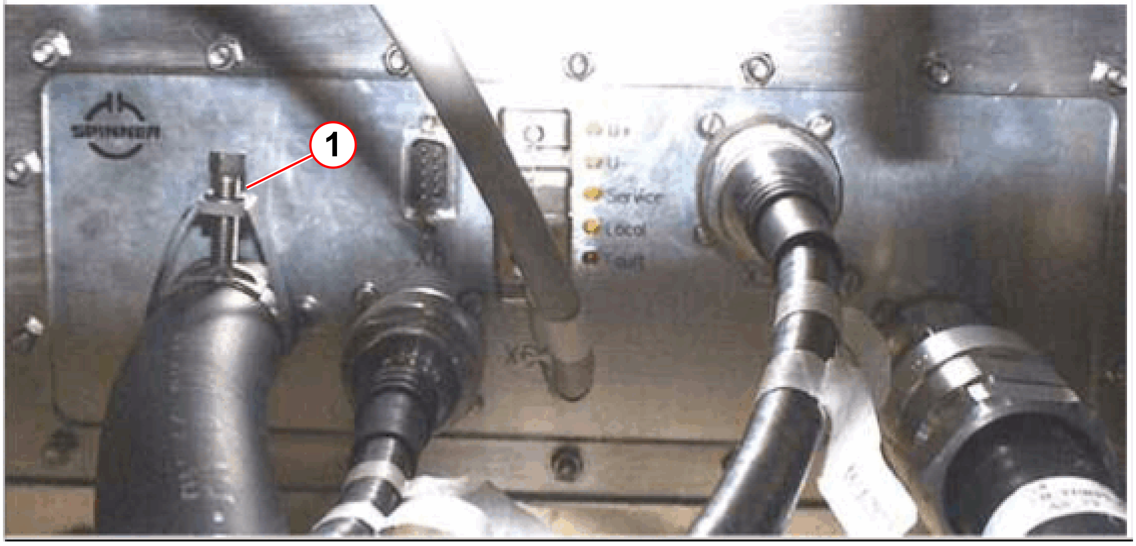
- Close the inflow and return valves to the gradient coil at the ACC roof top.
- At the magnet, disconnect the water inlet quick coupling to the gradient coil.

Fig. 31: TAS_3TM Equipment room side - connection X2 lower right

- In the equipment room, disconnect the water connection (X2) , holding the nut with a SW 42 and loosening the nut with a SW 36 open-ended wrench. Use plastic foils to protect electronics cabinets below and have a bucket ready to collect spilled water.
- Dismount the adapter X2 from the TAS (→ Fig. 31 Page 59).
- **TAS_3TM:**

  In the magnet room, disconnect the water hose from the pipe connection (X3), loosening the clamp with an open-ended wrench SW 7, or when it is not possible, cut it with a non-magnetic cutter. Pay attention not to scratch and damage the brass connection nozzle at the TAS, see (➔ Fig. 33 Page 61).

Fig. 32: TAS_3TM Magnet room side



(1) Pipe connection X3 - loosen clamp

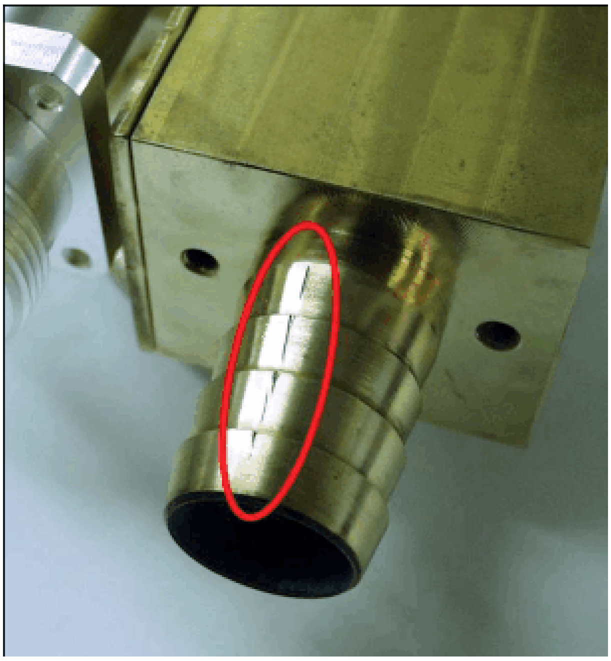
NOTICE

When cutting off the water hose to remove it from the nozzle, do not scratch and damage the brass nozzle! See also (➔ GC water distributor (Avanto/ Espree) / M6-020.841.16).

If the brass nozzle surface gets scratched, the connection will leak, leading to high repair costs.

» **Carefully scrape off the rubber hose layer by layer and parallel to the hose surface. Remove the black outer rubber layer, disrupt the white web and stop scraping when you reach the last black inner rubber layer to avoid scratching the nozzle.**

Fig. 33: TAS 3TM - scratched, damaged nozzle



- **TAS_CM:**

  In the magnet room, disconnect the water connection (X3), holding the nut with a SW 42 and loosening the nut with a SW 36 open-ended wrench. Use plastic foils to protect electronics cabinets below and have a bucket ready to collect spilled water.
- Wipe up residual water with paper tissue.
- In equipment- and magnet room, disconnect all RF-, control- and power cables. Be careful not to spill water to the connectors.
- Dismount the 18 nuts (SW 7), holding the TAS to the RF-filter plate in the magnet room.
- In the equipment room, remove the TAS from the RF-filter plate .

4.3.4 Assembly

- In the equipment room, push the threaded studs of the new TAS through the holes in the RF filter plate.
- In the examination room, fasten the TAS (with 18 nuts) to the RF filter plate.
- **TAS_3TM:**

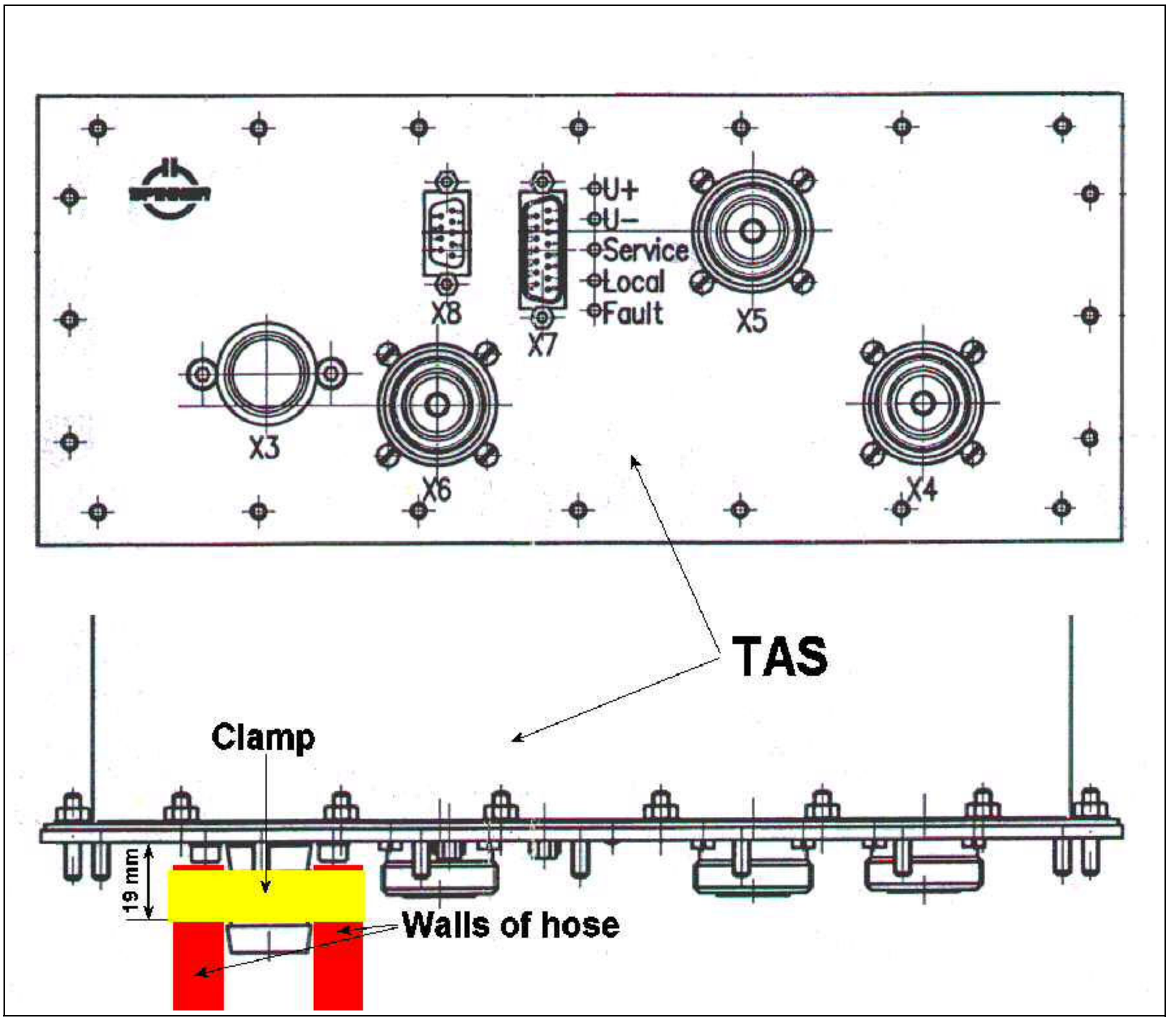
  In the examination room, connect the water hose to the pipe connection X3 of the TAS and install the clamp:

> i To avoid water leakages, consider the following guidelines

Rules to prevent water leakages:

- If possible, cutoff a sufficient part of the water hose so that a “new” part of the hose is used.
- Always cut perpendicularly, oblique cuts will result in a too short coverage for the connection.
- Push the water hose fully onto the fitting. Due to the Allen key screws close to it, the hose will not touch the housing of the TAS, but it should touch the screws.
- The hose clamp should be very close to the screws, but not touching them. A distance of one to three millimeters is OK, but it should not be more. Measuring from the front edge of the hose clamp to the housing of the TAS (not the screws), a distance of 19 millimeters should not be exceeded.
- Tighten the hose clamp as much as possible using a normal screwdriver. Additional tools to increase the torque should not be used.
- The use of a second hose clamp is usually counterproductive and will result in a connection that is not water-tight .

Fig. 34: TAS with water hose and clamp



- **TAS_CM:**

  In the magnet room, connect the water connection (X3), holding the nut with a SW 42 and mounting the nut with a SW 36 open-ended wrench. Use plastic foils to protect electronics cabinets below and have a bucket ready to collect spilled water.
- Mount the water hose adapter X2 at the new TAS.
- In the magnet and equipment room, reconnect all RF-, control- and power cables at the new TAS, using the following torques for the RF connectors:

  7/16: 25 Nm (SW 27 or 32)

  HN: 6 Nm (4 - 8 Nm, SW 22)

  N: 1.7 Nm (1 - 2 Nm, manually very tight)

4.3.5 Concluding Steps

- At the magnet, reconnect the water inlet quick coupling to the gradient coil.
- Open inflow and return valves to the gradient coil at the ACC roof top.

- Open the inflow and return valves (flow from SEP/IFP) at the ACC roof top.

4.3.5.1 Switching on and Refilling the Water Circuit

- To refill the water circuit, refer to Installation Instructions (➔ Starting magnet cooling / M6-020.812.02).
- To check the water pressure, refer to Startup description (➔ Check the water pressure / M6-020.815.01).

IFP

- Switch on F1 in the ACC
- Bleed the chiller water pump, refer to Startup description (➔ Bleeding the Chiller water pump / M6-020.815.01).

SEP

- Open the valve to the compensation vessel in the SEP.
- Switch on the cutter switch in the SEP (first remove the cover plate of the control cabinet).
- Switch on ACC/F102 (line connection box).

4.3.6 Adjustments

- Perform TU/ RF Characteristic
- Perform TU/ RF Characteristic LC (if MNO option is installed)
- Perform TU/ Coil Power Losses or BC Power Losses
- Perform TU/ Abs. Receive Path Calibration or BC Image Brightness

4.3.7 Final Checks

- Check tightness of cooling circuit connections at TAS.
- Perform QA/ Abs. Receive Path Calibration Check or BC Image Check
- Perform QA/ RF Verify
- Perform QA/ RF Verify LC (if MNO option is installed)
- Perform QA/ Coil Power Losses Check or BC Power Losses Check
- Perform QA/ Expert/ RF Path Attenuation Check.
- Perform an MR measurement with a local TX/RX coil (e.g., Extremity coil).

Description of LEDs at TAS_3TM/ TAS_CM

Tab. 7 LED at TAS_3TM/ TAS_CM

LED/ color	Label	Status On	Status Off
1 / green	U+	+ 15 V On	+ 15 V Off
2 / green	U-	- 15 V On	- 15 V Off

LED/ color	Label	Status On	Status Off
3 / yellow	Service	RFPA via S1 to dummy load	RFPA via S1 to BC or TX_LC (Imaging)
4 / yellow	Local	RFPA via S1/S2 to TX_LC via TALES	RFPA via S1/S2 to BC via BCCS
5 / red[1]	Fault	Discrepancy between relay position and relay control	Position of relay corresponds to control

1. LED not available for TAS_CM

CAUTION

Strong magnetic field.

Magnetic objects may cause injury. Magnetic objects are inside this unit.

» Maintain a safe distance to the magnetic field.

WARNING

High voltage (500 VDC) inside this unit.

Risk of electrical shock!

» Before opening the unit, switch off the power for the RFIS and the patient table and wait > 20 sec. (until voltage decreases to a harmless level).

5.1 RFIS Boards

Dyscon_BC_5A (material number 73 89 377)

Dyscon_16 (material number 73 88 775)

CAN_AND_CODE_CTRL (material number 73 89 351)

RFIS motherboard (material number 73 89 195)

5.1.1 Tools and time required

5.1.1.1 Time

45 min. for a single board

90 min. for motherboard (or complete RFIS)

5.1.1.2 Tools

- Non-magnetic tool kit
- Non-magnetic torque wrench
- **Espree:** Special short non-magnetic screwdriver (material number 51 37 083)

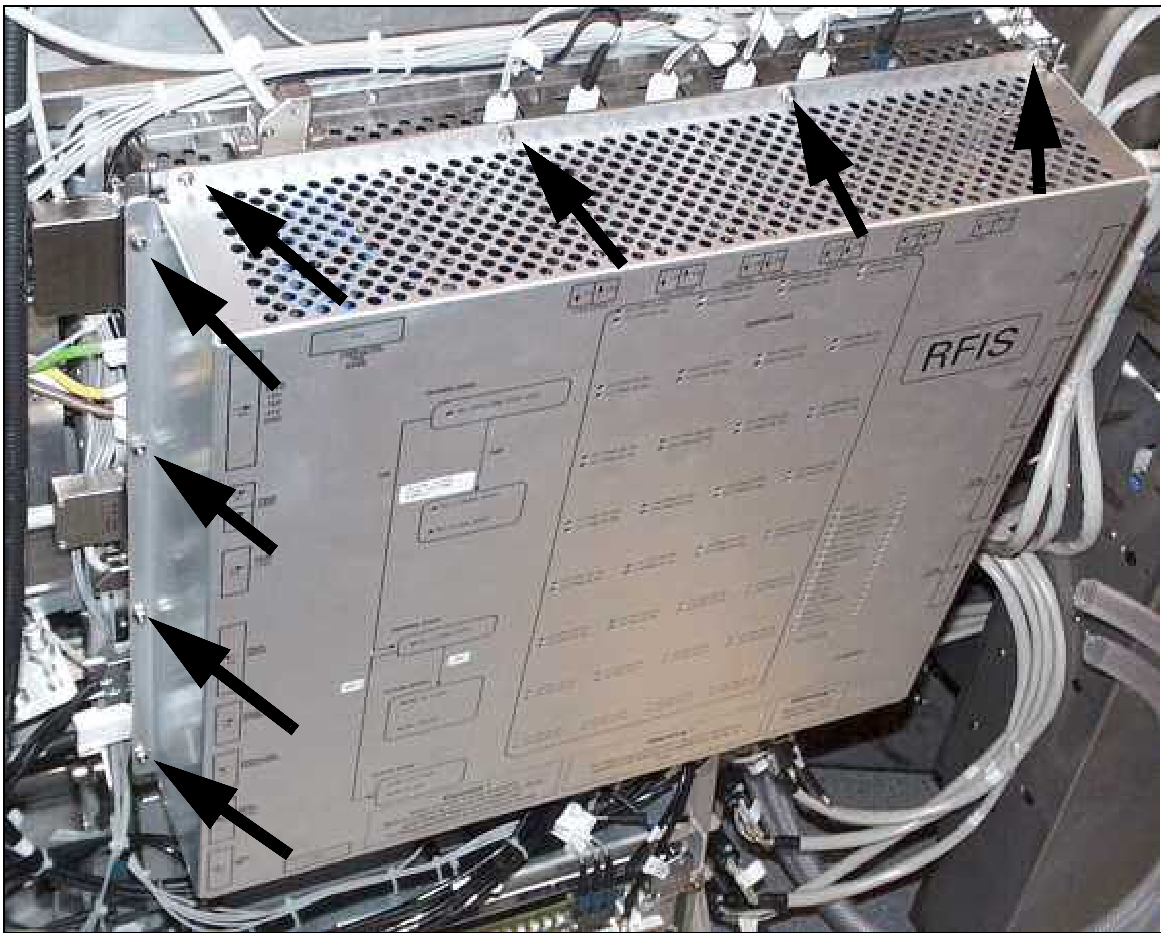
5.1.2 Preliminary

1. Switch off ACC/F24 (RFIS PSU) and ACC/F22 (PTAB).
2. Remove the left side of the magnet cover (seen from patient end).

5.1.3 Disassembly

- Remove the 16 nuts to remove the RFIS cover.

Fig. 35: RFIS with cover - mounting nuts

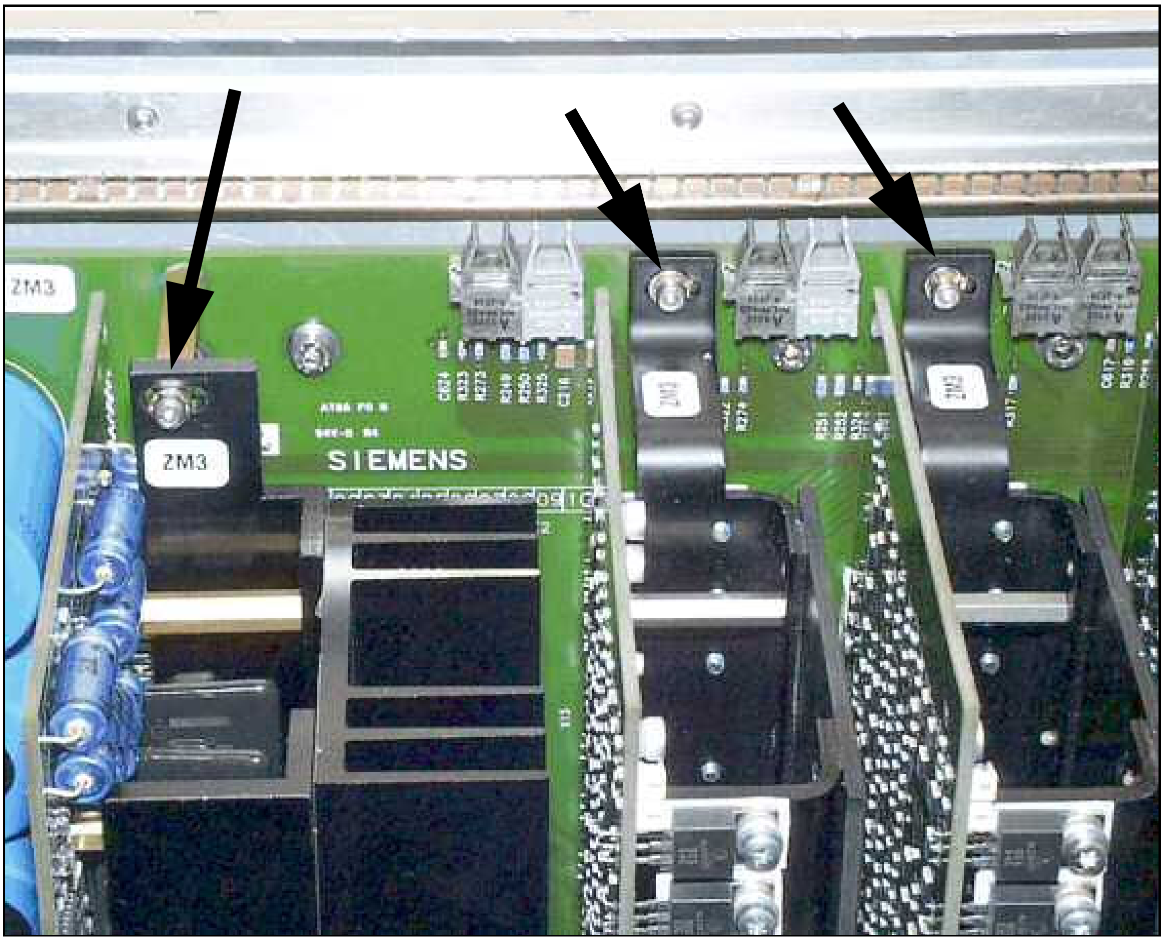


5.1.3.1 RFIS Boards

- Remove the screws (Allen screw size 2.5) on the upper and lower side of the plugged-in board (➔ Fig. 36 Page 69).

- **If removing the DYSCON_BC, disconnect the cable between DYSCON_BC and the motherboard.**

Fig. 36: Mounted boards on motherboard



> **Be careful when plugging in and removing the RFIS boards. Insert them parallel without tilting. Otherwise, the pins of the two 96-pin plugs may be damaged.**

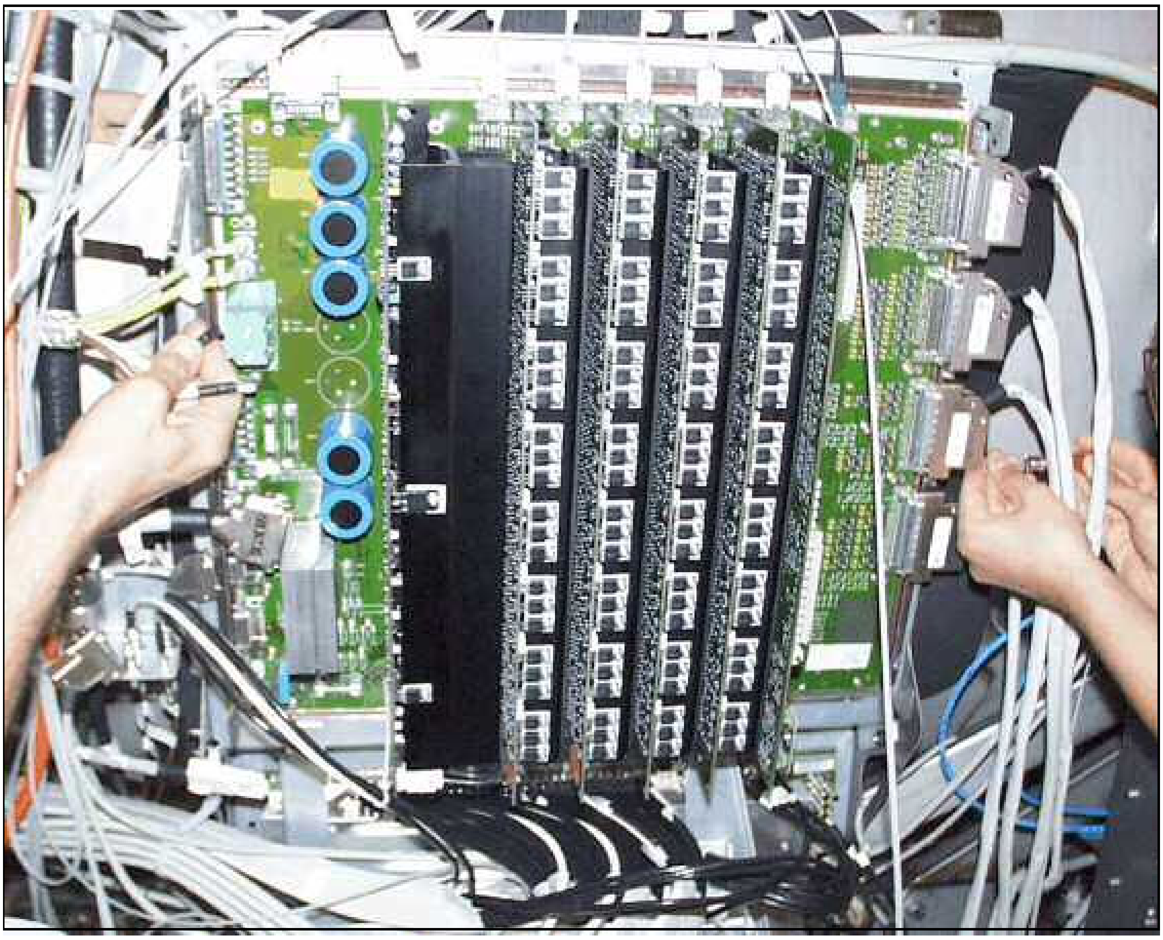
- Pull out the defective board from the 96-pin connector on the motherboard (→ Fig. 37 Page 70) (from left to right: DYSCON_BC - 4xDYSCON_16 - CAN_AND_COD_CTRL).

5.1.3.2 RFIS Motherboard

- **Disconnect the cable from DYSCON_BC to the motherboard.**

- Remove all boards from the motherboard.

Fig. 37: RFIS boards - Disassembling of connectors from motherboard



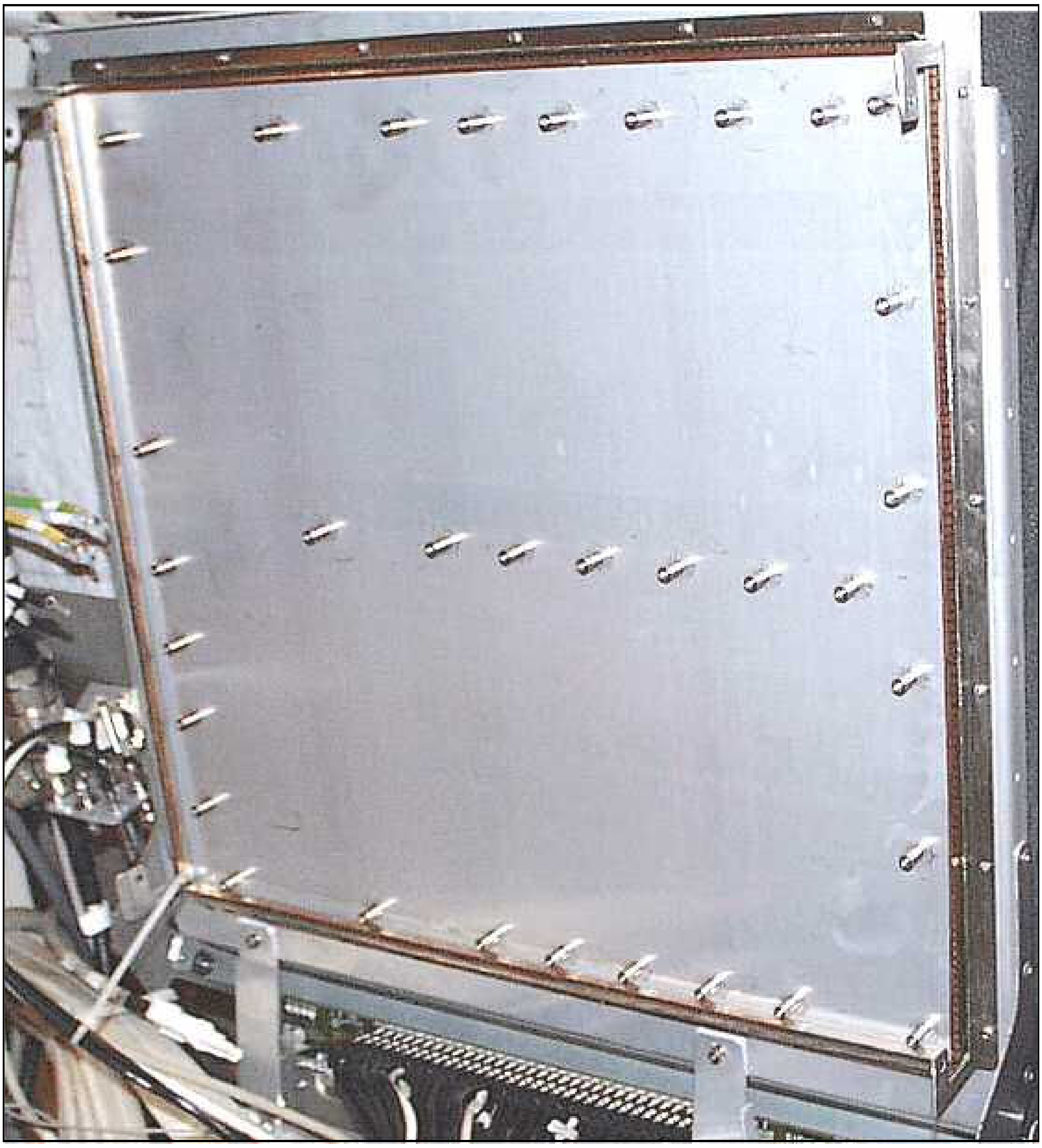
- Disconnect the local coil cable connectors on the right side.

  Espree:

  To remove connector X27/ X26/ X25, you will need a special short non-magnetic screwdriver (material number 51 37 083).
- Disconnect the power supply unit connections, the connection to the body coil at the left side of RFIS, and the FOCs on the top.

- Dismount the RFIS motherboard by loosening the 34 Allen screws that attach the motherboard via threaded studs to the mounting plate (➔ Fig. 38 Page 71).

Fig. 38: Mounting plate - RFIS motherboard



5.1.4 Assembly

5.1.4.1 RFIS Boards

- Carefully plug in the new board at its location on the motherboard.
- **If reassembling the DYSCON_BC, reconnect the cable between DYSCON_BC and motherboard.**

- Secure the board with the two screws on the upper and lower sides of the plugged-in board.

5.1.4.2 RFIS Motherboard

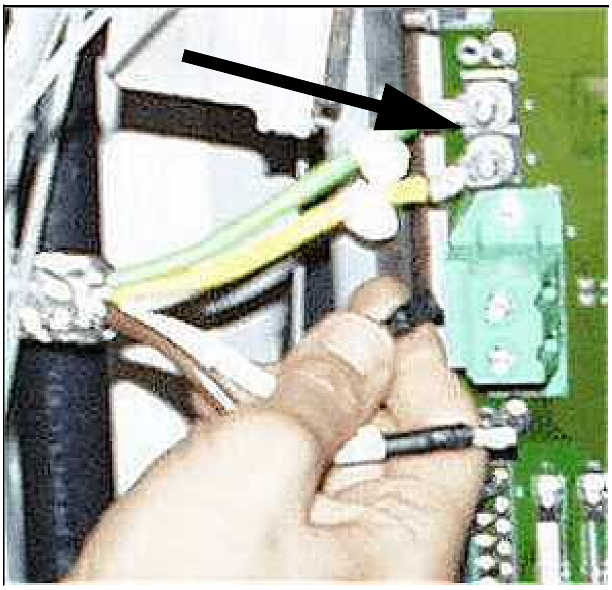
- Mount the motherboard on the mounting plate.
- Check the jumper setting on the motherboard:

  X51: Setting B/ enabled

  X52: Setting A/ disabled

  X53: Setting A/ disabled
- Reconnect all connectors on the motherboard.
- Fasten the connection studs X1/X2 (ground of +15V supply) with a torque of 4.8 +/- 10% Nm.

Fig. 39: P15V Ground studs X1/X2



> **i** Tighten the connector studs X1 and X2 (power ground P15V at RFIS motherboard with a torque of 4.8 +/- 10% Nm).
>
> Too low torque: Bad power connection.
>
> Too high torque: Danger of excessive mechanical stress to the thread.

- Plug in all boards at the appropriate location on the motherboard. (From left to right:

  DYSCON_BC_5A - 4x DYSCON_16 - CAN_AND_CODE_CTRL)
- Reconnect the cable between DYSCON_BC and motherboard.

5.1.5 Concluding Steps

- Mount the RFIS cover.
- Reassemble the magnet cover.

- Switch on ACC/F24 (RFIS PSU) and ACC/F22 (PTAB).

5.1.6 Adjustments

No adjustments necessary

5.1.7 Final Checks

- **Dyscon_16:** Run Coil QA/ Coil Check with a local coil connected to the appropriate LC Plug(s).
- **Dyscon_BC:** Run QA/ Coil Check for Body Coil.
- **Motherboard or complete RFIS:**

  Run QA/ Coil Check for Body Coil.

  Run Coil QA/ Coil Check with a local coil connected to the appropriate LC Plug(s).

5.2 Power Supply (E4A)

(material number 30 95 700)

5.2.1 Tools and time required

5.2.1.1 Time

45 min.

5.2.1.2 Tools

Normal tool set

5.2.2 Preliminary

- Switch off ACC/F24 (RFIS PSU).

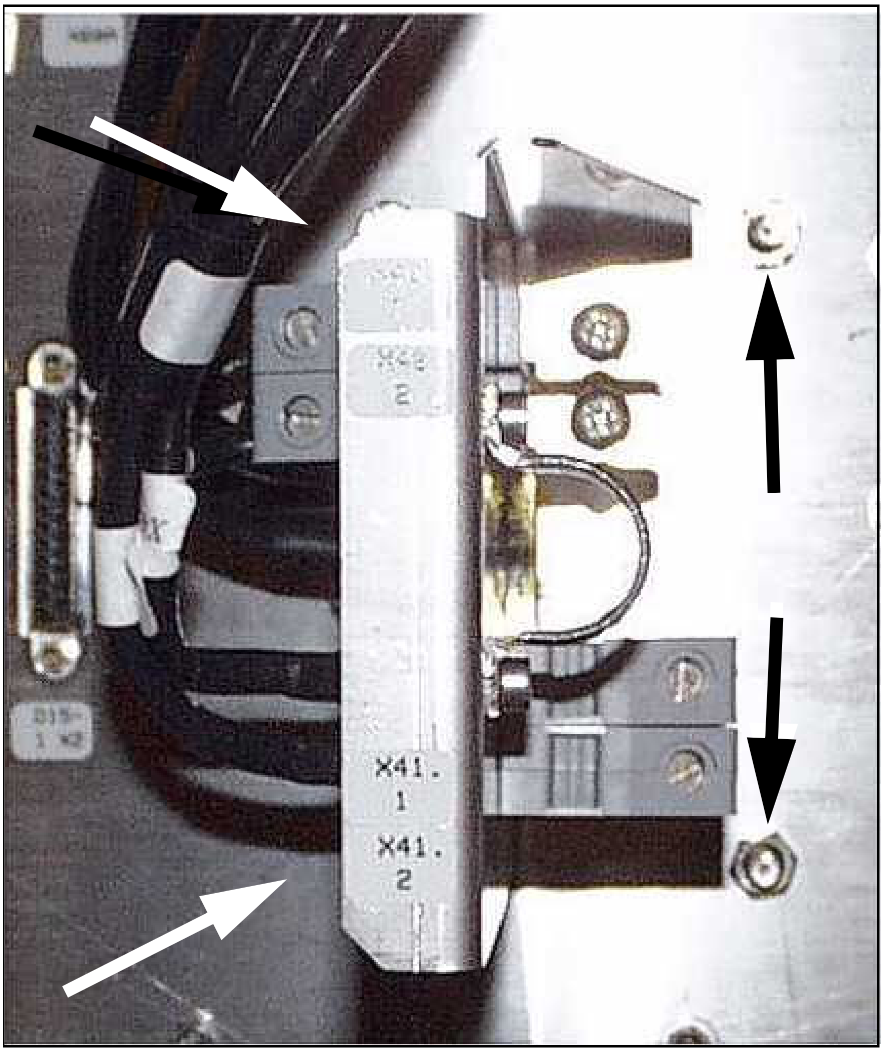
5.2.3 Disassembly

Fig. 40: Power supply E4A - Equipment room side

- Disconnect the line voltage plug (below the fan) from the E4A in the equipment room.
- Remove the 4 mounting nuts of the power supply unit on the RF-filter plate in the magnet room (➔ Fig. 41 Page 75).

- In the equipment room, pull E4A away from the RF-filter plate and disconnect the low voltage supply cables at E4A.

Fig. 41: Power supply E4A - Magnet room side, mounting nuts



5.2.4 Assembly

> **⚠ CAUTION**
>
> **Incorrect connections will destroy all components of the RFIS and RFAS**
>
> » **Before switching on power to the RFIS, measure the voltages at removed cable connections as described in the following.**

- Connect the low voltage supply cables to the new E4A in the equipment room. **Refer to the label on the power supply unit**.
- Position E4A at the RF-filter plate and push the threaded studs through the holes. Mount the four nuts in the magnet room again.

Tab. 8 Cable connections between PSU E4A and RFIS

PSU E4A	RF-filter plate Equipment room	RF-filter plate Magnet room	RFIS
+ 15 V	Z11	X40/1	D14/X3
+ 15 V	Z12	X40/2	D14/X3
GND (+15V)	Z13	X41/1	D14/X1
GND (+15V)	Z14	X41/2	D14/X2
+ 5 V	Z15	X42 (D-SUB)/A2	D14/X4 (D-SUB)/A2
GND (+5V)	Z16	X42 (D-SUB)/A1	D14/X4 (D-SUB)/A1
- 15 V	Z8	X42 (D-SUB)/A3	D14/X4 (D-SUB)/A3
GND (-15V)	Z16	X42 (D-SUB)/A1	D14/X4 (D-SUB)/A1
- 31 V	Z9	X42 (D-SUB)/A4	D14/X4 (D-SUB)/A4
GND (-31V)	Z16	X42 (D-SUB)/A1	D14/X4 (D-SUB)/A1

5.2.5 Concluding Steps

- Disconnect the plug D14/X4 and two cables D14/X3 from RFIS.
- Switch on ACC/F24.
- Measure the voltages of the power supply unit on the cables to:
  X3 - X1: P15V +/- 0.5V
  X4: A3 - A1, N15V +/- 0.75V
  X4: A2 - A1, P 5V +/- 0.5V
  X4: A4 - A1, -30V +/- 1V
- Switch off ACC/F24.
- Reconnect the cables at RFIS.

5.2.6 Adjustments

No adjustments necessary

5.2.7 Final Checks

- Check Error LEDs on RFIS
- Run QA/ Coil Check for Body Coil.
- Run Coil QA/ Coil Check with a local coil connected to the appropriate LC Plug(s).

6.1 Common Procedure

(Avanto: DORA, Part no.47 53 062; Espree: PORA, Part no. 73 92 470)

6.1.1 Tools and time required

6.1.1.1 Time

90 (110) min.

> Two persons are necessary to replace the PORA if no mechanical support is available.
> Weight of PORA with aluminium cooler: 32 kg, PORA with copper cooler: 38 kg

6.1.1.2 Tools

- Normal tool set
- only if MNO option is installed: Service Plug kit (07715720)

6.1.2 Preliminary

- Switch off ACC/F11 (RFPA).

Systems with SEP

- Switch off ACC/F102 (line connection box).
- Switch off cutter switch F3 in the SEP (water pump).

Systems with IFP

- Switch off ACC/F1 (water cooling circuit).

6.1.3 Disassembly

Fig. 42: Avanto: RFPA DORA front side, Allen head screws on the right side

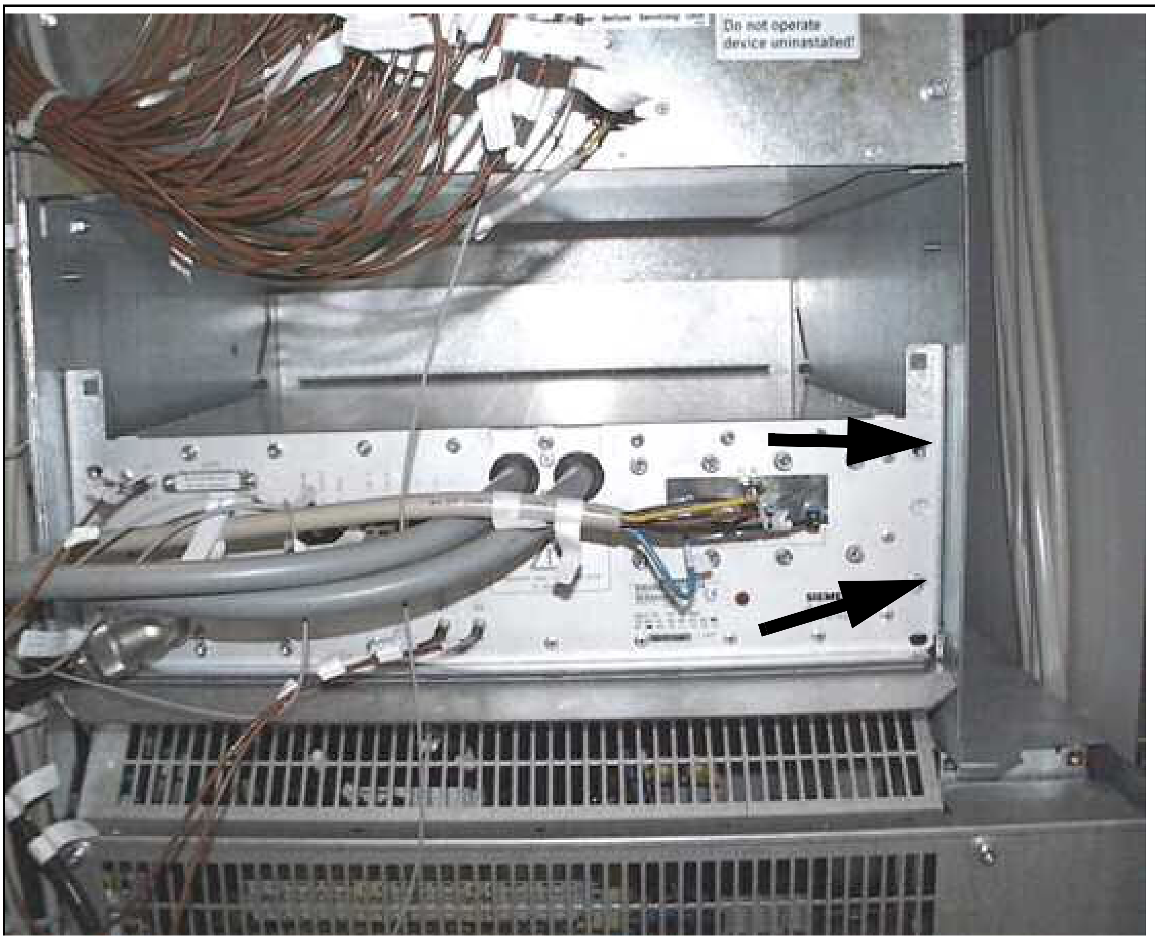


Fig. 43: Espree: RFPA PORA in the ACC

Fig. 44: ACS shutoff valve

(1) Open-end spanner

- Close the shutoff valves (see figure above) from and to SEP/IFP at the top of the ACS.
- Disconnect the quick connect water hose couplings.
- Remove the 3 phase power cable (5 jacks).
- Remove the QLA-cables and the RF output cable.
- Disconnect the FOCs.

> Pull out the FOC double connector U2/U3 straight up without applying unnecessary force.
>
> Disconnect the FOCs U1 and U6 by pressing the plastic release handle at the bottom of the connectors and by pulling the connectors straight up without turning or tilting.
>
> Do not use tools such as screwdrivers.

- Drain the water out of the RFPA.
- Remove the two Allen screws on the left and right of the front plate.

- Holding on to the RFPA (DORA) in back, and the RFPA (PORA) at the handles, pull it out of the ACC.

  For PORA, two persons are needed to lift the amplifier.

> Never use the water hoses to pull out the RFPA.

6.1.4 Assembly

- Push the new RFPA on the guide rails into the ACC.

> Be careful not to jam your hand or fingers between the RFPA and cabinet frame when inserting the new RFPA.

- Install the securing Allen screws on the front plate of the RFPA.
- Connect all cables and FOCs.

  For the 7/16 RF output cable, use a torque of 25 Nm (SW 27).
- Connect the quick connect water hose couplings.
- Open the shutoff valves from and to SEP/IFP at the top of the ACS.

6.1.5 Concluding Steps

- Switch on ACS/F1 or SEP/F3 (Water cooling circuit) and ACC/F102.
- Switch on ACC/F11 (RFPA).
- To check the water pressure, refer to Startup description (➔ Check the water pressure / M6-020.815.01).
- For systems with IFP: Bleed the chiller water pump, refer to Startup description (➔ Bleeding the Chiller water pump / M6-020.815.01).

CAUTION

Before returning or stocking a replaced RFPA, drain the water from the pipes of the device by opening both ends. Simultaneously press the center pins of the hose valves for both water hoses.

To prevent the water hoses from being damaged, they have to be coupled together.

» **There is the potential for irreversible damage to the device if the internal water pipes freeze during shipping.**

> Chilled water containing ethylene glycol should be collected while draining. This water should be refilled into the cooling circuit.

For shipment, use only the original packing.

Pay attention to ensure that the box is not upside down (check the fragile symbol on the outside of the box).

6.1.6 Adjustments

- Perform TU/ RF Characteristic.
- Perform TU/ RF Characteristic LC (if MNO option is installed)
- Perform TU/ Coil Power Losses or BC Power Losses
- Perform TU/ Abs. Receive Path Calibration or BC Image Brightness

6.1.7 Final Checks

- Perform QA/ Abs. Rec. Path Calibration Check or BC Image Check
- Perform QA/ RF Verify
- Perform QA/ RF Verify LC (if MNO option is installed)
- Perform QA/ Coil Power Losses Check or BC Power Losses Check
- Perform QA/ Stability_LongTermCheck,
  (only if license xxxBoldImaging is available).

7.1 Body Coil (BC Avanto)

(material number 81 20 540)

CAUTION

Strong magnetic field.

Magnetic objects may cause injury

» **Maintain a safe distance to the magnetic field.**

CAUTION

The foamed plastics and the plastic film around the body resonator serve to provide noise protection. Do not remove this cover.

» **Never deposit the body resonator on a sharp-edged surface. The plastic cover, damping material and the capacitors could be damaged.**

CAUTION

During the BC replacement, you have to work close to the non-isolated connections of the gradient cables at the gradient coil.

Hazardous voltages may be present at the connections of the gradient cables.

» **Switch off the gradient power amplifier via circuit breaker F2/SSU and F1/100 A contactor in the GPA. Wait at least 20 minutes before you start the BC tuning adjustments.**

7.1.1 Tools and time required

7.1.1.1 Time

5.5 hr. with two people.

7.1.1.2 Tools

- Non-magnetic tool kit
- Non-magnetic measuring tape or gauge
- Non-magnetic spirit level
- Service support (material number 8590841) - Foam pad for storing the BC at the ground
- Open-end wrench SW 18 (material number 8120904) - (delivered with new BC)
- Special non-magnetic screwdriver (material number 73 66 771) for BC trimmer capacitor adjustment.

- BNC 2m Service cable for step TU/ Tuning Calibration (8733479).
- Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration

7.1.2 Preliminary

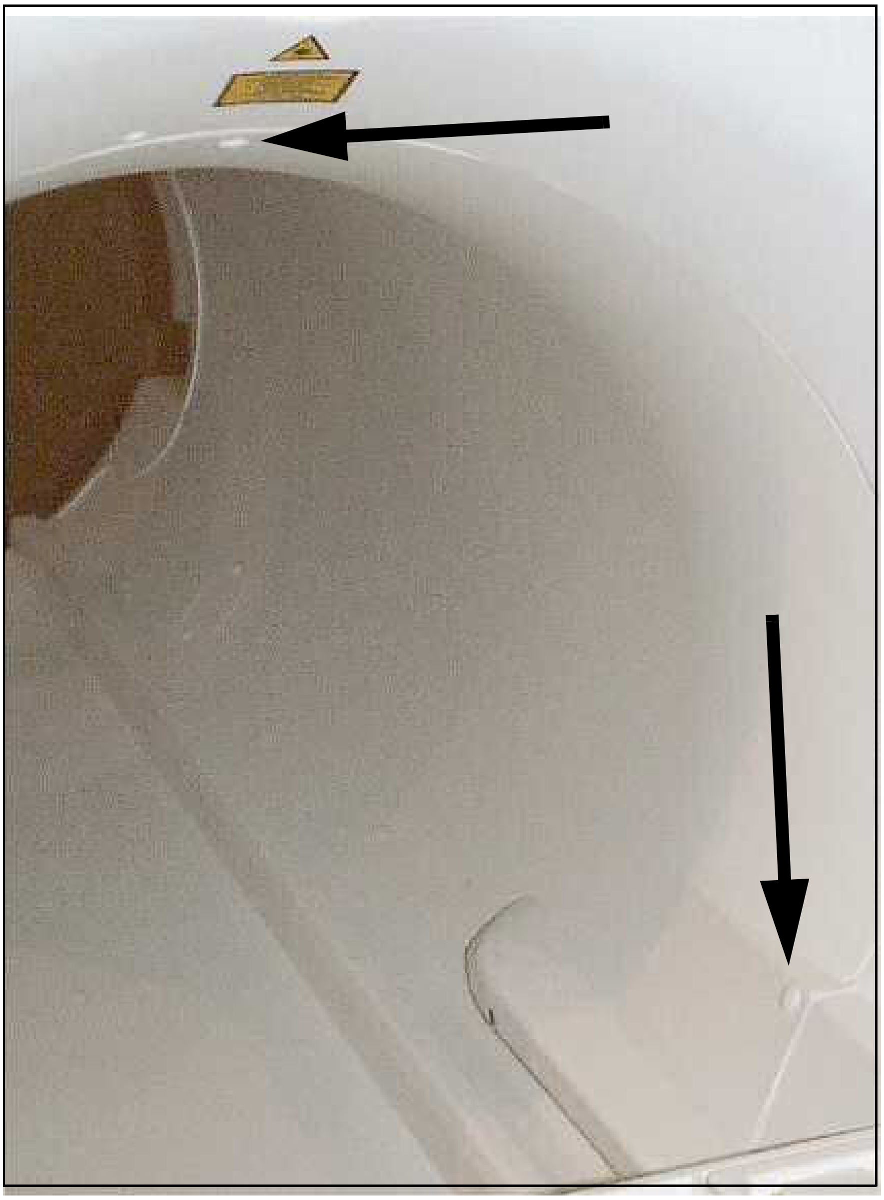
- Switch off GPA/GSSU/F2 and 100 A/F1 contactor in the GPA.
- Switch off ACC/F11 (RFPA).
- Move the patient table down.

7.1.3 Disassembly

7.1.3.1 Magnet Covers

- Remove the front side funnel:

Fig. 45: Front cover, position of grub screws at right and top



- Remove the four caps over the set screws securing the front funnel with plugs in lugs at the BC.
- Loosen the three set screws.
- Disconnect the plug of the light marker and remove the front funnel.

- Remove the back cover of the magnet:

Fig. 46: Cover - Back, position of grub screws, right side and below

- Remove the two caps over the set screws securing the upper third of the back cover and the three caps over the set screws securing the lower cover with plugs in lugs at the BC (➔ Fig. 46 Page 85).
- Loosen the five set screws.

Fig. 47: Mounting screws on back cover of magnet

- Loosen the three mounting (Allen) screws for the upper third of the cover and remove it.
- Loosen the four mounting (Allen) screws of the lower part of the magnet back cover.
- If necessary, remove the funnel light connectors and remove the cover.

7.1.3.2 BC

- Disconnect the two RF cables from the BC (SW 18), using the non-magnetic flat wrench (SW18) provided in the delivery volume.

Fig. 48: RF cable connection at BC

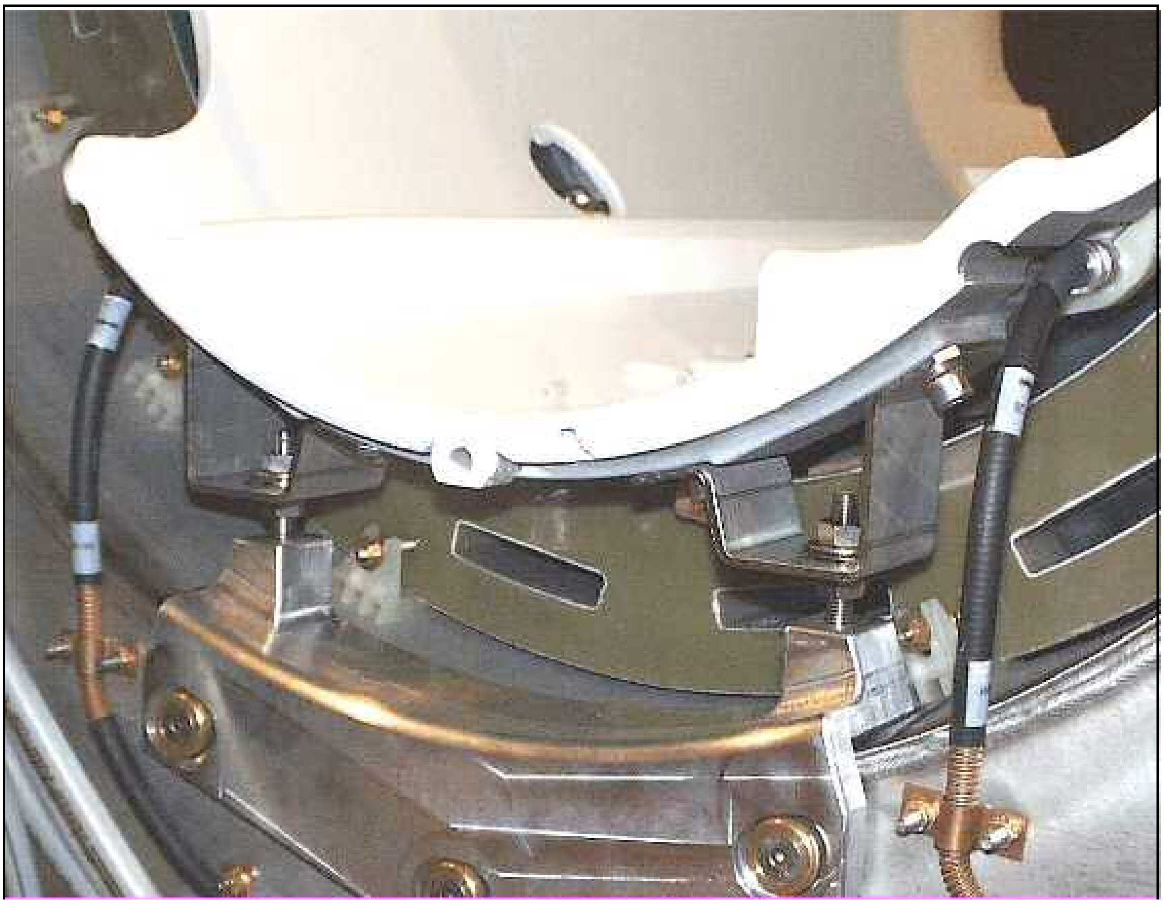


Fig. 49: RF pick-up probe cabling

- Disconnect the two RF pick-up coil cables from the BC.

  If necessary, loosen the cable collars at the magnet a little bit for easier reconnection later.

Fig. 50: Ground connection BC (left and right side)

- Disconnect the two grounds from the BC to the RF shield on the left and right sides and the one at the back end (bottom) of the RF shield.

Fig. 51: Ground connection BC (bottom side)

Fig. 52: Connector X3 at static detuning unit at BC

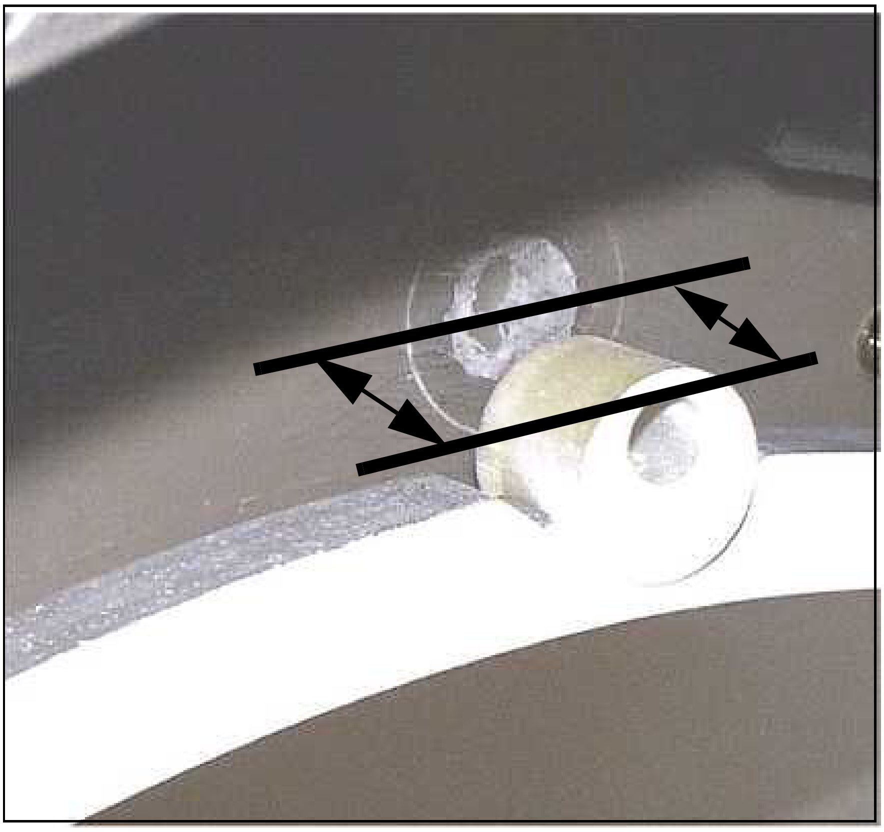
- Disconnect connector X3 from the BC. (Current/voltage supply of PIN diodes for dynamic/static (de)tuning of BC).

Measurements before removing the old BC

- Measure and note the Z-distance between front side of the upper lug at the BC and the edge of the gradient coil (➔ Fig. 53 Page 90). Nominal distance: 34 mm.

- Measure the radial distance of inner surface of BC at the front, to the magnet bore at three points.

Fig. 53: Z-distance BC - Gradient coil



Removing the old BC

- Remove the foam pieces (for noise damping) between the gradient coil and the BC. (Three or four pieces in back and one piece at front of the BC) (➔ Fig. 55 Page 92).
- Loosen the four mounting (8 mm Allen) bolts of the BC at the BC support (BC back) and the four mounting (8 mm Allen) bolts of the BC at the BC support (BC front).

  There is a 2 mm disc of Sylodyn between the BC surface and the support, as well as between the mounting bolt bushing and the support for noise dampening functions (➔ Fig. 54 Page 91). Ensure that these parts are not lost (➔ Fig. 56 Page 93).

Fig. 54: BC support with sylodyn insulations

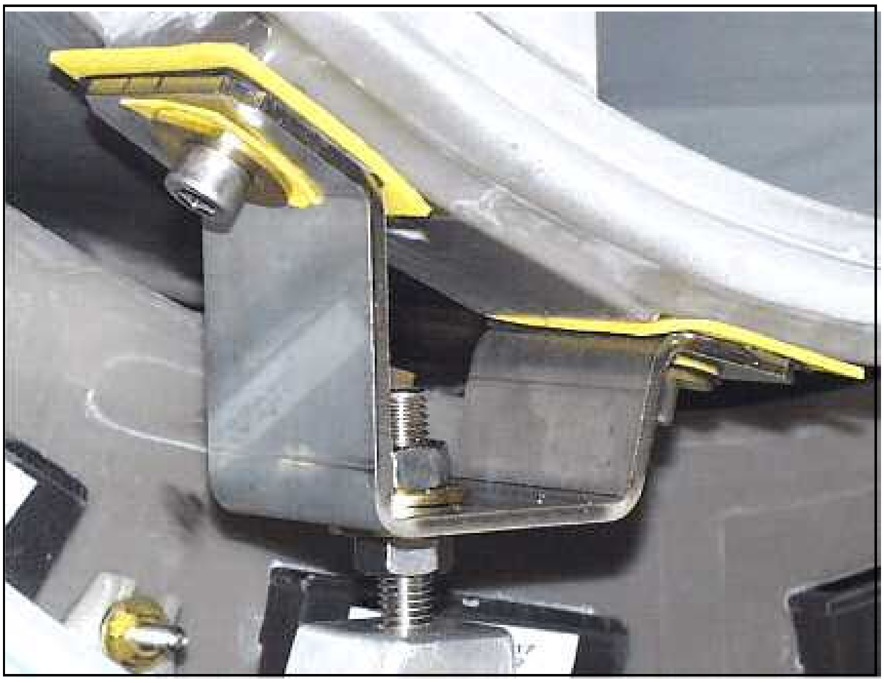


Fig. 55: Noise damping foam between BC and GC

Fig. 56: Sylodyn insulation, mounting bolt with insulated bush

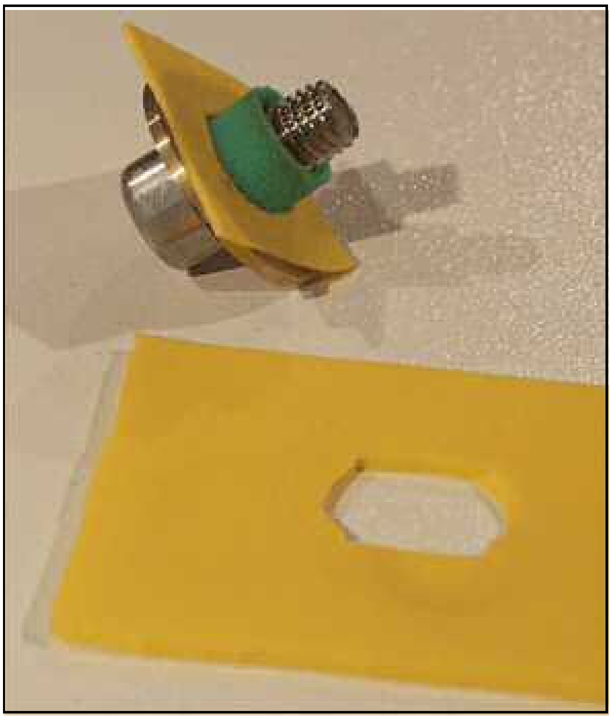


Fig. 57: Body coil support: Clamping nut upper side, support nut lower side

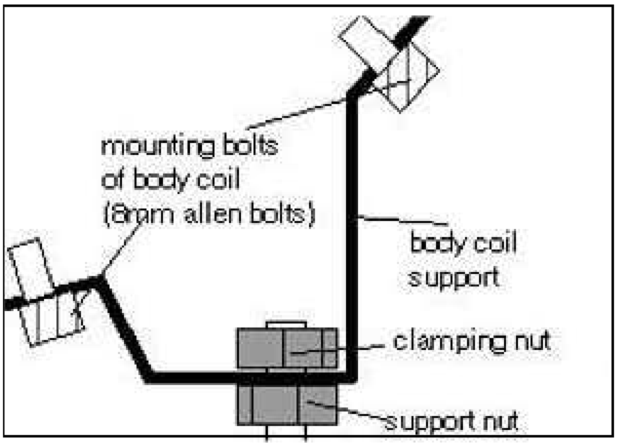
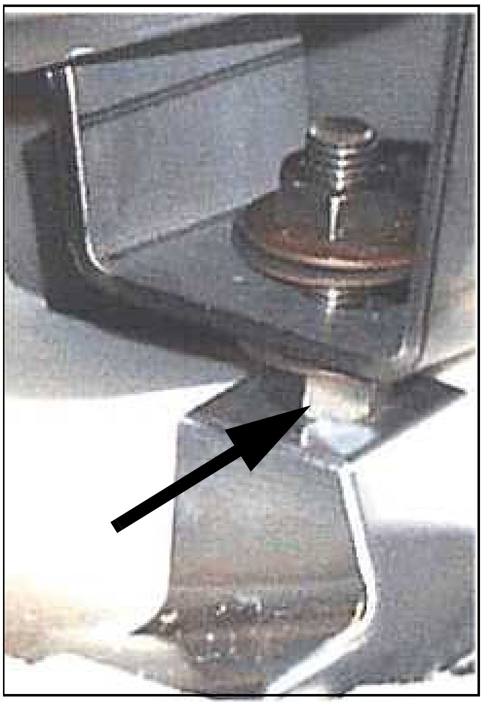


Fig. 58: BC support - Clamping nut above , support nut below



- Loosen the two lower support nuts at the supports in front and turn the nuts down. The upper clamping nuts remain fixed to assure the correct reassembling height (➔ Fig. 58 Page 94).

  All nuts on the back BC supports remain as is.

- Position the service support on the patient table and adjust the height of the patient table for easy removal of the BC (➔ Fig. 59 Page 95).

Fig. 59: Service support on the patient table

» The BC is now ready to be removed. At least two people are needed to remove the BC. The weight of the BC is approximately 90 kg.

Fig. 60: Aligning the BC to the service support

- Lift the BC a little bit at the magnet in back away from the supports.Then slowly push and pull the BC **centered on the rails in the gradient coil** out of the gradient coil. Keep the BC centered all times. Be careful not to destroy the plastic film and foamed plastic on the BC as well as block capacitors and ground connectors on the RF-shield if the BC sticks due to an uncentered position.
- Let the BC rest on the service support. (➔ Fig. 59 Page 95)When moving the BC out, shift the service support accordingly, so that the center of the BC rests on the center of the service support.
- Store the BC in a suitable place on the floor.

7.1.4 Assembly

- Place the new BC on the service support on the patient table.
- Move the new BC into the gradient coil. The BC will slide on the two rails inside the gradient coil.

Use extreme care when moving the BC into the magnet to avoid damage to the RF-shield inside the gradient coil.

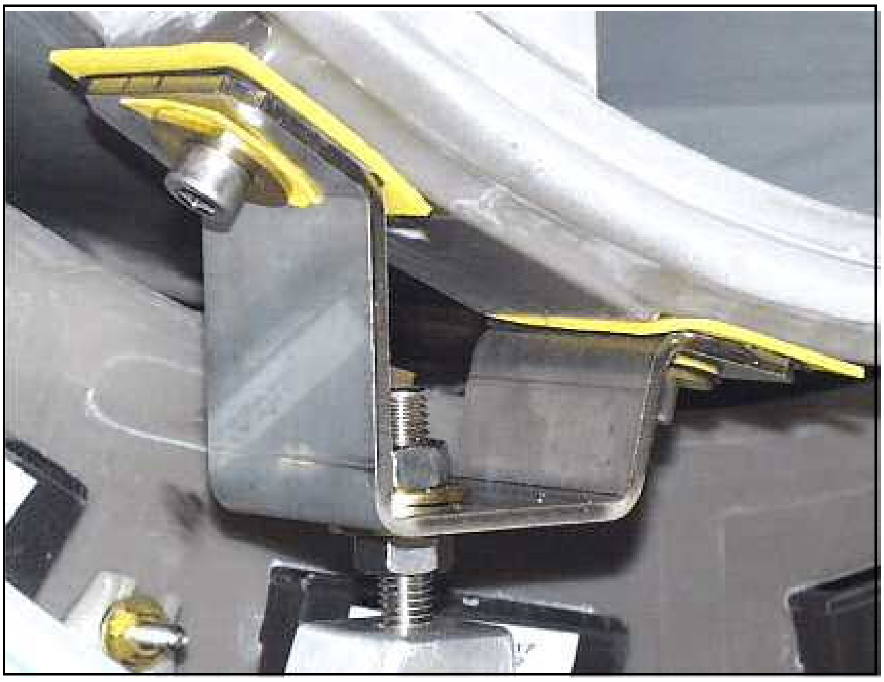
Keep the BC centered at all times on the slide rails in the gradient coil.

- Place the coil on the coil supports and insert all mounting (8 mm Allen) bolts hand tight. Turn the two support nuts of the front side support up in its old position.

  Ensure that the Sylodyn disc layer insulates the BC and the mounting bolt bushings from the support. The insulated bushings have to be centered on the holes of the support. Avoid hard contact between the bushings and the support (➔ Fig. 61 Page 97).

It is essential to decouple the BC with the Sylodyn damping layer from the supports. Any hard contact between the BC or mounting bolts and the supports would deteriorate the noise dampening beyond specifications.

Fig. 61: BC support with sylodyn insulations



- Connect the three grounding connectors.

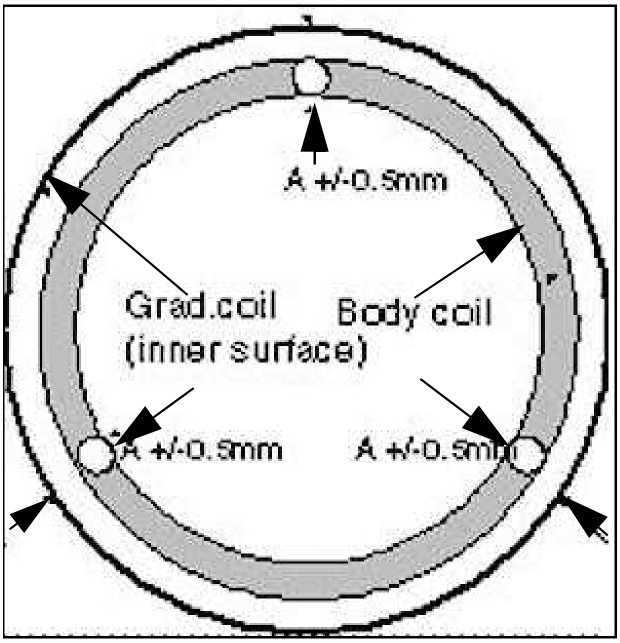
7.1.4.1 BC Alignment

- **Check adjustment in Z-direction:**

  Measure the distance between front of the upper lug at the BC and the edge of the gradient coil, and compare with the old measured value (➔ Fig. 53 Page 90).

  Adjust the Z-distance if necessary.

Fig. 62: Centric alignment to the gradient coil



- **Check the adjustment in the x and y-directions:**

  The body coil has to be centered to the gradient coil.

  The distance between the inner surface of the BC and the inner surface of the gradient coil must be the same all around (tolerance +/-0.5 mm).

  Hint: Use the 13 mm end of an Imprex non-magnetic wrench as a gauge.

  Adjust the x and y-distances if necessary using the support nuts.

  Measurement at the back of the magnet: Measure at three points: Top, left and right, shifted 12 degrees.

- **Measurement at front of the magnet:**

  Measure left and right at position shown in (→ Fig. 63 Page 99)

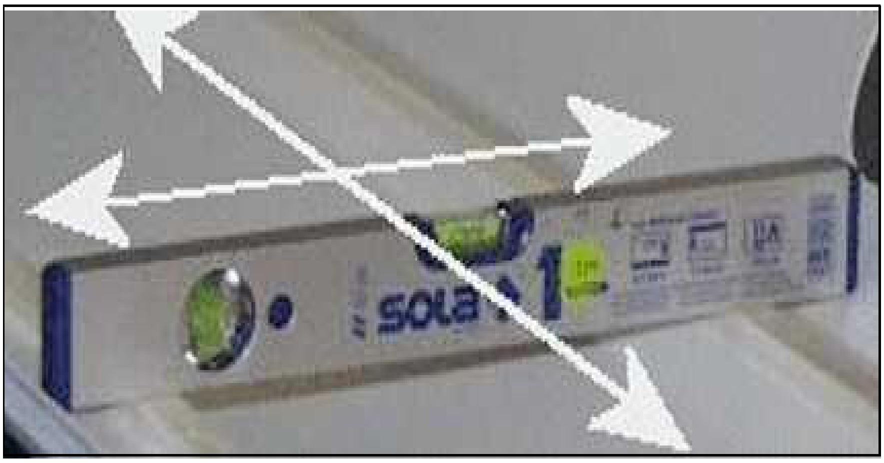
  Measure the radial distance of the inner surface of the new BC to the magnet bore at the same points as on the old BC.

  Adjust the distances using the support nuts.

Fig. 63: Measurement of radial distance BC -GC

- Use a non-magnetic spirit level to adjust the body coil horizontally in the X, Z-directions in front and in back.

Fig. 64: Leveling the BC in z and x direction



- Tighten all screws and nuts of the coil supports and check the alignment again.
- Reconnect the TX cables, the RF pick-up coil probe cables and the connector X3.
- Retighten the cable collars at the magnet.

- Attach the noise damping foam pieces again in their old position between the gradient coil and the BC. (Three or four pieces in back and one piece at the front of the BC).
- Switch on ACC/F11 (RFPA).

7.1.5 Adjustments

- Perform TU/ Tuning Calibration
- Perform TU/ BC Tuning
- Perform TU/ RF Characteristic
- Perform TU/ Coil Power Losses or BC Power Losses
- Perform TU/ Abs. Rec.Path Calibration or BC Image Brightness

7.1.5.1 BC Tuning

CAUTION

During BC Tuning you have to work close to the non-isolated connections at the gradient coil.

Risk of electrical shock! Hazardous voltages may be present at the gradient connections.

» **Switch off the gradient power amplifier.**

» **Follow the advice given by the program (pop-ups).**

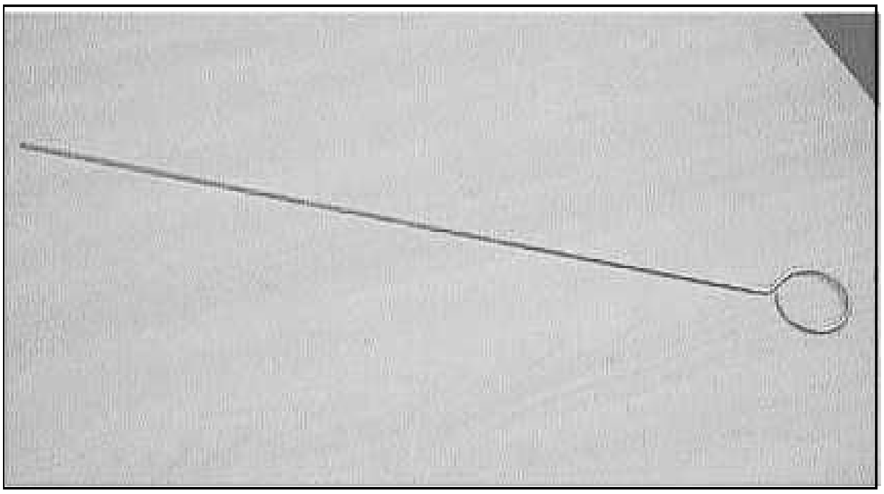
Prerequisites

Since BC Tuning uses the H and S matrices to perform calculations, the tuning calibration tune-up procedure has to be completed successfully before starting with the BC tuning.

To gain access to the trimmer capacitors at the back of the Body Coil, the back magnet cover has to be removed.

The special non-magnetic adjustment tool (i.e. long screwdriver, material number 73 66 771) is required to turn the trimmer capacitors (➔ Fig. 65 Page 101).

Fig. 65: Tool for trimmer capacitor adjustment



Frequency and Decoupling Adjustment

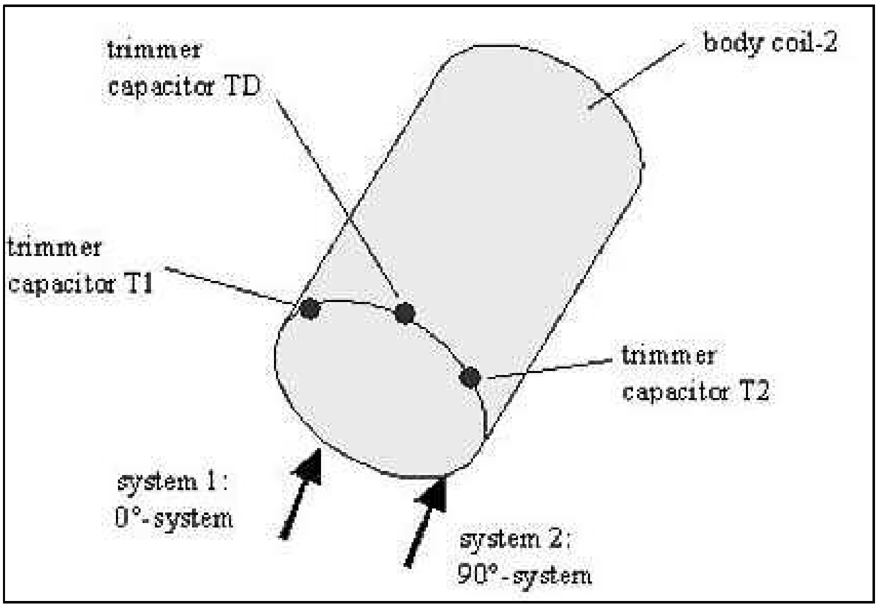
You will be guided through iterative fine adjustments. Between the adjustment steps, the resonance and decoupling of both systems (0 and 90 degree) can be varied manually using the three trimmer capacitors at the coil. The program stops when both resonance and decoupling are within specifications.

The trimmer capacitors T1 and T2 affect the resonance frequencies of systems 1 and 2. However, capacitor TD affects the decoupling between the two systems.

The decoupling has to be adjusted only if the measurement of the transmission value is not within specifications.

Two types of trimmer capacitors are used. They allow 60 respectively 25 turns with the same full range of capacitance). By default, they are set in the middle position, which means you can make 30 or 12 1/2 turns clockwise and 30 or 12 1/2 turns counter-clockwise.

Fig. 66: BC Tuning - trimmer capacitors, (T1 left, TD middle, T2 right)



To prevent damage to the trimmers, use caution when approaching the end position. Do not use excessive force.

Trimmer capacitor	Main influence on	Clockwise (cw)	Counterclockwise (ccw)
T1	Resonance frequency system 1	Decrease resonance frequency 1	Increase resonance frequency 1
T2	Resonance frequency system 2	Decrease resonance frequency 2	Increase resonance frequency 2
TD	Decoupling between systems 1 and 2	Increase capacitance	Decrease capacitance

If the values are slightly out of spec, the system will still work.

<table>
<tr><th>Step 1</th><th colspan="3">Reflection adjustment of delta frequency</th></tr>
<tr><td rowspan="6"></td><td colspan="3">delta < 100 kHz ?</td></tr>
<tr><td>Yes</td><td colspan="2">No</td></tr>
<tr><td rowspan="4">No action required</td><td colspan="2">Frequency 0 degree > frequency 90 degrees</td></tr>
<tr><td>Yes</td><td>No</td></tr>
<tr><td>Turn T1 2x cw
Turn T2 2x ccw</td><td>Turn T1 2x ccw
Turn T2 2x cw</td></tr>
<tr><td colspan="2">Go to step 1</td></tr>
</table>

<table>
<tr><th>Step 2</th><th colspan="2">Mechanical adjustment of delta decoupling (transmission)</th></tr>
<tr><td rowspan="4"></td><td colspan="2">Transmission 0... -12 dB</td></tr>
<tr><td>Yes</td><td>No</td></tr>
<tr><td>Adjust mechanical position of the BC in the gradient coil.</td><td rowspan="2">Go to Step 3</td></tr>
<tr><td>Go to step 2</td></tr>
</table>

<table>
<tr><th>Step 3</th><th colspan="3">Electrical adjustment of decoupling (transmission)</th></tr>
<tr><td rowspan="2"></td><td colspan="3">Transmission T = -12 dB .. -18 dB</td></tr>
<tr><td colspan="2">Yes</td><td>No</td></tr>
<tr><td rowspan="3">Step 3.1</td><td colspan="2">Turn TD 3x cw, press start button at panel</td><td rowspan="7">No action required</td></tr>
<tr><td colspan="2">T lower</td></tr>
<tr><td>Yes</td><td>No</td></tr>
<tr><td rowspan="3">Step 3.1.1</td><td>Turn TD 3x cw, press start button at panel</td><td>Turn TD 3x ccw, press start button at panel</td></tr>
<tr><td>T lower -> go to 3.1.1</td><td>T lower -> go to 3.1.1</td></tr>
<tr><td>No -> undo last step</td><td>No -> undo last step</td></tr>
<tr><td colspan="3">If necessary, repeat step 3.1</td></tr>
</table>

7.1.5.2 Remarks about BC Tuning

- T1 and T2 also influence the decoupling.
- TD also influences the frequency.
- The influence of TD relative to T1 and T2 on the decoupling is four times higher. However, it is only half as high with respect to the frequency of the two partial systems.

- To change the decoupling in the same direction, turn TD in the opposite direction as T1 and T2.

Freqency adjustment with constant decoupling:

0-degree system:

Turn T1 x turns and TD x/2 turns in the same direction.

90-degree system:

Turn T2 x turns and TD x/2 turns in the same direction.

Decoupling adjustment with constant frequency:

Turn TD x turns and T1,T2 x/2 turns in opposite direction.

7.1.6 Concluding Steps

- Mount magnet covers.
- Switch on GPA/GSSU/F2 and 100 A/F1 contactor in the GPA.

7.1.7 Final Checks

CAUTION

The width of the gaps between the patient table and the front cover on right and left side as well as the body coil and patient bore light rails should be of equal size and less than 8 mm. (Refer to the Start-up instructions)

Danger of injury to patient by jamming fingers.

» **Minimize the clearances by adjusting the covers relative to the Body coil. Refer to (➔ System Start-up / M6-020.815.01).**

- Perform QA/ Abs.Rec.Path Calibration Check or BC Image Check
- Perform QA/ BC Tuning Check
- Perform QA/ RF Verify
- Perform QA/ Coil Power Losses Check. or BC Power Losses Check
- Perform QA/ Spike Check.
- Perform QA/ SAR Monitor Test (VB17A and higher)

7.1.7.1 Checking the PIN Diodes of BC

(Only in case of problems)

For a quick check of the PIN-diodes, remove the connector X3 of the detuning unit at the BC and measure using the "diode test function" of a DVM in both (plus/minus) directions at the detuning unit.

Typical displayed values:	Forward direction: 0.4
	Reverse direction: “OL”

7.2 Upgrade Espree straight bore Gradient/Body Coil to Slimline

7.2.1 Worksteps - Overview

- Ramp magnet down.
- Remove front and rear cover.
- Disconnect all attached cables and remove old straight coil with the gradient coil lifting device.
- Remove shim trays and light bars, including securing materials, from the old coil.
- Install new Slimline Gradient coil and re-connect cabling.
- Install the empty shim trays.
- Install new Body Coil and re-connect cabling. Install light bars.
- Install the new divided rear cover and other covers that were removed, except for the front cover.
- Ramp magnet to field and let stabilize.

7.2.1.1 Software

Load SP1 Software patch for VB12A, if not already installed.

Go to configuration -> measurement settings -> set gradient to AS022SL

Reset gradient offsets and Super Conductive Shim Current.

Reboot the system to make the changes effective!

7.2.1.2 Work Steps - Overview continued

- Install Array Shim Device and shim magnet, starting with the bare plot.
- Re-install all covers.
- Perform all General TuneUp and QA steps including Body Coil Tuning.

7.2.1.3 Final Work Steps

- Perform final assembly, clean covers, and return system to customer.
- Complete Installation Protocol.
- The old straight bore coil (pn 100 47 942) must be returned to Erlangen.
- Scrap old rear cover and bore tube.

7.2.2 Tools and time required

7.2.2.1 Time

Total 5 days.

2 days require 4 people for replacing the hardware.

3 days require two people (one trained Espree CSE/ISE plus one assistant).

7.2.2.2 Tools

For USA, add TT before the material number when ordering from Tool&Test

Lifting device for GC/BC with accessory parts 1225	70 14 678
Short beam 4m	11 41 725
Coil support (sliding carriages with 2 support beams)	51 37 042
Magnet power supply	83 95 613
Espree Shim Array	83 68 313
Shim plates, left over from first installation	
Helium fill kit	
8-s. socket gradient coil non-mag.	103 99 445
Torque wrench 20 - 50 Nm	83 95 803
Accessory kit for torque wrench	83 96 116
Standard tool kit	
Nonmagnetic steel tool set	77 57 706
Service aid back cover (if needed)	100 18 422
Adjustment tool for body coil decoupling	73 66 771 (non T&T item)
Tools for hammering in the wedges (→ Fig. 76 Page 121)	
Measuring tape or gauge	
Ethylene glycol and antifreeze tester	81 15 094
Double hexagon 12-point socket	100 48 130
Plumber's tape (1 to 1.5" white sand cloth (local purchase))	
Scotch Brite pad (local purchase)	
Heat gun	

7.2.3 Preliminary

- SP1 Software patch installed, for VB12A.
- All upgrade equipment & tools on site.
- Measure the pathway from the exterior to examination room; minimum clearance for 1 m (37") for the gradient coil. Allow additional room to maneuver the lifting devices around corners.
- Liquid helium and helium gas are on site if refill is necessary.

7.2.3.1 Parts required

For **all** systems to be updated: GC/BC slim complete. 100 48 100

In addition

For systems <30099:	upgrade kit divided rear cover 100 18 727 (includes cable kit tunnel light)
For systems >30098:	cable kit tunnel light 100 18 757

7.2.3.2 Procedure

Ramp magnet down, refer to Startup Instructions.

For USA:

Download the MPS log file

To download the log file:

- In the ramping Software display, press the G key <Return>, this will ensure that you are at the log window.
- Select the following in the HyperTerminal: (top menu)

  Capture Text/start

  Type the location and name of the log file e.g. C:\Medcom\Temp \MR25123_51099RDlog.txt (Note: capture all ramp logfiles both above and below)
- Press the L key <Return>, the data window appears.

  The data window contains seven columns. Each line displays the data stored at a specific period of time during magnet ramping.
- After the data has stopped scrolling, select Capture Text/Stop button, transfer the file to a memory stick, and send the log to TSE Andy Wren andy.wren@siemens.com & Jerry White jerry.white@siemens.com

> **CAUTION**
>
> **During the GC/BC replacement you have to work rather close to the non-isolated connections of the gradient cables at the GC/BC.**
>
> **Hazardous voltages may be present at the connections of the gradient cables.**
>
> **» Switch off the gradient power amplifier via circuit breaker F2/SSU and F1/100 A contactor in the GPA. Wait at least 20 minutes before you start working at the GC/BC.**

- Move the patient table down.
- Switch off GPA/GSSU/F2 and 100 A/F1 contactor in the GPA.
- Switch off ACC/F11 (RFPA).
- Switch off ACC/F24 (RFIS).
- Switch off electrical shim F21.
- Disassemble the front funnel and ring, the back covers of the magnet, as well as the bore cover (with the foam rubber strips) together with its supports at the magnet.

  (Refer to (→ Covers / M6-020.841)).

  Remove the light bars and their securing materials from the BC.

7.2.4 Disassembly of the GC/BC

Prerequisites: The GC/BC is freely accessible.

7.2.4.1 Disconnecting the Water Hoses

Systems with SEP

- Switch off cutter switch F3 in the ACS (water pump).
- Disconnect the water couplings.

Systems with IFP

- Switch off ACC/F1 (water cooling circuit).
- Disconnect the water couplings.

7.2.4.2 Disconnecting Cables

- Remove the cover of the gradient cable connections at the magnet.
- Loosen the gradient cables; pull the shim plug and the plugs of the coil temperature monitoring.
- Remove the gradient coaxial link cables from the magnet.
- Loosen the ground braids from GC/BC in back of the magnet.
- Disconnect the two TX-cables from the BC; disconnect RFIS/ X9 and loosen the cable ties which route the cable <W5103> at the magnet.

7.2.4.3 Disassembling mechanical Components

- Remove the shim trays from the GC. (Attach any missing numbering). Remove all shim plates.
- Loosen the front nuts and washers on the threaded bolts of the wedges (using a 3-mm Allen key) securing the GC/BC.
- Carefully remove the foam rubber strips between the GC/BC and the magnet bore (front and back of magnet).

7.2.4.4 Measurements before removing the GC/BC

Fig. 67: Z-alignment - measure "A"

- Measure the distance **"A"** between the **inner front face** of the **BC/GC (“straight bore”)** or between the **front face** of the **GC (“slim line”)** and the front edge of the magnet at four positions.

  The following table shows the specified distance A dependent on **measure M**, printed on the front of the magnet.

- Put a level on the edge at the magnet front bore that does not show welding seams and measure the distance to the inner edge of the gradient coil.

"A" (mm) “straight bore” Distance inner front face GC/BC to magnet edge	“A” (mm) “slim line” Distance front face GC to magnet edge	Measure M on magnet
70	35	-5 S
69	34	-4 S
68	33	**-3 S**
67	32	-2 S
66	31	-1 S
65	30	**0**

"A" (mm) “straight bore” Distance inner front face GC/BC to magnet edge	“A” (mm) “slim line” Distance front face GC to magnet edge	Measure M on magnet
64	29	+1 P
63	28	+2 P
62	27	+3 P
61	26	+4 P
60	25	+5 P

The new GC has to be adjusted in the Z-direction at the same distance **"A"** as shown in the conversion table above.

Example 1:

The old straight coil has a distance "A" of 63 mm to the front edge of the magnet. This corresponds to measure "M" of +2P.

Measure "M" is printed on front of the magnet for verification.

Install the new Slimline gradient coil at a distance of 28 mm to the front edge of the magnet, according the conversion table. This corresponds to measure "M" of +2P.

In other words:

Old straight coil "A" = 63mm

New slim coil "A" = 28 mm.

This is the correct positioning in example 1.

Example 2:

The old straight coil has a distance "A" of 66 mm to the front edge of the magnet. This corresponds to measure "M" of -1S.

Install the new Slimline gradient coil at a distance of 31 mm to the front edge of the magnet, according the conversion table. This corresponds to measure "M" of -1S.

In other words:

Old straight coil "A" = 66 mm

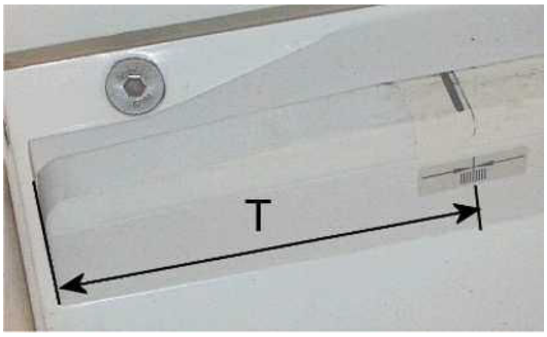
New slim coil "A" = 31 mm

This is the correct positioning in example 2.

For any other combinations please refer to the conversion table above!

- Measure and note the Z-distance (T) of the marker for the shim device inside the body coil

Fig. 68: Position of marker for shim device

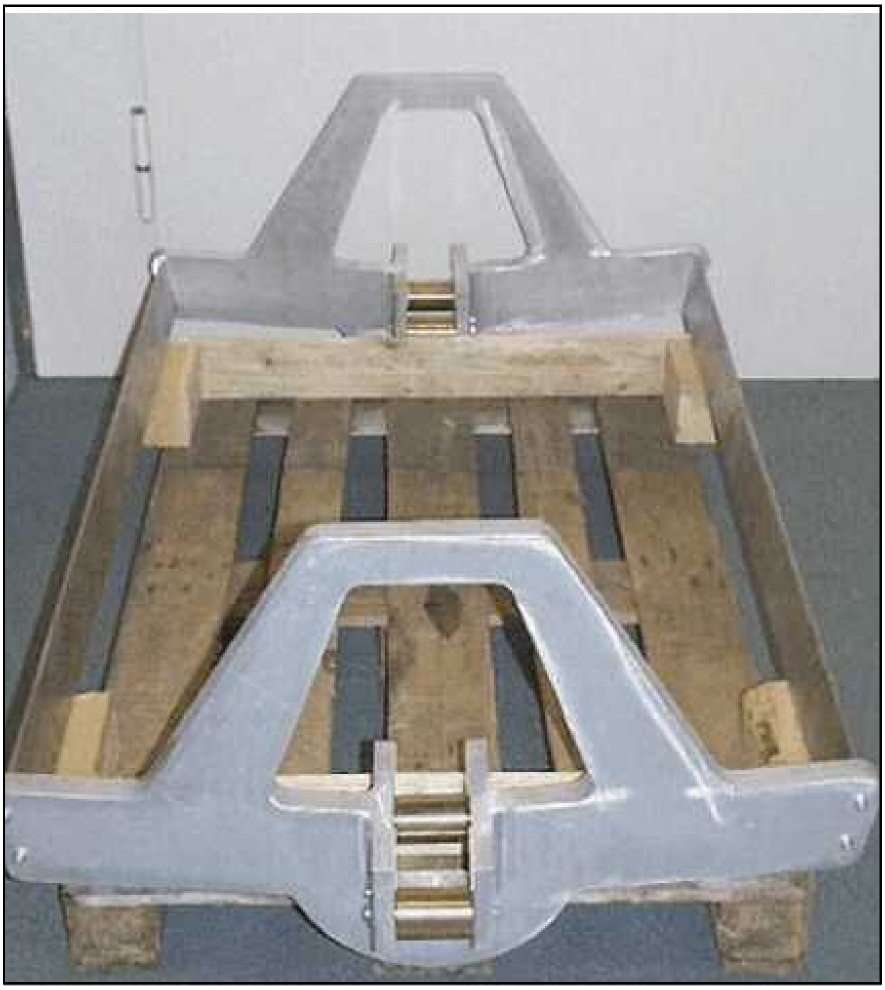


7.2.4.5 Removing the GC/BC using the Lifting Device

Lifting Device Assembly Configuration

- Mobile support 1 in front of the patient table
- Mobile support 2 under the patient table in front of magnet
- Stationary support behind the magnet (**Caution: Use foot extensions!)**
- Two connected beams
- Two sliding supports running on the connected beams.

Fig. 69: Carriages of the GC/BC lifting device



Procedure

- Disassemble the sliding carriage and put the two support beams through shim pockets 1 and 8.
- Mount the back/front sliding carriage on the support beams (on shim pocket positions 1 and 8) at the magnet back/front.
- Position the stationary beam support behind the magnet.
- To protect the RF shield, put a large piece of foam or similar inside the GC/BC.

- Put the rear (short) support beam through the GC/BC and sliding carriages and place its end on the stationary support.

Fig. 70: Beam on stationary support - GC with mounted sliding carriages

- Position the front support beam using both mobile supports 1, 2.

  If necessary adjust the height of the patient table.
- Put the two beams together and secure the connection with two bolts.

- Adjust the height of the beam to the position of the sliding carriages.

Fig. 71: GC/BC mounted to the carriages on the beam

- Remove the mobile support stand 2.
- Deinstall all wedges:

  First remove the upper wedges; carefully loosen the wedges (by lightly hammering against both sides of the wedge with a plastic hammer).

  For removing the lower wedges, raise the beam to lift the GC/BC in the magnet bore. Then remove the lower wedges.
- Put the water hoses with quick connections and the cable to the RFIS into the gradient coil. Proceed carefully. Do not damage the shim and temperature sensor connection cables and connectors.
- Remove the back washers, nuts and threaded bolts from the GC.

- Push out the GC/BC. Move it up to the front of mobile support 1 using the sliding carriages.

Fig. 72: GC/BC removed - mobile support 2

- Reconnect mobile support stand 2 with the foremost support beam; separate the back beam.
- Transport the GC/BC out of the magnet room using mobile support stand 1 and 2.

Fig. 73: Removal of GC/BC using carriages on the beam

7.2.4.6 Preparing the GC/BC for Shipment

- Place the GC/BC on the transport pallet, disassemble the sliding carriages and support beams. Remove the support beam of the lifting device.
- Attach the GC/BC to the transport pallet with transport safety items.

CAUTION

Prior to shipping the GC/BC, any remaining water must be emptied from the gradient coil.

Otherwise, the internal water channels may freeze.

» **Open the quick connectors using the delivered adapters or by pressing in the quick connects with a piece of hose or a screwdriver. If available, use compressed air for emptying the GC/BC. The fluid should be refilled in the cooling circuit.**

7.2.4.7 Return of parts

The old straight GC/BC must be returned to Germany.

Material number 100 47 942

Make sure that this material number is listed on the transport box.

Remove all other labels from the transport box.

Procedure for the USA:

For additional information, please contact the USC.

7.2.5 Assembly

7.2.5.1 Installing the new Gradient Coil

- To protect the RF shield, put a large piece of foam or similar inside the GC.
- Connect the sliding carriages to the two support beams in shim pockets 1 and 8 of the new GC.
- Mount the beam (routed through the GC and sliding carriages) on mobile support stands 1 and 2.
- Transport the GC using mobile transport stands 1 and 2 in the magnet room over the patient table.
- Reattach both support beams, secure with two bolts and remove mobile support stand 2.
- Prior to pushing the GC into the magnet, clean the magnet bore with a vacuum cleaner.
- Center the GC to the magnet bore by adjusting the height of the support beam and repositioning the mobile support. Adjust the support beam position in the Z-direction with a level.
- Push the GC into the magnet until the Z-distance **"A"** is reached. Measure **"A"** at several points between the inner front face of the GC to the front edge of the magnet (small edge without welding seams). (→ Measurements before removing the GC/ BC / Page 110)

7.2.5.2 Adjusting the GC and assembling the Wedges

- Screw in the threaded bolts for the wedge attachment and attach the back nuts and washers (up to a minimum distance of approximately 10 mm from the edge of the gradient coil).
- Verify the correct Z-adjustment and radial adjustment of the GC:

  The horizontal position of the support beams is required for the correct radial position of the GC.

  Check the position of the beam with a level and adjust the height of the supports, if required.

- Using the level, which is placed on the two support beams in the shim pockets, check the correct position of the GC (regarding the X-axis).

  Alternatively, you can mount two screws securing the wedge in threaded holes of equal height as a support base for level measurement.

Fig. 74: X-alignment of the BC/GC using a level

- Central adjustment:

  Place the GC in exactly the Center Position by either raising or lowering the support beam or by adjusting the position of mobile support stand 1:

  The distance of the outer surface in the center of a shim pocket to the bore tube of the magnet "R" has to be measured in front and back of the magnet at several

locations around the coil. All measurements of "R" must be within a tolerance margin of +/- 1 mm (i.e. R +/- 1 mm).

Fig. 75: Radial alignment - Measure "R"

- Attach all wedges on both sides by hand (lightly press them in) and test for appropriate fit:

  Begin with the two lower wedges, then add two upper wedges etc. (Total: on the front side 12 wedges, in back 10 wedges).

  Note:

  The two lower wedges at the magnet front must be harmonized with the bore cover supports.

  The lower wedge in back of the magnet as well as the wedge below the RF-connectors are secured to a threaded bolt at the magnet. (➔ Fig. 80 Page 124)

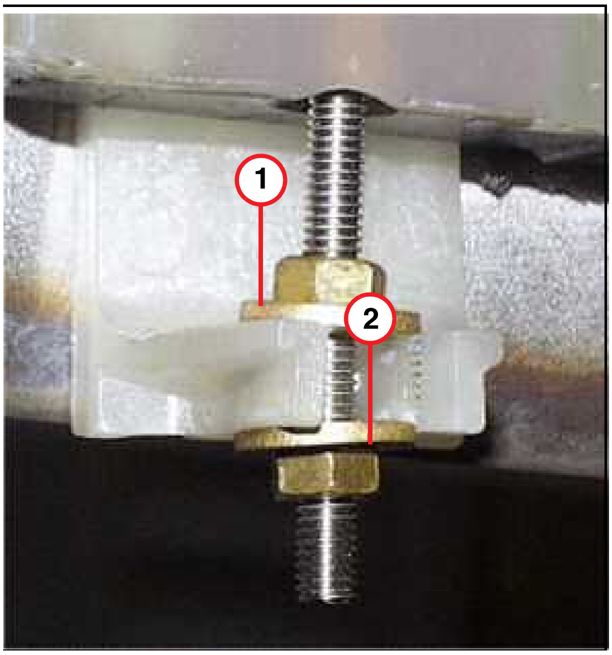
  The length of the threaded bolts can be adjusted to attach the safety washers and nuts, if the wedge cannot be pushed in far enough. (Use a 3-mm Allen key).

Fig. 76: Example of tools: Plastic hammer and synthetic holders

- Using a plastic hammer and synthetic or delivered brass holders, lightly hammer in all wedges.
- Verify the correct Z-adjustments repeatedly.
- Recheck the horizontal position (i.e. possible shifting) of the GC by using a level across the support beams connected to the sliding rods.
- Recheck the radial distances of the GC **"R"**.
- If all distances are correct, hammer in the wedges as follows:
  - Simultaneously hammer in the wedges that are located in the same position in back and front of the magnet.

    Start with those in the lower back and front of the magnet, then move on to the upper back and front of the magnet.
  - Measure the radial distance **"R"** repeatedly.

Fig. 77: Example of mounted wedge with threaded rod



(1) Back nut and washer
(2) Front nut and washer

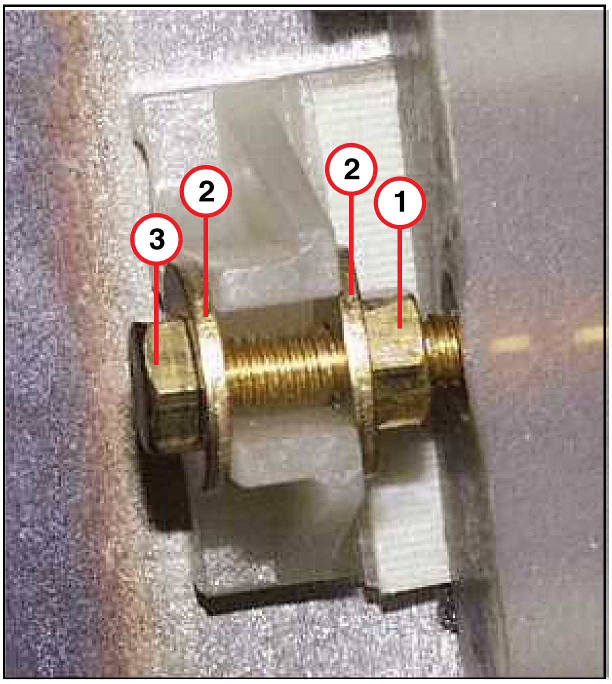
If all distances are correct:

- Finally, hammer in all wedges crosswise in front and back of the magnet.
- Place washers and nuts on the threaded bolts for the wedges or mount the screws with washers and nut. For securing apply Loctite 221.
- If threaded bolts are used:

  Twist the back nuts to the front until the washers touch the wedge. Then firmly tighten the front nuts.

  Secure the rods and nuts with Loctite 221.

Fig. 78: Example of mounted wedge with screw



(1) Back nut
(2) Washer
(3) Screw

- If screws are used:

  Tighten the screws and then the back nut. Secure the screws and nuts with Loctite 221.
- The wedge should have a distance of 15 mm to the coil, so you can insert nut (1) and washer (2).

  Tighten the screw (3) until the wedge is preloaded and the distance to the coil is approx. 10 mm.

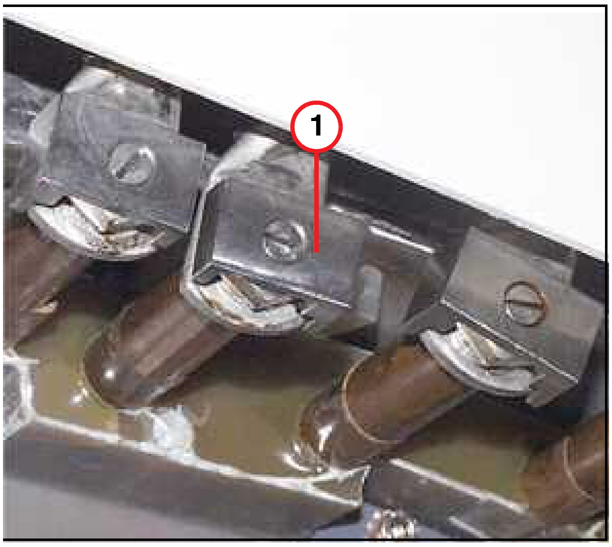
7.2.5.3 Replacement of coaxial link Cables at Magnet

(The coaxial link cables are part of the GC/BC slim complete)

> Please follow the detailed mounting procedure for the coaxial link cables under (→ Gradient Cables (Espree) / M6-020.841.16).

7.2.5.4 Concluding Steps of the GC/BC Replacement

Fig. 79: Gradient cable connections at GC



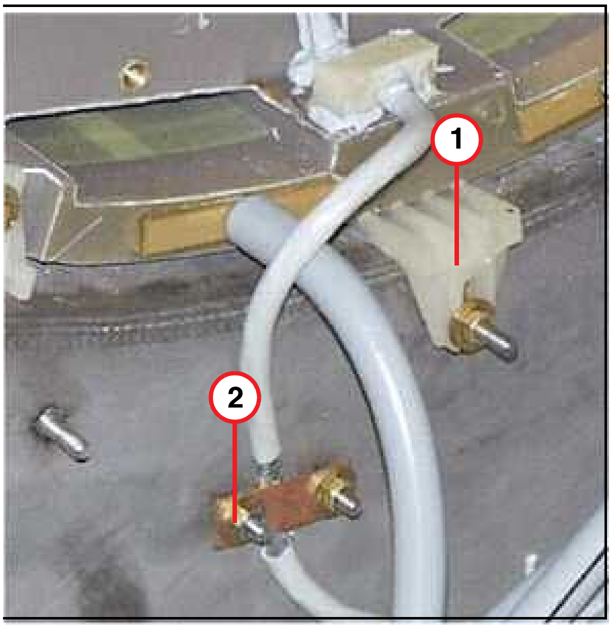
(1) Securing clamp

- Disassemble the support beams of the lifting device and the sliding carriages.
- Push in and fasten the shim trays.
- **Note:**

  Attach the foam rubber strips between the magnet bore and the GC.

  The foam rubber strips come with the new coils.
- Reconnect the gradient cables. Use a torque of 22 +/- 1 Nm for the screw connections. Mount the securing clamps on top, parallel to the cable terminal. (Minimum distance of non-insulated parts: 4 mm). Secure the screws of the securing clamps with Loctite 221.

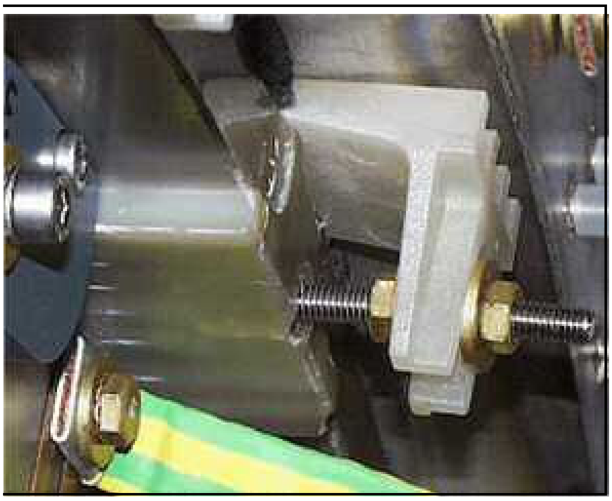
Fig. 80: Bottom of the GC/BC, back view



(1) Lower wedge secured at the magnet
(2) Clamp for grounding cable shield at the magnet (cable <W 5103>)

- Mount the cover above the gradient coil connections.
- Reconnect electrical shim and temperature sensor cables.

Fig. 81: Ground braid at the GC and an example of a mounted wedge



- Reconnect the grounding braid to GC in back of the magnet.

Fig. 82: Mounted wedge with screw

- Reconnect the water quick couplings.

7.2.5.5 Installing the new body coil

CAUTION

The foamed plastics and the plastic film around the body resonator serve to provide noise protection. Do not remove this cover.

» Never deposit the body resonator on a sharp-edged surface. The plastic cover, damping material and the capacitors could be damaged.

Place a sheet of foam on the patient table to protect the BC from damage.

- **Connect the QLA cables to the RF pick-up coils at the BC:**

Fig. 83: Position of QLA connector of pick-up coil on the BC; distance to back edge of BC is 180 mm.

(1) QLA connector (one of the two connectors to the right and left on the top of the BC)

- Place the new BC on the patient table and put the cabling into the BC bore.
- Slide the new BC on the two rails into the gradient coil.

Use extreme care when moving the BC into the magnet to avoid damaging the RF shield inside the gradient coil as well as the BC cabling.

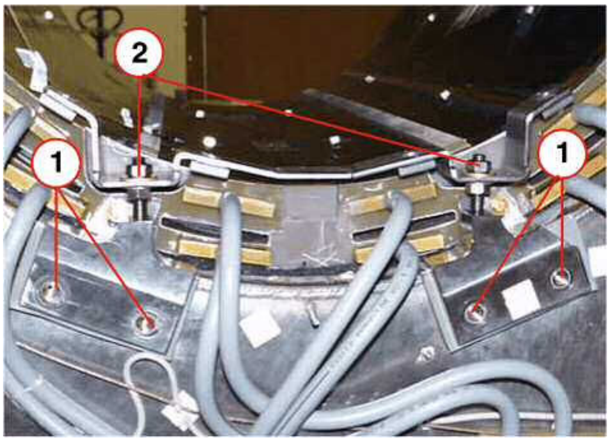
Keep the BC centered at all times on the slide rails in the gradient coil.

Be careful with the RF cables and detuning cable when pulling the BC.

- Assemble the back BC support (→ Fig. 84 Page 127) and route the detuning cable under the support. Place the BC on the coil support, insert all mounting head socket hex screws, and tighten by hand.

  Assemble the front BC support as well.

Fig. 84: BC support at back of the magnet



(1) Assemble BC support at outer vacuum chamber
(2) Clamping nuts

7.2.5.6 BC Alignment

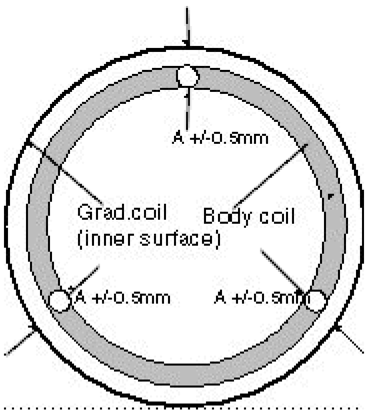
- **Check adjustment in x/y/z -direction:**

  The body coil has to be centered to the gradient coil in x- and y- direction. The distance between the inner surface of the BC and the inner surface of the gradient coil as well as the distance to the magnet bore must be the same all around (tolerance +/- 0.5 mm).

  The distance in z- direction from the front edge of the body coil to the front edge of the magnet has to be 40.25 mm +- 1 mm.

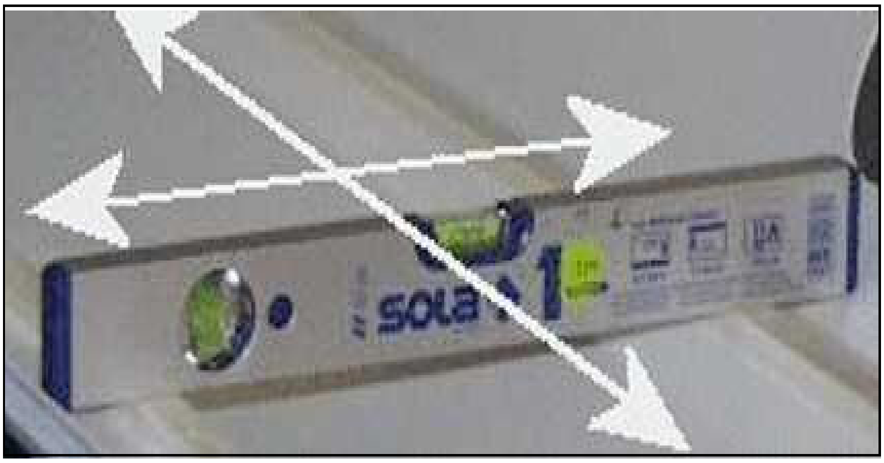
  The BC is NOT centered to the GC in z- direction.

Fig. 85: Centered alignment of BC to GC



Use a spirit level to adjust the body coil horizontally in Z/ X direction at the front and back.

Fig. 86: Leveling the BC in z and x direction



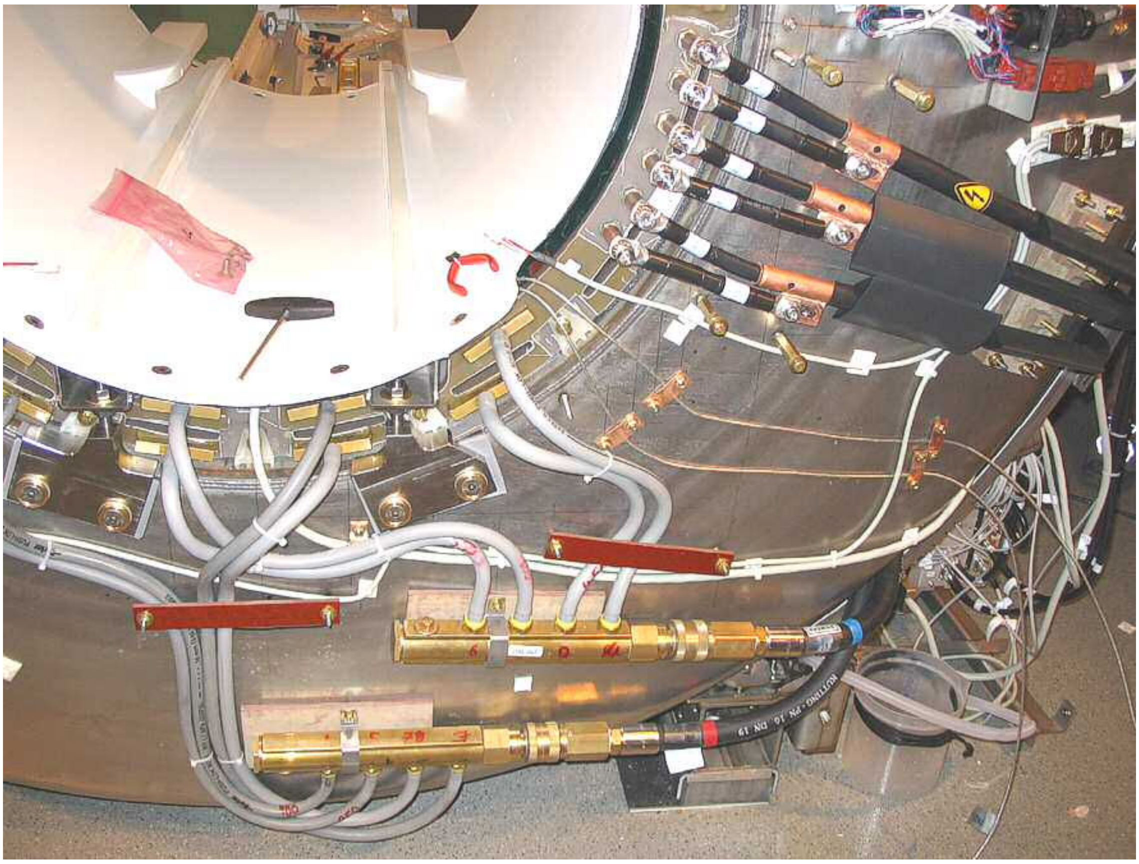
Adjust the X/Y-distances if necessary by loosening the support nuts. Measure at the back and front of the magnet at positions shifted 120 degrees.

In (➔ Fig. 85 Page 127) the three measurement points "A" are shifted in intervals of 120 degrees.

- Concluding steps:
  - Tighten all screws and nuts of the coil support and check alignment again.
  - Route the RF-cables to TALES/X3/X4 and the PIN-diode cable to RFIS/ X9 and tighten the corresponding cable collars at the magnet.

- Install the light bars. Route the bore light cables as shown in (→ Fig. 87 Page 129) Very important: use many cable ties!

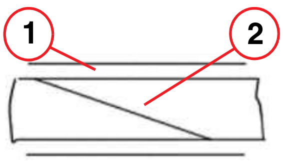
Fig. 87: Cable routing



- Attach the noise damping foam pieces between the gradient coil and the BC.

- Attach the supplied rubber sealing band (Vitolen 120) centrically onto the front and back edge of the BC. The start and end of the band should be cut at an acute angle and overlap when attached at the edge on top of the BC.

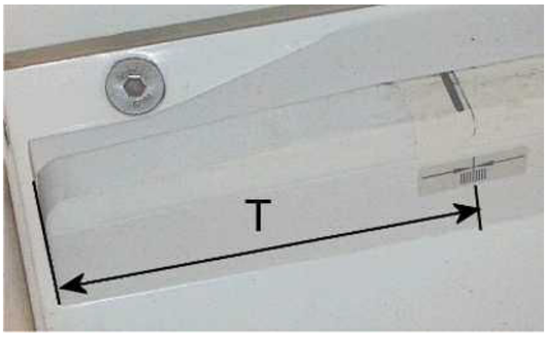
Fig. 88: Overlapping and attached start and end of sealing band on the edge of the BC



(1) Edge of the BC
(2) Attached sealing band

- Attach (if not yet completed) the marker for the shim device at the same Z-distance **T** as measured at the old BC. The marker is delivered in a plastic bag attached at the new BC's bore.

Fig. 89: Position of marker for shim device



7.2.5.7 Restart of MR System

- Refill the water cooling system (for systems with KKT chiller: 35% - 38% ethylene glycol-water mixture); use the fluid drained from the old GC/BC.
- Refill LHe, if necessary.
- Restart the cooling system (water pump), switch on fuses:

  SEP: F3 in ACS.

  IFP: F1 in ACC
- Ramp up the magnet and let stabilize.
- Start-up system/GPA/RF system: Switch on 100A/F1 contactor and GPA/GSSU/F2.

WARNING

Hazardous voltages may be present at the connections of the gradient cables.

Risk of electrical shock!

» **If working at the GC/BC, e.g. changing shim trays, switch off the gradient power amplifier via circuit breaker F2/SSU and F1/100A contactor in the GPA. Wait at least 20 minutes before you start working at the GC/BC.**

- Switch on F11/RFPA in the ACC.
- Switch on F24/RFIS in the ACC.

7.2.5.8 Software

- Install the SP1 Software patch for VB12A, if not already installed.
- Go to Configuration -> Measurement settings -> set gradient to AS022SL
- Click SAVE
- Set the gradient offsets and the Super Conductive Shim Current to zero

7.2.5.9 Procedure

- Switch to Advanced User and open a command tool.
- Type in modifymeaspara
- Set GradOffsets to zero (Refer to (→ Fig. 90 Page 132)).

- Set superConductivityShimCurrent to zero (Refer to (→ Fig. 90 Page 132)).

Fig. 90: ModifyMeasPara

```
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\meduser>modifymeaspara

----------------------------------------------------------------------
ModifyMeasPara V2.0
----------------------------------------------------------------------
   1. LM-MagnCenter               Change LightMarkerDistToMagnetCenterZ
   2. GradOffsets                 Change gradient offsets
   3. SuperConductiveShimCurrent  Change SuperConductiveShimCurrent
----------------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
----------------------------------------------------------------------
MMP> 2
----------------------------------------------------------------------
ATTENTION: The values you entered will be rounded with corresponding
           DAC steps!
----------------------------------------------------------------------
----------------------------------------------------------------------
Enter gradient offset for x-axis [-0.0331128 mT/m]:0
Value not changed.
----------------------------------------------------------------------
Enter gradient offset for y-axis [-0.386078 mT/m]:0
Value not changed.
----------------------------------------------------------------------
Enter gradient offset for z-axis [0.783131 mT/m]:0
Value not changed.

----------------------------------------------------------------------
ModifyMeasPara V2.0
----------------------------------------------------------------------
   1. LM-MagnCenter               Change LightMarkerDistToMagnetCenterZ
   2. GradOffsets                 Change gradient offsets
   3. SuperConductiveShimCurrent  Change SuperConductiveShimCurrent
----------------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
----------------------------------------------------------------------
MMP> 3
----------------------------------------------------------------------
Enter value for superConductivityShimCurrent [-9833 mA]:0
Value not changed.

----------------------------------------------------------------------
ModifyMeasPara V2.0
----------------------------------------------------------------------
   1. LM-MagnCenter               Change LightMarkerDistToMagnetCenterZ
   2. GradOffsets                 Change gradient offsets
   3. SuperConductiveShimCurrent  Change SuperConductiveShimCurrent
----------------------------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)q
----------------------------------------------------------------------
MMP>
```

- Quit the menu with q and close the command tool window.

- **Reboot the system.**

7.2.5.10 Reshim the Magnet

For shimming refer to the Installation Instructions -> Startup -> Ramping the magnet -> Shim

Start with the **bare** plot.

Make sure the shim trays are empty.

Install the Array Shim Device

7.2.5.11 Installing the front cover/funnel

- Install the covers in reverse order as described in(→ Removing the front covers/ funnel / Page 141) .
- Measure the width of the gaps between BC and patient table. It should be equal on both sides and less than 8 mm; for details of the procedure, refer to UI MR036/05/SI.

7.2.6 Adjustments

- Perform a full TuneUp.

> If the TuneUp step Coil Power Losses fails, contact HSC.

Title:	TuneUp/QA measurement step Coil Power Losses fails	
Problem:	**What:**	After slim line coil upgrade, the TuneUp/QA step Coil Power Losses fails
	Where:	n.a.
	When:	After slim line coil upgrade
	Extent:	All Magnetom Espree systems running with VB12A and VB12A-SP1

Corrective Actions

Verification: n.a.

Solution:

Perform the following workaround in order to get the TuneUp/QA measurement step coil power loss working:

1. Enable Advanced User
2. Open a cmd tool window (Start -> Run and type in cmd and press <ENTER>)
3. Type in the following command:

   `C:\MedCom\bin\InstallCoils.exe REGISTRY BARE and press <ENTER>`
4. Wait until the command has finished.
5. Close the cmd tool window
6. Now perform the Tune up/QA measurement step coil power loss
7. Reboot the system (MRC/MRIR and MPCU)

7.2.6.1 BC Tuning

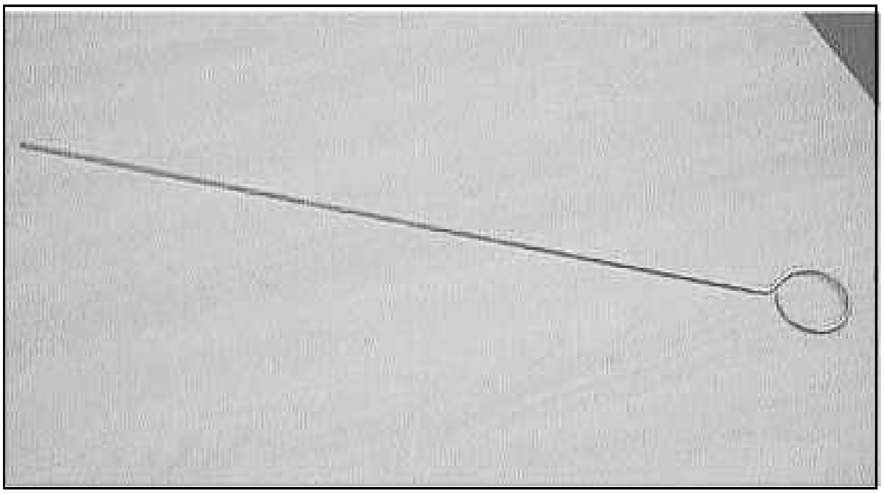
Prerequisites

Since BC Tuning uses the H and S matrices to perform calculations, the tuning calibration tune-up procedure has to be completed successfully before starting with the BC tuning.

To gain access to the trimmer capacitors at the back of the Body Coil, the back magnet cover has to be removed.

The special non-magnetic adjustment tool (i.e. long screwdriver, material number 73 66 771) is required to turn the trimmer capacitors. Refer to (➔ Fig. 91 Page 134).

Fig. 91: Tool for trimmer capacitor adjustment



- Switch on ACC/ F11 (RFPA) and F24 (RFIS).

- Check under SeSo / Reports / MR Configuration: The correct gradient coil under “Measurement Settings" has to be shown: **“AS022SL”**. If not , e.g., after replacement of an old GC/BC, the coil has to be configured.

The following measurement Tune-up / BC Tuning should be done with the back funnel removed and repeated with the back funnel installed.

Frequency and Decoupling Adjustment

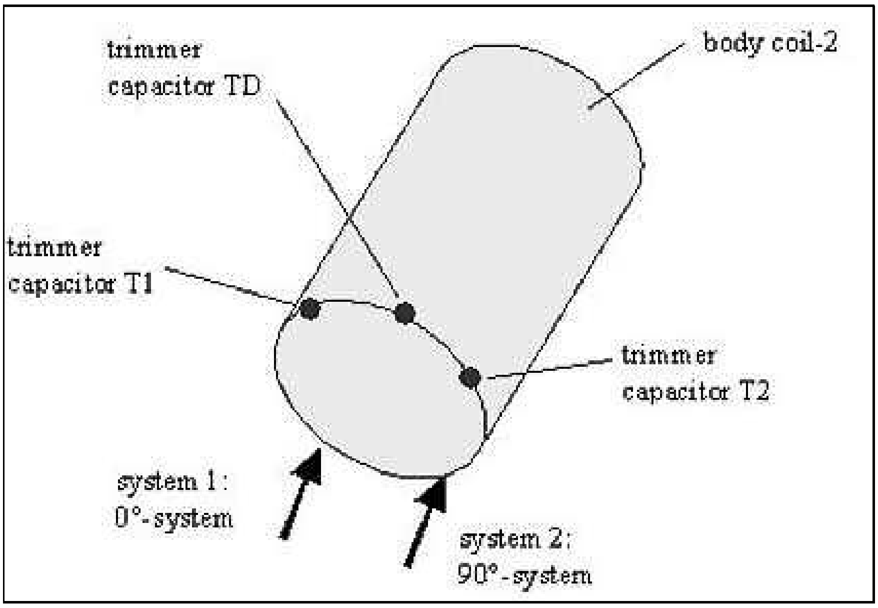
You will be guided through iterative fine adjustments. Between the adjustment steps, the resonance and decoupling of both systems (0 and 90 degree) can be varied manually using the three trim capacitors at the coil. The program stops when both resonance and decoupling are within specifications.

The trim capacitors T2 and T1 affect the resonance frequencies of systems 1 and 2. However, capacitor TD affects the decoupling between the two systems.

The decoupling has to be adjusted only if the measurement of the transmission value is not within specifications.

The trim capacitors allow 53 turns. By default, they are set in the middle position, which means you can make 26+1/2 turns clockwise or 26+1/2 turns counter-clockwise.

Fig. 92: BC Tuning - trimmer capacitors, (T1 left, TD middle, T2 right)



To prevent damage to the trimmers, use caution when approaching the end position.

Do not use excessive force.

Trimmer capacitor	Main influence on	Clockwise (cw)	Counterclockwise (ccw)
T2	Resonance frequency system 1	Decrease resonance frequency 1	Increase resonance frequency 1

Trimmer capacitor	Main influence on	Clockwise (cw)	Counterclockwise (ccw)
T1	Resonance frequency system 2	Decrease resonance frequency 2	Increase resonance frequency 2
TD	Decoupling between systems 1 and 2	Increase capacity	Decrease capacity

If the values are slightly out of spec, the system will still work.

Be aware that trimmer capacitor T2 influences system 1 and vice versa.

<table>
<tr><th>Step 1</th><th colspan="3">Reflection adjustment of delta frequency</th></tr>
<tr><td rowspan="6"></td><td colspan="3">delta < 100 kHz ?</td></tr>
<tr><td>Yes</td><td colspan="2">No</td></tr>
<tr><td rowspan="4">No action required</td><td colspan="2">Frequency 0 degree > frequency 90 degrees</td></tr>
<tr><td>Yes</td><td>No</td></tr>
<tr><td>Turn T2 2x cw
Turn T1 2x ccw</td><td>Turn T2 2x ccw
Turn T1 2x cw</td></tr>
<tr><td colspan="2">Go to step 1</td></tr>
</table>

<table>
<tr><th>Step 2</th><th colspan="2">Mechanical adjustment of delta decoupling (transmission)</th></tr>
<tr><td rowspan="4"></td><td colspan="2">Transmission 0... -12 dB</td></tr>
<tr><td>Yes</td><td>No</td></tr>
<tr><td>Adjust mechanical position of the BC in the gradient coil.</td><td rowspan="2">Go to Step 3</td></tr>
<tr><td>Go to step 2</td></tr>
</table>

<table>
<tr><th>Step 3</th><th colspan="2">Electrical adjustment of decoupling (transmission)</th></tr>
<tr><td rowspan="2"></td><td colspan="2">Transmission T = -12 dB .. -18 dB</td></tr>
<tr><td>Yes</td><td>No</td></tr>
</table>

Step 3	Electrical adjustment of decoupling (transmission)		
Step 3.1	Turn TD 3x cw, press start button at panel		
	T lower		
	Yes	No	
Step 3.1.1	Turn TD 3x cw, press start button at panel	Turn TD 3x ccw, press start button at panel	No action required
	T lower -> go to 3.1.1	T lower -> go to 3.1.1	
	No -> undo last step	No -> undo last step	
If necessary, repeat step 3.1			

7.2.6.2 Remarks about BC tuning

- T1 and T2 also influence the decoupling.
- TD also influences the frequency.
- The influence of TD relative to T1 and T2 on the decoupling is four times higher. However, it is only half as high with respect to the frequency of the two partial systems.
- To change the decoupling in the same direction, turn TD in the opposite direction as T1 and T2.

Frequency adjustment with constant decoupling:

0-degree system:

Turn T1 x turns and TD x/2 turns in the same direction.

90-degree system:

Turn T2 x turns and TD x/2 turns in the same direction.

Decoupling adjustment with constant frequency:

Turn TD x turns and T1,T2 x/2 turns in opposite direction.

7.2.7 Concluding Steps

- Install the remaining magnet covers (➔ Magnet Covers / Page 84)
- Perform TU/ Coil Power Losses again

7.2.8 Final Checks

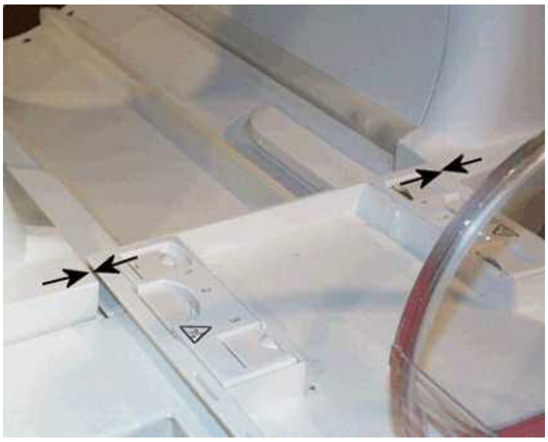
CAUTION

The width of the gaps between the patient table and the front cover on right and left side as well as the body coil and patient bore light bars should be of equal size and less than 8 mm. (Refer to the Start-up instructions)

Danger of injury to patient by jamming fingers.

» **Minimize the gaps by adjusting the covers relative to the Body coil. Refer to (➔ System Start-up / M6-020.815.01).**

Fig. 93: Gap between patient table and cover / bore



- Perform full QA.

If QA/ Image Orientation Check fails, contact HSC.

Title:	Image Orientation Check failed on new Espree installations	
Problem:	**What:**	Image Orientation Check fails
	Where:	n.a.
	When:	Performing the QA Checks
	Extent:	On newly delivered MR Espree systems and on slim line upgrades with VB12A-SP1 installed.
Corrective Actions	**Verification:**	Check the following files on the MRC:

- C:\MedCom\MriSiteData\GradientCoil\coeff.grad
- Q:\MriSiteData\GradientCoil\coeff.grad

They should be the same size. If the size of Q:\MriSiteData\GradientCoil\coeff.grad is smaller than the file under C:\MedCom\MriSiteData\GradientCoil\coeff.grad perform the following workaround:

The file on share drive Q is on the MRIR.

Other possible causes: n.a.

Solution:

1. enable Advanced User
2. start Explorer
3. rename file Q:\MriSiteData\GradientCoil\coeff.grad to Q:\MriSiteData\GradientCoil\coeff.grad.old
4. copy the correct gradient file (e.g. coeff_AS022SL.grad) from C:\MedCom\MriProduct\GradientCoils to Q:\MriSiteData\GradientCoil\
5. rename it coeff.grad.

This problem will be fixed in syngo MR B13

- **Perform Backup / Restore to save the TuneUp data.**

7.3 Body Coil (Slim Line) Espree

(material number 10048071)

CAUTION

Strong magnetic field.

Magnetic objects may cause injury

» **Maintain a safe distance to the magnetic field.**

CAUTION

The foamed plastics and the plastic film around the body resonator serve to provide noise protection. Do not remove this cover.

» **Never deposit the body resonator on a sharp-edged surface. The plastic cover, damping material and the capacitors could be damaged.**

WARNING

During the BC replacement, you have to work close to the non-isolated connections of the gradient cables at the gradient coil.

Hazardous voltages may be present at the connections of the gradient cables.

Risk of electrical shock!

» **Switch off the gradient power amplifier via circuit breaker F2/SSU and F1/100 A contactor in the GPA. Wait at least 20 minutes before you start working at the BC.**

7.3.1 Tools and time required

7.3.1.1 Time

5.5 hr.

7.3.1.2 Tools

- Non-magnetic tool kit
- Non-magnetic measuring tape or gauge
- Non-magnetic spirit level
- Special non-magnetic screwdriver (material number 73 66 771) for BC trimmer capacitor adjustment.
- BNC 2m Service cable for step TU/ Tuning Calibration (8733479).

- Set of RF adapters including short and 50 ohm terminators for step TU/ Tuning Calibration

7.3.2 Preliminary

- Switch off GPA/GSSU/F2 and the F1/100A contactor in the GPA.
- Switch off ACC/ F11 (RFPA) and F24 (RFIS).
- Move the patient table down.

7.3.3 Disassembly

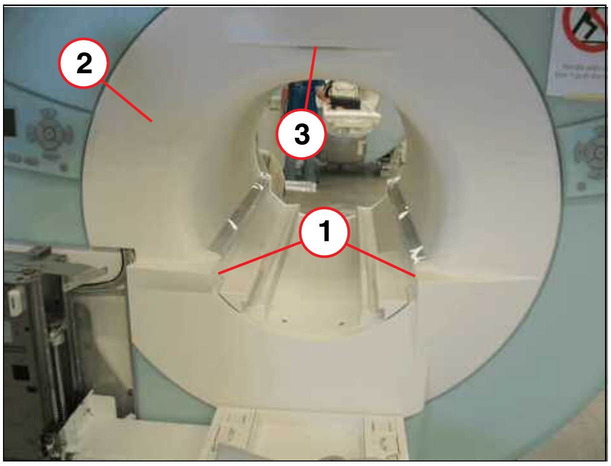
7.3.3.1 Magnet Covers

For details, refer to (→ Espree covers / M6-020.841).

Removing the front covers/funnel

Removing the outer front cover

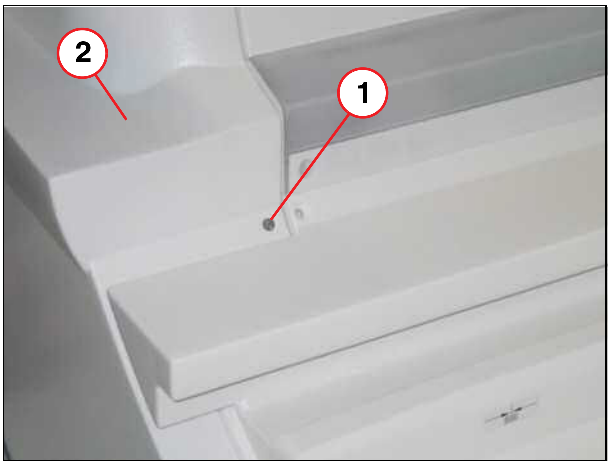
Fig. 94: Funnel



(1) Screws
(2) Funnel
(3) Laser localizer faceplate

- Remove the two caps over the set screws securing the outer cover at the inner front cover (left and right side).

Fig. 95: Funnel, left side



(1) Screw
(2) Funnel

- Loosen the two set screws.
- Remove the two screws at the light marker.

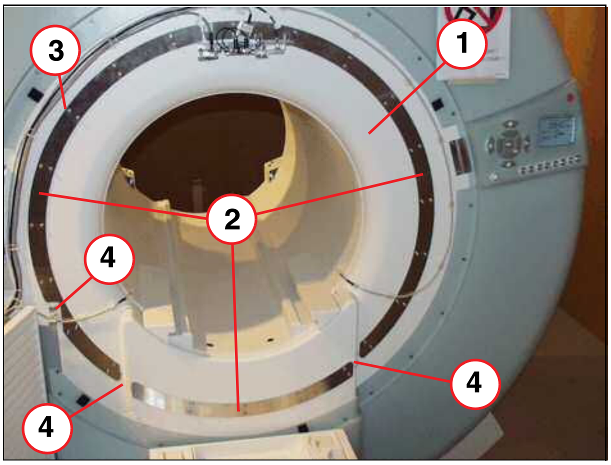
Fig. 96: Two screws at light marker

(1) Screws

- Pull the front cover from the magnet (attached via Velcro).

Removing the front insert funnel

Fig. 97: Front insert funnel (example)



(1) Front insert funnel
(2) Connection rings
(3) Screw
(4) Seal

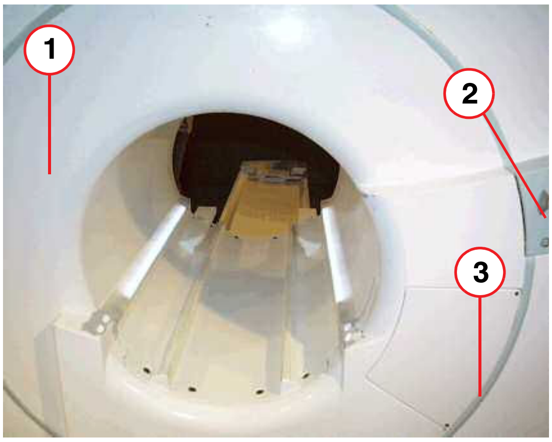
- Disconnect the plug of the light marker.

Fig. 98: Disconnecting the light marker plug

- Remove the outside screws (3) on the connection ring.
- Remove the lifting column top covering.
- Remove the front insert funnel.

Removing the back Funnel

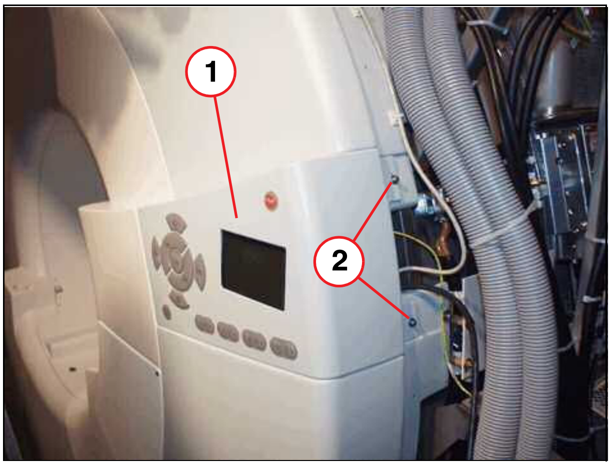
Fig. 99: Removing the back funnel



(1) Back funnel
(2) Second or dummy operating unit
(3) Silicon sealing ring

- Slide up and remove the service access cover (left side of magnet seen from patient end).

Fig. 100: Rear operating unit 1



(1) Rear operating unit
(2) Screws

Remove the dummy unit / rear operating unit (1):

- Remove the two screws (2) reachable from the service access cover side.
- Tilt out the back operating unit and disconnect the cables.
- Remove the back operating unit.

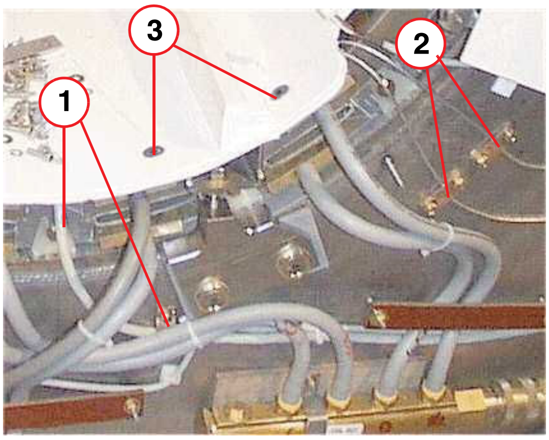
- Remove the silicon sealing ring.
- Remove the eight screws (M8, secured with Omnifit 100M) on the perimeter of the back funnel.
- Pull the funnel out 20 mm in horizontal direction; the funnel then lowers 10 mm.Tilt the back cover slightly with the top out of the bore and loosen the securing strap on

the top. Tilt the funnel at approx. 30 degrees outwards and remove the funnel by lifting it upward and out.

- Disconnect the plugs of the patient bore light.
- Remove the two patient bore light rails (mounted with three plastic screws on both sides below the rails at the BC bore).

7.3.3.2 BC

Fig. 101: BC cabling

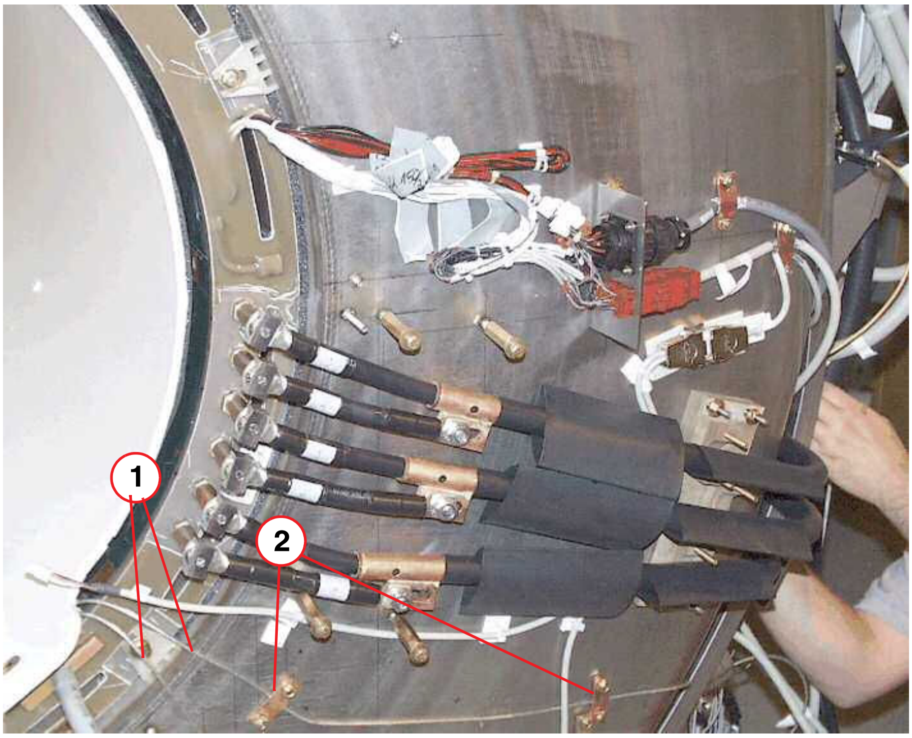


(1) PIN-diode / cable and collar at magnet
(2) Two thin RF-cables / cables and collars at magnet
(3) Head socket screws at BC

- Disconnect the RF cables from TALES/X3/X4 and loosen the four cable collars (ground fixings of the cable shield) at the back at the magnet.

Two of the RF cable collars are positioned under the outer back cover and may be not so easily accessed.

Fig. 102: GC/BC cabling at magnet service side



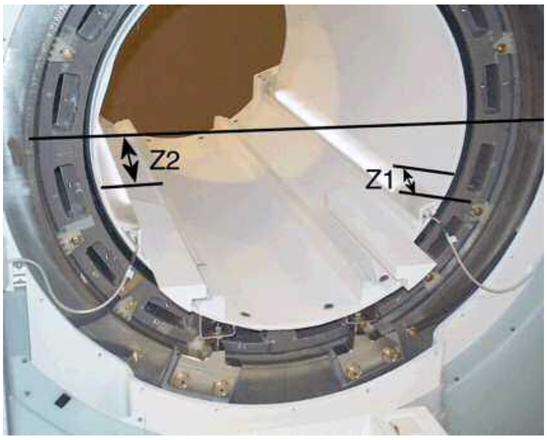
(1) RF cables
(2) Upper RF cable collars at magnet

- Disconnect X9 at RFIS and loosen the cable collar (ground fixing of the cable shield) at the back of the magnet.

  Carefully pull the cabling out of the cover.

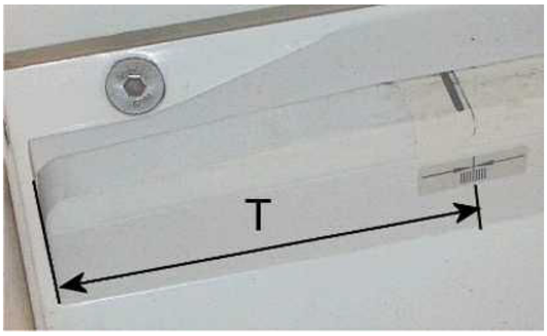
Measurements before removing the old BC

Fig. 103: BC alignment in z-direction



- Measure and note the Z-distance between front edge of the body coil and the front edge of the gradient coil (**Z1**).
- Measure and note the Z-distance between front edge of the body coil and the edge of magnet bore (**Z2**): Measure from the edge of a non-magnetic spirit level held across the magnet bore to the front edge of the BC. Z2 should be 40.25 mm +/- 1 mm.
- Measure and note the Z-distance (**T**) of the marker for the shim device.

Fig. 104: Position of marker for shim device



Removing the old BC

- If there, remove the foam pieces (noise damping) between the gradient coil and BC.
- Loosen and remove the four head socket hex screws at the BC supports on each side.
- **Before removing the BC, disconnect the QLA cables from the RF pick-up coils at the BC:**

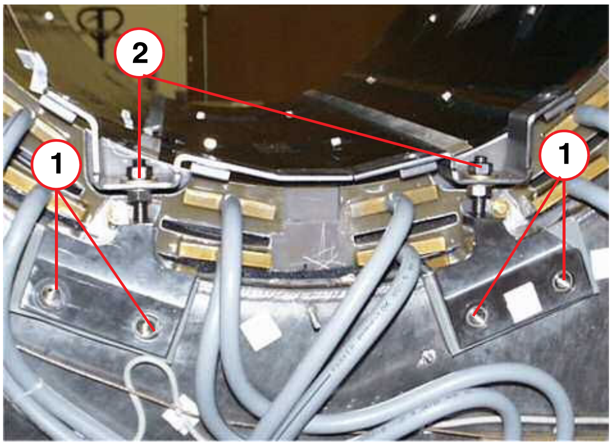
  To access the connectors (➔ Fig. 105 Page 148), pull the BC approximately 20 cm backwards out of the gradient coil:

  For this purpose, disassemble the BC support at the back completely. The upper clamping nuts of the support remain fixed to assure the correct reassembling position of the BC(➔ Fig. 106 Page 148) .

Fig. 105: Position of QLA connector of pick-up coil on the BC; distance to back edge of BC is 180 mm.

(1) QLA connector (one of the two connectors to the right and left on the top of the BC)

Fig. 106: BC support at back of the magnet

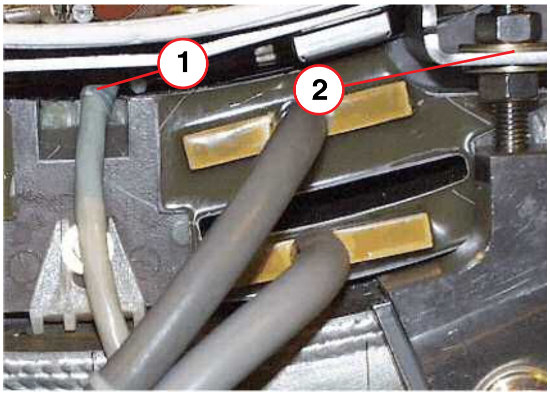


(1) Disassemble BC support from OVC
(2) Clamping nuts remain fixed

» The BC is now ready to be removed. At least two people are needed to remove the BC. The weight of the BC is approximately 90 kg.

> Put the RF cables and the detuning cable in the BC bore and secure with adhesive tape.
> Be careful with the RF cables and detuning cable when pulling the BC.

Fig. 107: Routing of the PIN-diode cable



(1) PIN-diode cable between the BC support and the gradient coil
(2) BC support / Loosen the nuts to lift the support

- Lift the BC slightly at magnet service end away from the supports. Slowly push and pull the BC, which is centered on the rails in the gradient coil, out of the gradient coil. Keep the BC centered all times.

  Lift the service end support to route the PIN diode cable connector.

  Be careful not to destroy the cabling, the plastic film and foamed plastic at the BC, as well as block capacitors on the RF shield, should the BC get caught due to not being centered.
- Store the BC in a suitable place on the floor:

Place a sheet of foam under the BC to protect it from damage.

7.3.4 Assembly

Place a sheet of foam on the patient table to protect the BC from damage.

- Place the new BC on the patient table and put the cabling into the BC bore.
- Slide the new BC on the two rails into the gradient coil.

Use extreme care when moving the BC into the magnet to avoid damaging the RF shield inside the gradient coil as well as the BC cabling.

Keep the BC centered at all times on the slide rails in the gradient coil.

- **Connect the QLA cables to the RF pick-up coils at the BC:**

  To access the connectors (→ Fig. 105 Page 148), pull the BC approximately 20 cm backwards out of the gradient coil.
  Be careful with the RF cables and detuning cable when pulling the BC.

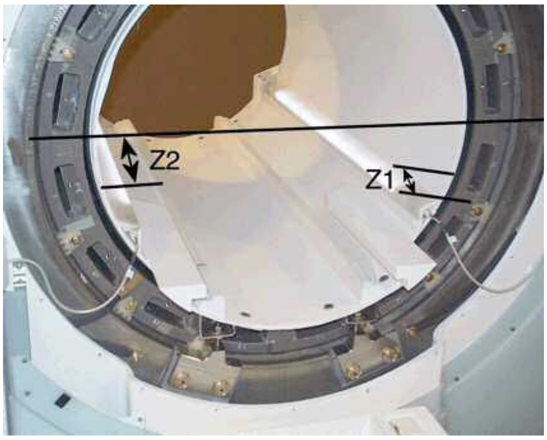
- Reassemble the back BC support (→ Fig. 106 Page 148) and route the detuning cable under the support. Place the BC on the coil support, insert all mounting head socket hex screws, and tighten by hand.

7.3.4.1 BC Alignment

- **Check adjustment in Z-direction:**

  Measure the distances **Z1** and **Z2** between the BC and gradient coil front edges and the magnet front bore to the BC as described in (→ Measurements before removing the old BC / Page 147) and compare with the previous measurements at the old coil.

Fig. 108: BC alignment in z-direction

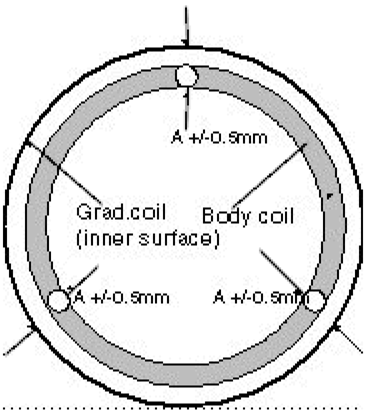


Adjust the Z-distance if necessary.

- **Check adjustment in x/y-direction:**

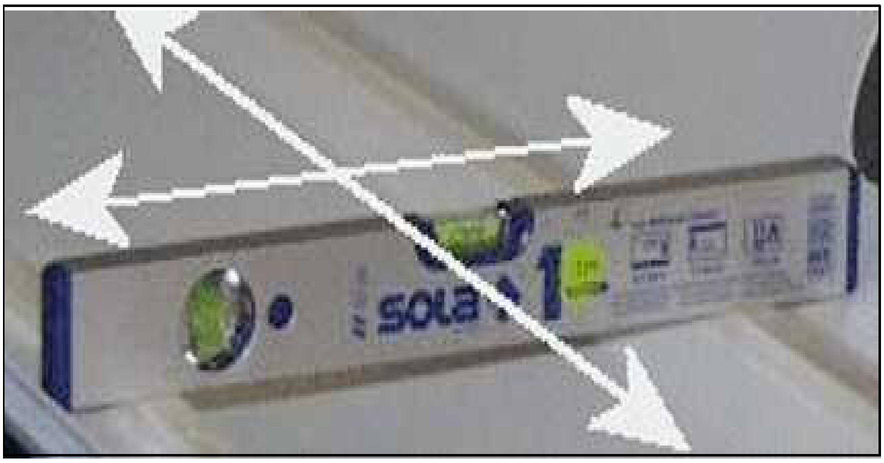
  The body coil has to be centered to the gradient coil. The distance between the inner surface of the BC and the inner surface of the gradient coil as well as the distance to the magnet bore must be the same all around (tolerance +/- 0.5 mm).

Fig. 109: Centered alignment of BC to GC



Use a non-magnetic spirit level to adjust the body coil horizontally in X/Y direction at the front and back.

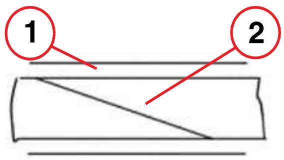
Fig. 110: Leveling the BC in z and x direction



Adjust the X/Y-distances if necessary by loosening the support nuts. Measure at the back and front of the magnet at positions shifted 120 degrees.

- Concluding steps:
  - Tighten all screws and nuts of the coil support and check alignment again.
  - Route the RF-cables to TALES/X3/X4 and the PIN-diode cable to RFIS/ X9 and tighten the corresponding cable collars at the magnet.
  - If there, attach the noise damping foam pieces again in their previous positions between the gradient coil and the BC.
- Attach the supplied rubber sealing band (Vitolen 120) centrically onto the front and back edge of the BC. The start and end of the band should be cut at an acute angle and overlap when attached at the edge on top of the BC.

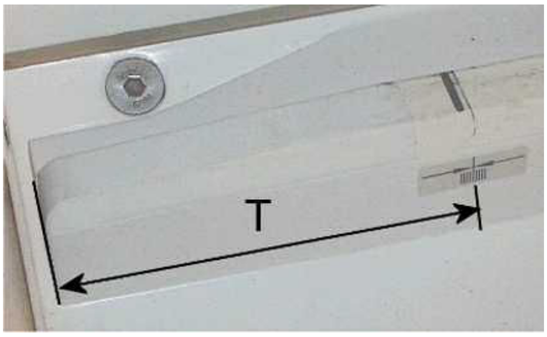
Fig. 111: Overlapping and attached start and end of sealing band on the edge of the BC



(1) Edge of the BC
(2) Attached sealing band

- Attach (if not yet completed) the marker for the shim device at the same Z-distance **T** as measured at the old BC. The marker is delivered in a plastic bag attached at the new BC's bore.

Fig. 112: Position of marker for shim device



- Mount the patient lighting bars.

7.3.4.2 Installing the front cover/funnel

- Install the covers in reverse order as described in(→ Removing the front covers/ funnel / Page 141) .
- Measure the width of the gaps between BC and patient table. It should be equal on both sides and less than 8 mm; for details of the procedure, refer to (→ System Start-up / M6-020.815.01).

7.3.5 Adjustments

- Perform TU/ Tuning Calibration.
- Perform TU/ BC Tuning.
- Perform TU/ RF Characteristic
- Perform TU/ Coil Power Losses.
- Perform TU/ Abs.Rec.Path Calibration or BC Image Brightness

7.3.5.1 BC Tuning

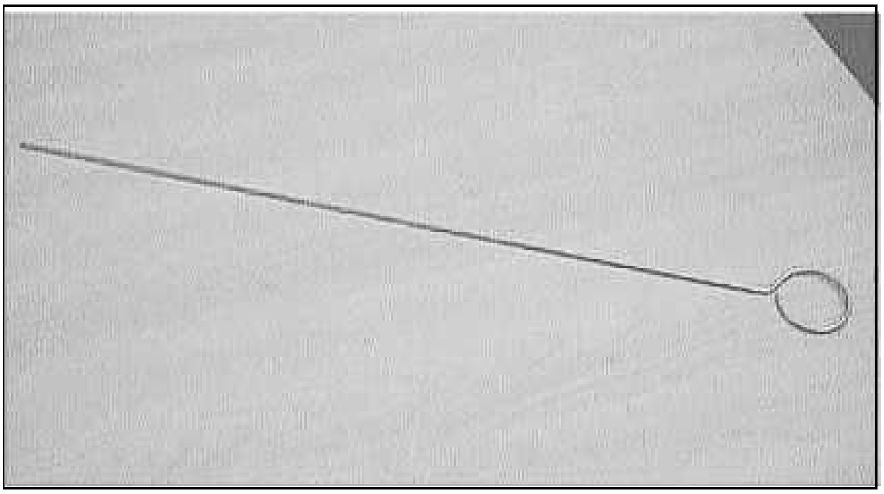
Prerequisites

Since BC Tuning uses the H and S matrices to perform calculations, the tuning calibration tune-up procedure has to be completed successfully before starting with the BC tuning.

To gain access to the trimmer capacitors at the back of the Body Coil, the back magnet cover has to be removed.

The special non-magnetic adjustment tool (i.e. long screwdriver, material number 73 66 771) is required to turn the trimmer capacitors. Refer to (→ Fig. 113 Page 153).

Fig. 113: Tool for trimmer capacitor adjustment



- Switch on ACC/ F11 (RFPA) and F24 (RFIS).
- Check under SeSo / Reports / MR Configuration: The correct gradient coil under “Measurement Settings" has to be shown: **“AS022SL”**. If not , e.g., after replacement of an old GC/BC, the coil has to be configured.

The following measurement Tune-up / BC Tuning should be done with the back funnel removed and repeated with the back funnel installed.

Frequency and Decoupling Adjustment

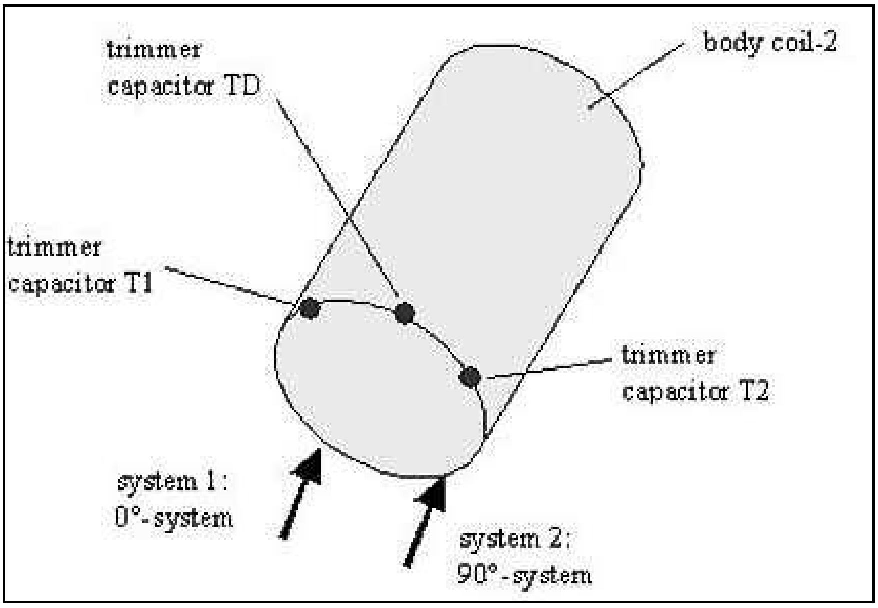
You will be guided through iterative fine adjustments. Between the adjustment steps, the resonance and decoupling of both systems (0 and 90 degree) can be varied manually using the three trimmer capacitors at the coil. The program stops when both resonance and decoupling are within specifications.

The trimmer capacitors T2 and T1 affect the resonance frequencies of systems 1 and 2. However, capacitor TD affects the decoupling between the two systems.

The decoupling has to be adjusted only if the measurement of the transmission value is not within specifications.

The trimmer capacitors allow 53 turns. By default, they are set in the middle position, which means you can make 26+1/2 turns clockwise or 26+1/2 turns counter-clockwise.

Fig. 114: BC Tuning - trimmer capacitors, (T1 left, TD middle, T2 right)



To prevent damage to the trimmers, use caution when approaching the end position.

Do not use excessive force.

Trimmer capacitor	Main influence on	Clockwise (cw)	Counterclockwise (ccw)
T2	Resonance frequency system 1	Decrease resonance frequency 1	Increase resonance frequency 1
T1	Resonance frequency system 2	Decrease resonance frequency 2	Increase resonance frequency 2
TD	Decoupling between systems 1 and 2	Increase capacitance	Decrease capacitance

If the values are slightly out of spec, the system will still work.

Be aware that trimmer capacitor T2 influences system 1 and vice versa.

<table>
<tr><th>Step 1</th><th colspan="3">Reflection adjustment of delta frequency</th></tr>
<tr><td rowspan="6"></td><td colspan="3">delta < 100 kHz ?</td></tr>
<tr><td>Yes</td><td colspan="2">No</td></tr>
<tr><td rowspan="4">No action required</td><td colspan="2">Frequency 0 degree > frequency 90 degrees</td></tr>
<tr><td>Yes</td><td>No</td></tr>
<tr><td>Turn T2 2x cw
Turn T1 2x ccw</td><td>Turn T2 2x ccw
Turn T1 2x cw</td></tr>
<tr><td colspan="2">Go to step 1</td></tr>
</table>

<table>
<tr><th>Step 2</th><th colspan="2">Mechanical adjustment of decoupling (transmission)</th></tr>
<tr><td rowspan="4"></td><td colspan="2">Transmission 0... -12 dB</td></tr>
<tr><td>Yes</td><td>No</td></tr>
<tr><td>Adjust mechanical position of the BC in the gradient coil.</td><td rowspan="2">Go to Step 3</td></tr>
<tr><td>Go to step 2</td></tr>
</table>

<table>
<tr><th>Step 3</th><th colspan="3">Electrical adjustment of decoupling (transmission)</th></tr>
<tr><td rowspan="2"></td><td colspan="3">Transmission T = -12 dB .. -18 dB</td></tr>
<tr><td colspan="2">Yes</td><td>No</td></tr>
<tr><td rowspan="3">Step 3.1</td><td colspan="2">Turn TD 3x cw, press start button at panel</td><td rowspan="7">No action required</td></tr>
<tr><td colspan="2">T lower</td></tr>
<tr><td>Yes</td><td>No</td></tr>
<tr><td rowspan="3">Step 3.1.1</td><td>Turn TD 3x cw, press start button at panel</td><td>Turn TD 3x ccw, press start button at panel</td></tr>
<tr><td>T lower -> go to 3.1.1</td><td>T lower -> go to 3.1.1</td></tr>
<tr><td>No -> undo last step</td><td>No -> undo last step</td></tr>
<tr><td colspan="3">If necessary, repeat step 3.1</td></tr>
</table>

7.3.5.2 Remarks about BC tuning

- T1 and T2 also influence the decoupling.
- TD also influences the frequency.
- The influence of TD relative to T1 and T2 on the decoupling is four times higher. However, it is only half as high with respect to the frequency of the two partial systems.

- To change the decoupling in the same direction, turn TD in the opposite direction as T1 and T2.

Frequency adjustment with constant decoupling:

0-degree system:

Turn T1 x turns and TD x/2 turns in the same direction.

90-degree system:

Turn T2 x turns and TD x/2 turns in the same direction.

Decoupling adjustment with constant frequency:

Turn TD x turns and T1,T2 x/2 turns in opposite direction.

7.3.6 Concluding Steps

- Install the remaining magnet covers (→ Magnet Covers / Page 84).
- **Perform Backup / Restore to save the TuneUp data.**
- Perform TuneUp/ Coil Power Losses again.

7.3.7 Final Checks

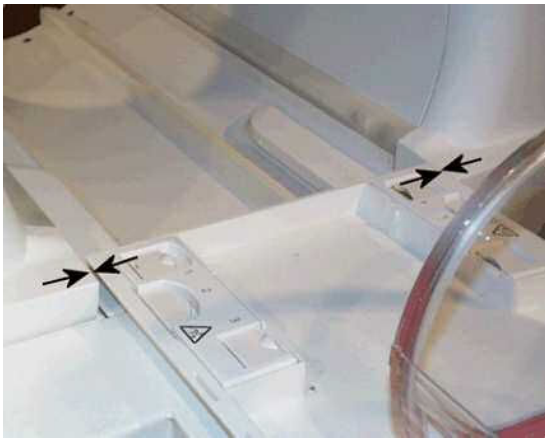
CAUTION

The width of the gaps between the patient table and the front cover on right and left side as well as the body coil and patient bore light rails should be of equal size and less than 8 mm. (Refer to the Start-up instructions)

Danger of injury to patient by jamming fingers.

» **Minimize the clearances by adjusting the covers relative to the Body coil. Refer to (→ System Start-up / M6-020.815.01).**

Fig. 115: Gap between patient table and cover / bore



- Perform QA/ Abs.Rec.Path Calibration or BC Image Check.
- Perform QA/ BC Tuning Check.

- Perform QA/ RF Verify.
- Perform QA/ Coil Power Losses Check.
- Perform QA/ Spike Check.
- Perform QA/ SAR monitor test (VB17A and higher).

8.1 Common Procedure

8.1.1 Tools and time required

8.1.1.1 Time

approx. 30 min., depending on the coil

8.1.1.2 Tools

Specific phantoms and phantom holders

8.1.2 Preparation

- Position the coil with the appropriate phantom set-up.

8.1.3 Final Checks

- Perform QA/ Coil Quality Assurance/ Coil Check for the specific coil.

Version 08

- New names for Avanto-Dot (VD13A) TU/ QA-procedures added.

Version 09

- added graphic (➔ Replacement of coaxial link Cables at Magnet / Page 123)
- Chapter "RFAS", section "TAS_3TM - TAS_CM" - water hose removal procedure improved to avoid damaged brass nozzle (➔ Fig. 33 Page 61).
- Chapter "RFAS", section "TALES, BCCS" - eliminated need to execute TestTools after TALES replacement (➔ Final Checks / Page 45).

Version 10

- All chapters, minor editorial changes.

10 List of Hazard IDs

There are no Hazard IDs in this document.

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